

Target

LPDDR5X Stack Specification

245FBGA, 8.2x12.4
32Gb (16Gb*2) LPDDR5X SDRAM, 2-Channels
16Gb (64Mb x16DQ x4banks x4BGs) per Channel

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Datasheet

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LPDDR5X SDRAM SPECIFICATION**32Gb (16Gb*2) LPDDR5X SDRAM, 2-Channels****245FBGA_8.2x12.4 Package****Per Channel : 16Gb (64Mb x16DQ x4Banks x4BGs)****1.0 KEY FEATURE**

- Double-data rate architecture; two data transfers per WCK clock cycle
- Differential data strobes (WCK_t, WCK_c), These are received with data to be used in capturing data at the receiver.
- Differential data strobes (RDQS_t, RDQS_c), These are transmitted with data to be used in capturing data at the receiver.
- Differential clock inputs (CK_t and CK_c)
- Differential data strobes WCK_t and WCK_c
- Commands & addresses entered on both positive and negative CK edges; data and data mask referenced to both edges of WCK
- 1 channel composition
- 4 internal banks and 4 internal bank-groups at bank-group mode, 16 internal banks at 16 bank mode.
- DMI Pin : DBI (Data Bus Inversion) when normal write and read operation, Data mask (DM) for masked write when DBI off
 - Counting # of DQ's 1 for masked write when DBI on
- Burst Length: 16, 32 (OTF)
- Burst Type: Sequential
- Read & Write latency : Refer to LPDDR5 AC Timing Table.
- Auto Precharge option for each burst access
- Configurable Drive Strength
- Refresh and Self Refresh Modes
- Partial Array Self Refresh and Temperature Compensated Self Refresh
- WCK2CK Leveling
- CA Calibration
- Internal VREF and VREF training
- FIFO based write/read training
- MPC (Multi Purpose Command)
- LVSTL_05 (Low Voltage Swing Terminated Logic) IO
- VDD1/VDD2H/VDD2L/VDDQ : 1.8V/1.05V/0.9V/0.5V
- VSSQ Termination
- No DLL : WCK to RDQS is not synchronized
- Edge aligned data output, write training for data input center align
- Refresh rate : 3.9us

2.0 ORDERING INFORMATION

Part No.	Org.	Package	Temperature	Max Frequency	Interface
K3KL8L80DM-TGCT	x32	8.2x12.4 245FBGA	Tc = -25 ~ 85°C	7500Mbps (tWCK=0.2667ns)	LVSTLE_05

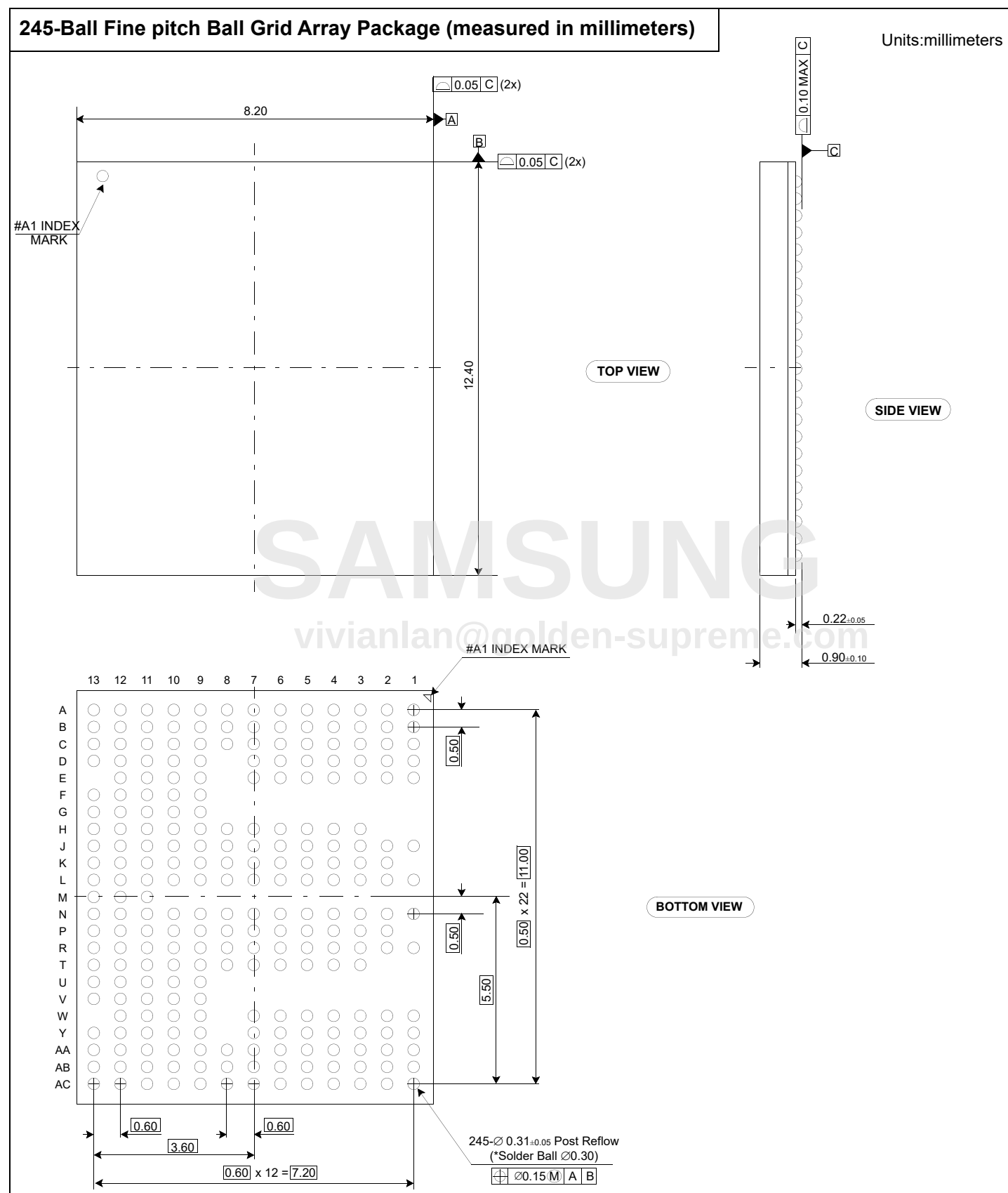
NOTE :

1) 7500Mbps is backward compatible to lower frequency.

Samsung Mobile DRAM Stack Memory	K3	K	L8	L8	0	D	M	-	I	G	CT	Speed CT:0.2667ns
Device Type K : LPDDR5X + LPDDR5X												Temp G : -25 ~ 85°C
A-Port I/F & Density & Vcc & Org. L8 : LPDDR5X & Ch.A 16Gb & VDD1=1.8V, VDD2H=1.05V, VDD2L=0.9V, VDDQ=0.5V & Ch.A : x16 & Ch.A 1CS												Package T : 245-FBGA (8.2x12.4)
B-Port I/F & Density & Vcc & Org. L8 : LPDDR5X & Ch.B 16Gb & VDD1=1.8V, VDD2H=1.05V, VDD2L=0.9V, VDDQ=0.5V & Ch.B : x16 & Ch.B 1CS												Generation M : 1'st Generation
C-Port I/F & Density & Vcc & Org. 0 : Reserve for future use												Version D : D-die

3.0 PACKAGE DIMENSION & PIN DESCRIPTION

3.1 LPDDR5X SDRAM Package Dimension



3.2 LPDDR5X SDRAM Package Ballout

245-Ball FBGA													
	1	2	3	4	5	6	7	8	9	10	11	12	13
A	DNU	DNU	VDD1	VDD2L	VDD2H	VSS	VSS	VDD2H	VSS	VDD2H	VSS	DNU	DNU
B	DNU	DQ8_A	VSS	DQ15_A	VSS	RDQS1_C_A	RDQS1_T_A	VSS	CA2_A	CA1_A	VDD2H	VSS	DNU
C	DQ10_A	VSS	DQ9_A	VDDQ	DMI1_A	VDDQ	VSS	VSS	VDD2H	VSS	CA6_A	CA4_A	VDD1
D	VSS	DQ12_A	VSS	VDDQ	VSS	WCK1_C_A	WCK1_T_A		CA0_A	CA5_A	VDD2H	VSS	VDD2L
E	DQ13_A	VSS	DQ14_A	VDDQ	DQ11_A	VSS	VDD2L		VDD2H	VSS	CA3_A	VSS	
F									VSS	RFU	VSS	CK_C_A	VSS
G									NC	CS0_A	VSS	CK_T_A	VSS
H			DQ6_A	VSS	DQ5_A	VSS	DQ7_A	VSS	VDDQ	VDDQ	VSS	VSS	VSS
J	VSS	DQ0_A	VSS	DMI0_A	VSS	VDDQ	VDDQ	WCK0_C_A	WCK0_T_A	VSS	VDD2H	VDD2H	ZQ_A
K		VDD2H	VDD2H	VSS	DQ4_A	VSS	DQ3_A	VSS	RDQS0_C_A	RDQS0_T_A	VDDQ	VSS	VDD2L
L	VSS	VSS	VSS	DQ1_A	VSS	DQ2_A	VSS	VDDQ	VSS	VSS	VDD2H	VDD2H	VDD1
M											VSS	VSS	VSS
N	VSS	VSS	VSS	DQ9_B	VSS	DQ10_B	VSS	VDDQ	VSS	VSS	VDD2H	VDD2H	VDD1
P		VDD2H	VDD2H	VSS	DQ12_B	VSS	DQ11_B	VSS	RDQS1_C_B	RDQS1_T_B	VDDQ	VSS	VDD2L
R	VSS	DQ8_B	VSS	DMI1_B	VSS	VDDQ	VDDQ	WCK1_C_B	WCK1_T_B	VSS	VDD2H	VDD2H	RESET_n
T			DQ14_B	VSS	DQ13_B	VSS	DQ15_B	VSS	VDDQ	VDDQ	VSS	VSS	VSS
U									NC	CS0_B	VSS	CK_C_B	VSS
V									VSS	RFU	VSS	CK_T_B	VSS
W	DQ5_B	VSS	DQ6_B	VDDQ	DQ3_B	VSS	VDD2L		VDD2H	VSS	CA6_B	VSS	
Y	VSS	DQ4_B	VSS	VDDQ	VSS	WCK0_C_B	WCK0_T_B		CA0_B	CA2_B	VDD2H	VSS	VDD2L
AA	DQ2_B	VSS	DQ1_B	VDDQ	DMI0_B	VDDQ	VSS	VSS	VDD2H	VSS	CA5_B	CA4_B	VDD1
AB	DNU	DQ0_B	VSS	DQ7_B	VSS	RDQS0_C_B	RDQS0_T_B	VSS	CA3_B	CA1_B	VDD2H	VSS	DNU
AC	DNU	DNU	VDD1	VDD2L	VDD2H	VSS	VSS	VDD2H	VSS	VDD2H	VSS	DNU	DNU

[Top View]

	Channel A		Channel B
	Power		Ground
	ZQ		RESET_n
	RFU/DNU/NC		

NOTE :

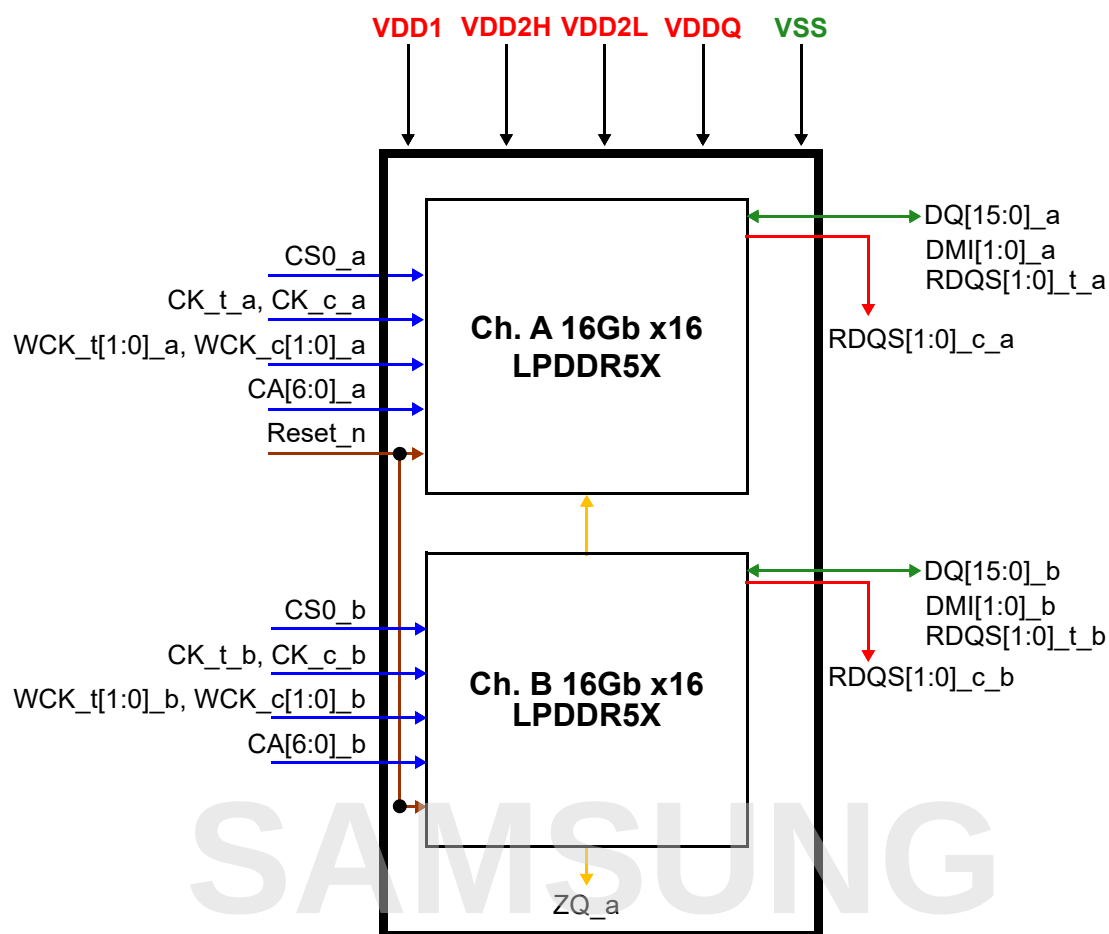
1) A 240 ohm, ±1% tolerance external resistor per ZQ pin shall be connected between the each ZQ pin and VDDQ.

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3.3 Functional Block Diagram



3.4 Pad definition and Description

[Table 1] Pad Definition and Description

Symbol	Type	Description	NOTE
CK_t, CK_c	Input	Clock: CK_t and CK_c are differential clock inputs. All Double Data Rate (DDR) Command/Address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single Data Rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).	
CS	Input	Chip Select: CS is part of the command code and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or Deep Sleep mode where it becomes an asynchronous signal.	
CA[6:0]	Input	Command/Address Inputs: CA signals provide the Command and Address input according to the Command Truth Table.	
DQ[15:0]	I/O	Data Input/Output: Bi-direction data bus.	
WCK[1:0]_t WCK[1:0]_c	Input	Data Clocks: WCK_t and WCK_c are differential clocks used for WRITE data capture and READ data output.	
RDQS [1:0]_t RDQS[1:0]_c	RDQS_t: I/O, RDQS_c: Output	Read Data Strobe: RDQS_t and RDQS_c are the differential output clock signals used to strobe data during a READ operation. And RDQS_t is also used as a Parity pin at Write with Link Protection enabled.	1
DMI[1:0]	I/O	Data Mask Inversion: DMI achieves multiple function such as Data Mask (DM), Data Bus Inversion (DBI), and Parity at read with ECC operation by setting the Mode Register and DMI is a bi-directional signal and each byte of data has a DMI signal.	1
ZQ	Reference	ZQ: ZQ is used to calibrate the output drive strength and the termination resistance as calibration reference. There is one ZQ pad per die. The ZQ pin shall be connected to VDDQ through a 240Ω ±1% resistor.	
VDDQ, VDD1, VDD2H, VDD2L	Supply	Power Supplies: Isolated on the die for improved noise immunity.	
VSS	GND	Ground Reference: Power supply ground reference	
RESET_n	Input	RESET: When asserted LOW, the RESET_n signal resets the die. RESET_n is an asynchronous signal.	
NC	-	No Connect: Not internally connected.	
RFU	-	Reserved Future Use: Not internally connected.	

NOTE :

- 1) See Table Pins Per-Byte Signal List/Description for Link Protection disabled and Pins Per-Byte Signal List/Description for Link Protection enabled Per-Byte Signal List/Description for multi-function definition for these pins.

4.0 FUNCTIONAL DESCRIPTION

LPDDR5/LPDDR5X SDRAM is a high-speed synchronous SDRAM device internally configured with 1 channel containing either 16 or 8 DQ signals. The bank architecture is user-selectable and can be either four banks with four bank groups (BG Mode), or sixteen banks (16B Mode) for LPDDR5X SDRAM. See LPDDR5/LPDDR5X Bank Architecture in LPDDR5 operation and timing datasheet for more information. Density can range from 2Gb to 32Gb.

These LPDDR5 SDRAM devices contain the following number of bits:

2Gb has 2,147,483,648 bits
3Gb has 3,221,225,472 bits
4Gb has 4,294,967,296 bits
6Gb has 6,442,450,944 bits
8Gb has 8,589,934,592 bits
12Gb has 12,884,901,888 bits
16Gb has 17,179,869,184 bits
24Gb has 25,769,803,776 bits
32Gb has 34,359,738,368 bits

LPDDR5 SDRAM devices use a command clock (CK) that operates at a reduced rate from per-byte data clock (WCK). There are seven (DDR) command/address (CA) pins that the memory controller uses to transmit command, address, bank, configuration, and training information to the SDRAM. CA signals are latched on both the rising and falling CK edges when indicated by a high signal on the single (SDR) CS pin. Most commands are 1 nCK in duration. See Table 7 for details.

The WCK:CK ratio is user-selectable as either 2:1 or 4:1. For low-power operation the WCK is generally designed to operate only when read or write data needs to be transmitted on the bus. Due to the high speeds required for WCK, the LPDDR5 SDRAM generally implements circuitry to divide the WCK frequency immediately after the WCK receiver. This leads to a requirement to synchronize WCK to CK when the WCK needs to be re-started from an idle time. See WCK2CK Synchronization operation in LPDDR5 Operation and Timing for details.

The double-data-rate I/O signalling in high-speed modes is designed to operate with on-die termination on both the memory controller and the LPDDR5 SDRAM. The high-speed I/O was designed to operate with a VDDQ of 0.5V nominal. Signals from the memory controller to the SDRAM were designed assuming a V_{oh} of $0.5 \times VDDQ$, or 250mV. Signals from the SDRAM to the memory controller are calibrated to $0.5 \times VDDQ$, or 250mV nominal. In low-speed modes the signalling may operate unterminated with nominal VDDQ in a range from 0.3V - 0.5V both to and from the SDRAM, with some limitations.

Data access size depends on the bank configuration, with 16n and 32n prefetches supported. A single BL16 Read or Write access for a x16, BG mode LPDDR5 SDRAM, typically consists of a single 256-bit wide data transfer at the internal SDRAM core, and sixteen corresponding 16-bit wide, one half - WCK clock-cycle data transfers at the I/O pins.

Read and Write accesses to LPDDR5 SDRAMs are burst oriented; accesses start at a selected location and continue for a programmed number of locations in a programmed sequence. Accesses begin with the registration of an Activate command. Prior to issuing a Read, Write, or Mask Write command, WCK must be synchronized to CK by issuing a CAS command with (WS_RD, WS_WR or WS_FS) followed by a Read, Write, or Mask Write command.

The address and BA bits registered coincident with the Activate command are used to select the row and the bank to be accessed. The address bits registered coincident with the Read, Write or Mask Write command are used to select the bank and the starting column location for the burst access.

Prior to normal operation, the LPDDR5 SDRAM is required to be initialized. The Power-Up, Initialization, and Power-off Procedure section provides detailed information covering device initialization.

4.1 LPDDR5X SDRAM ADDRESSING

[Table 2] LPDDR5X SDRAM x16 mode Addressing for BG Mode (4Banks/4Bank Groups)

Memory Density	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb
Configuration	8Mb x 16DQ x 4 BG x 4 banks	12Mb x 16DQ x 4 BG x 4 banks	16Mb x 16DQ x 4 BG x 4 banks	24Mb x 16DQ x 4 BG x 4 banks	32Mb x 16DQ x 4 BG x 4 banks	48Mb x 16DQ x 4 BG x 4 banks	64Mb x 16DQ x 4 BG x 4 banks	96Mb x 16DQ x 4 BG x 4 banks	128Mb x 16DQ x 4 BG x 4 banks
Number of Banks in BG	4	4	4	4	4	4	4	4	4
Number of Bank Groups	4	4	4	4	4	4	4	4	4
Array Pre-Fetch	256	256	256	256	256	256	256	256	256
Number of Rows	8,192	12,288	16,384	24,576	32,768	49,152	65,536	98,304	131,072
Number of Columns (fetch boundaries)	64	64	64	64	64	64	64	64	64
Page Size (Bytes)	2,048	2,048	2,048	2,048	2,048	2,048	2,048	2,048	2,048
Density	2,147,483,648	3,221,225,472	4,294,967,296	6,442,450,944	8,589,934,592	12,884,901,888	17,179,869,184	25,769,803,776	34,359,738,368
Bank Address	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1
Bank Group Addresses	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1
Row Addresses	R0 - R12	R0 - R13 (R12=0 when R13=1)	R0 - R13	R0 - R14 (R13=0 when R14=1)	R0 - R14	R0 - R15 (R14=0 when R15=1)	R0 - R15	R0 - R16 (R15=0 when R16=1)	R0 - R16
Column Addresses	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5
Burst Addresses	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3
Burst Starting Address Boundary	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit
Native Burst Length	16	16	16	16	16	16	16	16	16

NOTE :

- 1) The lower three burst addresses (B0-B2) are assumed to be "zero" and are not transmitted on the CA bus.
- 2) Row and Column address values on the CA bus that are not used for a particular density must be at valid logic levels.
- 3) For non - binary memory densities, only quarter of the row address space is invalid. When the MSB address bit is "HIGH", then the MSB - 1 address bit must be "LOW".
- 4) A row address input that violates the restriction described in note 3 may result in undefined or vendor specific behavior. Consult the memory vendor for more information.

[Table 3] LPDDR5X SDRAM x8 mode Addressing for BG Mode

Memory Density	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb
Configuration	16Mb x 8DQ x 4 BG x 4 banks	24Mb x 8DQ x 4 BG x 4 banks	32Mb x 8DQ x 4 BG x 4 banks	48Mb x 8DQ x 4 BG x 4 banks	64Mb x 8DQ x 4 BG x 4 banks	96Mb x 8DQ x 4 BG x 4 banks	128Mb x 8DQ x 4 BG x 4 banks	196Mb x 8DQ x 4 BG x 4 banks	256Mb x 8DQ x 4 BG x 4 banks
Number of Banks in BG	4	4	4	4	4	4	4	4	4
Number of Bank Groups	4	4	4	4	4	4	4	4	4
Array Pre-Fetch	128	128	128	128	128	128	128	128	128
Number of Rows	16,384	24,576	32,768	49,152	65,536	98,304	131,072	196,608	262,144
Number of Columns (fetch boundaries)	64	64	64	64	64	64	64	64	64
Page Size (Bytes)	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024
Density	2,147,483,648	3,221,225,472	4,294,967,296	6,442,450,944	8,589,934,592	12,884,901,888	17,179,869,184	25,769,803,776	34,359,738,368
Bank Address	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1	BA0 - BA1
Bank Group Addresses	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1	BG0 - BG1
Row Addresses	R0 - R13	R0 - R14 (R13=0 when R14=1)	R0 - R14	R0 - R15 (R14=0 when R15=1)	R0 - R15	R0 - R16 (R15=0 when R16=1)	R0 - R16	R0 - R17 (R16=0 when R17=1)	R0 - R17
Column Addresses	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5
Burst Addresses	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3
Burst Starting Address Boundary	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit
Native Burst Length	16	16	16	16	16	16	16	16	16

- NOTE :**
- 1) The lower three burst addresses (B0-B2) are assumed to be "zero" and are not transmitted on the CA bus.
 - 2) Row and Column address values on the CA bus that are not used for a particular density must be at valid logic levels.
 - 3) For non - binary memory densities, only quarter of the row address space is invalid. When the MSB address bit is "HIGH", then the MSB - 1 address bit must be "LOW".
 - 4) A row address input that violates the restriction described in note 3 may result in undefined or vendor specific behavior. Consult the memory vendor for more information.

[Table 4] LPDDR5X SDRAM x16 mode Addressing for 16B Mode

Memory Density	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb
Configuration	8Mb x 16DQ x 16 banks	12Mb x 16DQ x 16 banks	16Mb x 16DQ x 16 banks	24Mb x 16DQ x 16 banks	32Mb x 16DQ x 16 banks	48Mb x 16DQ x 16 banks	64Mb x 16DQ x 16 banks	96Mb x 16DQ x 16 banks	128Mb x 16DQ x 16 banks
Number of Banks	16	16	16	16	16	16	16	16	16
Array Pre-Fetch	256	256	256	256	256	256	256	256	256
Number of Rows	8,192	12,288	16,384	24,576	32,768	49,152	65,536	98,304	131,072
Number of Columns (fetch boundaries)	64	64	64	64	64	64	64	64	64
Page Size (Bytes)	2,048	2,048	2,048	2,048	2,048	2,048	2,048	2,048	2,048
Density	2,147,483,648	3,221,225,472	4,294,967,296	6,442,450,944	8,589,934,592	12,884,901,888	17,179,869,184	25,769,803,776	34,359,738,368
Bank Address	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3
Row Addresses	R0 - R12	R0 - R13 (R12=0 when R13=1)	R0 - R13	R0 - R14 (R13=0 when R14=1)	R0 - R14	R0 - R15 (R14=0 when R15=1)	R0 - R15	R0 - R16 (R15=0 when R16=1)	R0 - R16
Column Addresses	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5
Burst Addresses	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3
Burst Starting Address Boundary	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit
Native Burst Length	16	16	16	16	16	16	16	16	16

NOTE :

- 1) The lower three burst addresses (B0-B2) are assumed to be "zero" and are not transmitted on the CA bus.
- 2) Row and Column address values on the CA bus that are not used for a particular density must be at valid logic levels.
- 3) For non-binary memory densities, only quarter of the row address space is invalid. When the MSB address bit is "HIGH", then the MSB-1 address bit must be "LOW".
- 4) A row address input that violates the restriction described in note 3 may result in undefined or vendor specific behavior. Consult the memory vendor for more information.

[Table 5] LPDDR5X SDRAM x8 mode Addressing for 16B Mode

Memory Density	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb
Configuration	16Mb x 8DQ x 16 banks	24Mb x 8DQ x 16 banks	32Mb x 8DQ x 16 banks	48Mb x 8DQ x 16 banks	64Mb x 8DQ x 16 banks	96Mb x 8DQ x 16 banks	128Mb x 8DQ x 16 banks	192Mb x 8DQ x 16 banks	256Mb x 8DQ x 16 banks
Number of Banks	16	16	16	16	16	16	16	16	16
Array Pre-Fetch	128	128	128	128	128	128	128	128	128
Number of Rows	16,384	24,576	32,768	49,152	65,536	98,304	131,072	196,608	262,144
Number of Columns (fetch boundaries)	64	64	64	64	64	64	64	64	64
Page Size (Bytes)	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024
Density	2,147,483,648	3,221,225,472	4,294,967,296	6,442,450,944	8,589,934,592	12,884,901,888	17,179,869,184	25,769,803,776	34,359,738,368
Bank Address	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3	BA0 - BA3
Row Addresses	R0 - R13	R0 - R14 (R13=0 when R14=1)	R0 - R14	R0 - R15 (R14=0 when R15=1)	R0 - R15	R0 - R16 (R15=0 when R16=1)	R0 - R16	R0 - R17 (R16=0 when R17=1)	R0 - R17
Column Addresses	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5	C0-C5
Burst Addresses	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3	B0-B3
Burst Starting Address Boundary	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit	128-bit
Native Burst Length	16	16	16	16	16	16	16	16	16

NOTE :

- 1) The lower three burst addresses (B0-B2) are assumed to be "zero" and are not transmitted on the CA bus.
- 2) Row and Column address values on the CA bus that are not used for a particular density must be at valid logic levels.
- 3) For non - binary memory densities, only quarter of the row address space is invalid. When the MSB address bit is "HIGH", then the MSB - 1 address bit must be "LOW".
- 4) A row address input that violates the restriction described in note 3 may result in undefined or vendor specific behavior. Consult the memory vendor for more information.

5.0 SPEED GRADES

[Table 6] LPDDR5X Speed Grades

OPERATING MODE / SPEED GRADE				LPDDR5X-7500	Unit
DVFS ²⁾	Enhanced DVFS ⁶⁾	WCK:CK Ratio ³⁾	Banks ⁴⁾	Maximum Data Rate ¹⁾	
Disabled	Disabled	4:1	BG	7500	Mbps
Disabled	Disabled	4:1	16B	3200	
Disabled	Disabled	2:1	BG	Not Supported	
Disabled	Disabled	2:1	16B	3200	
Enabled	Disabled	4:1	BG	Not Supported	
Enabled	Disabled	4:1	16B	1600	
Enabled	Disabled	2:1	BG	Not Supported	
Enabled	Disabled	2:1	16B	1600	
Disabled	Enabled	4:1	BG	Not Supported	
Disabled	Enabled	4:1	16B	3200	
Disabled	Enabled	2:1	BG	Not Supported	
Disabled	Enabled	2:1	16B	3200	
Link ECC Feature support by DRAM				Required	

NOTE :

- 1) Speed grades represent the maximum data rate capability of the device. Higher speed grade devices can be operated at lower data rates. Achieving maximum data rates requires appropriate latency settings.
- 2) DVFS is enabled/disabled in MR19 OP[1:0] and MR13 OP[7].
- 3) WCK:CK ratio is set in MR18 OP[7].
- 4) Bank organization (BG, or 16B) is set in MR3 OP[4:3].
- 5) The LPDDR5X SDRAM supports the data-rate of DVFSQ up to 3200Mbps. As DVFSQ is enabled without an ODT, signal integrity of the channel between the memory and the DRAM controller should be guaranteed in advance. Please contact the vendor for enabling DVFSQ Data Rate over 3200Mbps.
- 6) Enhanced DVFS is enabled/disabled in MR19 OP[1:0] and MR13 OP[7]. Refer to MR41 OP[2:1] for Enhanced DVFS mode support.

6.0 SIMPLIFIED LPDDR5 STATE DIAGRAM

LPDDR5 SDRAM state diagram provides a simplified illustration of allowed state transitions and the related commands to control them. For a complete definition of the device behavior, the information provided by the state diagram should be integrated with the truth tables and timing specification.

The truth tables provide complementary information to the state diagram; they clarify the device behavior and the applied restrictions when considering the actual state of all banks.

For the command definition, see 7.0.

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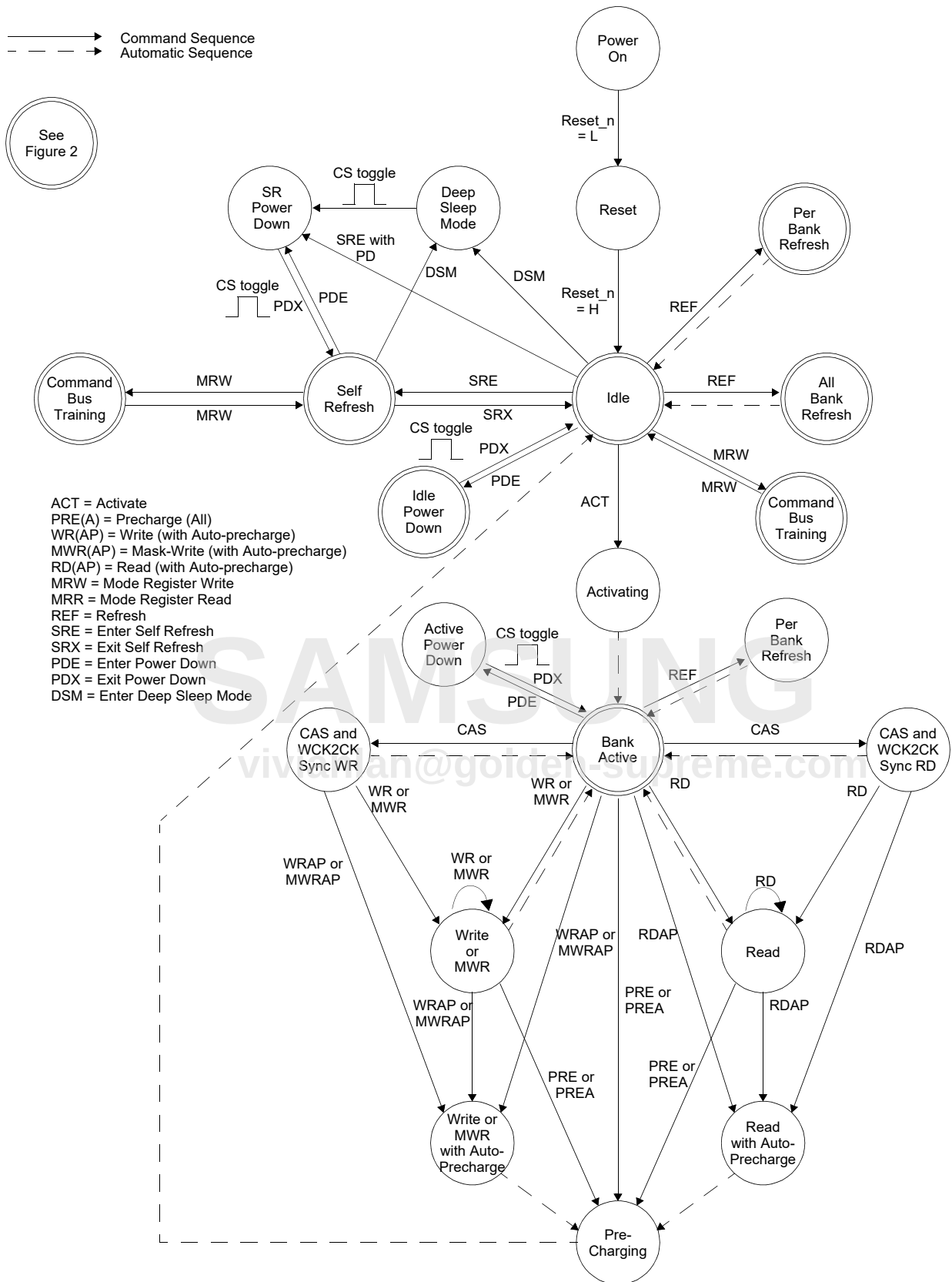


Figure 1. LPDDR5: Simplified Bus Interface State Diagram

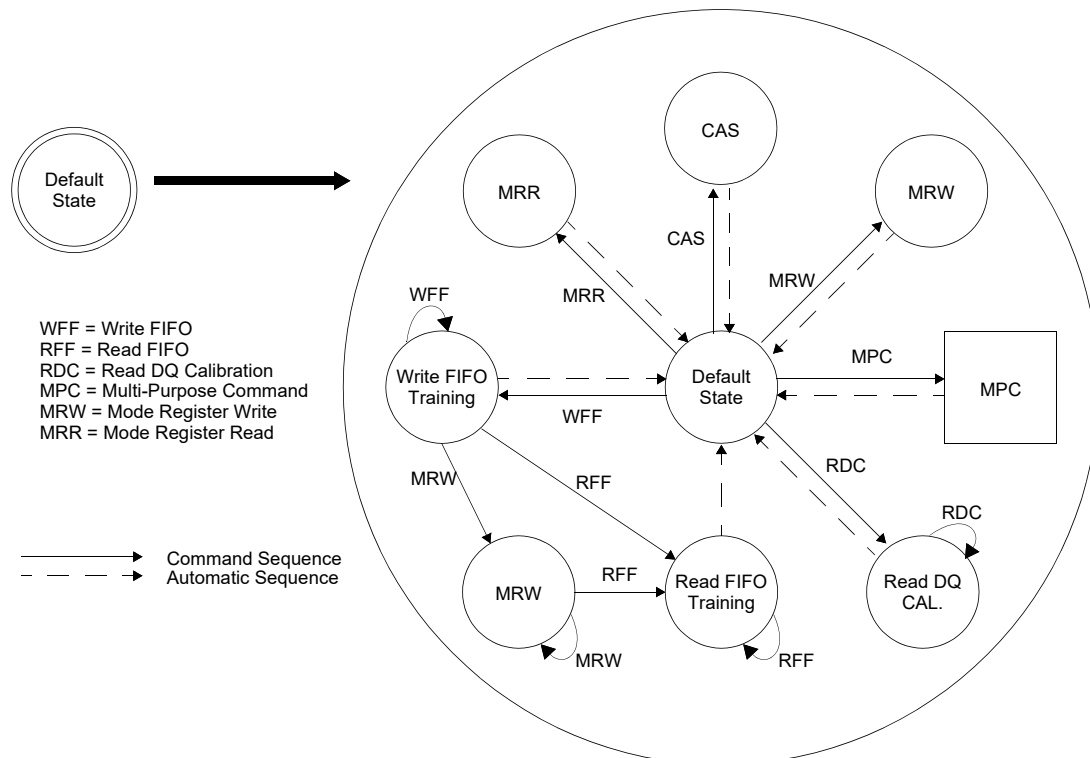


Figure 2. Sub-State Diagram-1 related with MRR, MRW, CAS, WFF, RFF, RDC and MPC Command

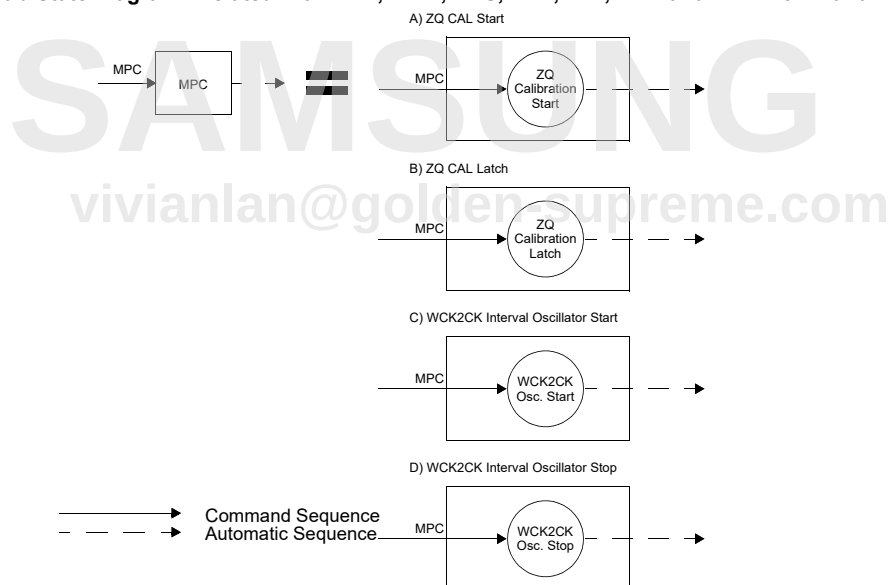


Figure 3. Sub-State Diagram-2: related with MPC State

NOTE :

- 1) From the Self-Refresh state the device can enter Power-Down, MRR, MRW, MPC and Deep Sleep Mode states. See on Self-Refresh for more information.
- 2) In IDLE state, all banks are pre-charged.
- 3) In the case of a MRW command to enter a training mode, the state machine will not automatically return to the IDLE state at the conclusion of training. See on Mode Register Write (MRW), for more information.
- 4) In the case of a MPC command to enter a training mode, the state machine may not automatically return to the IDLE state at the conclusion of training. See on Multi-Purpose Command (MPC), for more information.
- 5) This simplified State Diagram is intended to provide an overview of the possible state transitions and the commands to control them. In particular situations involving more than one bank, the enabling or disabling of on-die termination, and some other events are not captured in full detail.
- 6) States that have an "automatic return" and can be accessed from more than one prior state(e.g., MRW from either idle of Active states) will return to the state from when they were initiated (e.g., MRW from Idle will return to Idle).
- 7) The RESET_n pin can be asserted from any state and will cause the SDRAM to go to the Reset State. The diagram shows RESET state applied from the Power-On as an example, but the Diagram should not be construed as a restriction on RESET_n.
- 8) Bank Active state can enter "CAS and WCK2CK Sync WR", "CAS and WCK2CK Sync RD" State for WCK-to-CK Synchronization, Non-Target ODT setting or Burst Length setting if it is needed.
- 9) Deep Sleep Mode state can enter Self-Refresh Power Down state toggling CS (VDD2H level). See Deep Sleep Mode for more information.
- 10) "Bank Active" to "Per-Bank Refresh" transition only refers to different banks not the same bank.
- 11) Only MRW commands for MR16 OP[1:0]: FSP-WR, OP[3:2]: VRCG and MR14 OP[6:0]:VREF(DQ[7:0]) and MR15 OP[6:0]:VREF(DQ[15:8]) are allowed from WRITE FIFO command to READ FIFO command.

7.0 COMMAND TRUTH TABLE

Operation or timing that is not specified is illegal, and after such an event, in order to guarantee proper operation, the LPDDR5 device must be reset or power-cycled and then restarted through the specified initialization sequence before normal operation can continue.

[Table 7] Command Truth Table

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
DESELECT (DES)	Any	L	X	X	X	X	X	X	X	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
NO OPERATION (NOP)	Any	H	L	L	L	L	L	L	L	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
POWER DOWN ENTRY (PDE)	Any	H	L	L	L	L	L	L	H	R1	1, 2, 12
		L	X	X	X	X	X	X	X	R2	
ACTIVATE-1 (ACT-1)	Any	H	H	H	H	R14	R15	R16	R17	R1	1, 2, 3, 4
	BG	X	BA0	BA1	BG0	BG1	R11	R12	R13	F1	
	16B		BA0	BA1	BA2	BA3					
ACTIVATE-2 (ACT-2)	Any	H	H	H	L	R7	R8	R9	R10	R1	1, 2, 4
		X	R0	R1	R2	R3	R4	R5	R6	F1	
PRECHARGE (PRE) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	H	R1	1, 2, 3, 5
	BG	X	BA0	BA1	BG0	BG1	V	V	AB	F1	
	16B		BA0	BA1	BA2	BA3					
PRECHARGE (PRE) (Per Bank, w/o address sample)	Any	H	L	L	L	H	H	H	H	R1	1,2,3
	BG	X	BA0	BA1	BG0	BG1	V	L	L	F1	
	16B		BA0	BA1	BA2	BA3					
PRECHARGE (PRE) (Per Bank, w/ address sample)	Any	H	L	L	L	H	H	H	H	R1	1,2,3
	BG	X	BA0	BA1	BG0	BG1	V	H	L	F1	
	16B		BA0	BA1	BA2	BA3					
REFRESH (REF) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	L	R1	1, 2, 3, 5,14,15,16
	BG	X	BA0	BA1	BG0	RFM	SB0	V	AB	F1	
	16B		BA0	BA1	BA2		SB0	V			
MASK WRITE (MWR)	Any	H	L	H	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B		BA0	BA1	BA2	BA3					
WRITE (WR16 or WR)	Any	H	L	H	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B		BA0	BA1	BA2	BA3					
WRITE32 (WR32)	BG/16B	H	L	L	H	L	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B		BA0	BA1	BA2	BA3					
READ (RD16 or RD)	Any	H	H	L	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B		BA0	BA1	BA2	BA3					
READ32 (RD32)	BG/16B	H	H	L	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B		BA0	BA1	BA2	BA3					

[Table 7] Command Truth Table

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
CAS	Any	H	L	L	H	H	WS_WR	WS_RD	WS_FS	R1	1, 2, 7, 13
		X	DC0	DC1	DC2	DC3	WRX	WXSA	WXSB/B3	F1	
MULTI PURPOSE COMMAND (MPC)	Any	H	L	L	L	L	H	H	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
SELF REFRESH ENTRY (SRE)	Any	H	L	L	L	H	L	H	H	R1	1, 2, 9
		X	V	V	V	V	V	DSM	PD	F1	
SELF REFRESH EXIT (SRX)	Any	H	L	L	L	H	L	H	L	R1	1, 2
		X	V	V	V	V	V	V	V	F1	
MODE REGISTER WRITE-1 (MRW-1)	Any	H	L	L	L	H	H	L	H	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
MODE REGISTER WRITE-2 (MRW-2)	Any	H	L	L	L	H	L	L	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
MODE REGISTER READ (MRR)	Any	H	L	L	L	H	H	L	L	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
WRITE FIFO (WFF)	Any	H	L	L	L	L	L	H	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ FIFO (RFF)	Any	H	L	L	L	L	L	H	L	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ DQ CALIBRATION (RDC)	Any	H	L	L	L	L	H	L	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	

NOTE :

- 1) LPDDR5 commands are one clock cycles long and defined by the states of CS at the rising edge (R1) of clock and CA[6:0] at the rising edge (R1) and the falling edge (F1) of clock. Note that some operations such as ACTIVATE, MODE REGISTER WRITE or WCK2CK SYNC (WS_RD, WS_WR, WS_FS) require two commands to initiate.
- 2) "V" means "H" or "L" (a defined logic level). "X" means don't care in which case CK_t, CK_c and CA[6:0] can be floated.
- 3) Bank Group Address BG[1:0] and Bank Address BA[1:0] determine which bank is to be operated upon.
- 4) An ACTIVATE-1 command must be followed by an ACTIVATE-2 command. Only CAS, WRITE, MASK WRITE, READ, PRECHARGE (to a different bank) commands can be issued between ACTIVATE-1 and ACTIVATE-2 commands. The maximum timing gap between ACTIVATE-1 and ACTIVATE-2 commands are 8 clock cycles. An ACTIVATE-1 command must be issued first before issuing an ACTIVATE-2 command. Once an ACTIVATE-1 command is issued, an ACTIVATE-2 command must be issued before issuing another ACTIVATE-1 command.
- 5) AB "HIGH" during PRECHARGE or REFRESH command indicates that command must be applied to all banks and bank address is a don't care.
- 6) AP "HIGH" during WRITE, MASK WRITE or READ commands indicates that an auto-precharge will occur to the bank associated with the WRITE, MASK WRITE or READ command.
- 7) CAS command is only required for WRITE, MASK WRITE or READ operation if optional low power features (DC0-3 or WRX/WXSA/WXSB) of these commands are required or if Burst Start Address (B3) is non-zero to READ or if WCK2CK SYNC (WS_RD, WS_WR) or FAST SYNC (WS_FS) is applied to a target device. CAS operands (DC0-3, WRX/WXSA/WXSB and B3) are valid to immediately following WRITE or READ only. In case of CAS-FAST_SYNC(WS_FS="H" AND WS_WR="L" AND WS_RD="L") or CAS-SYNC_OFF(WS_FS="H" AND WS_WR="H" AND WS_RD="H"), other CAS operands (DC0-3, WRX, WXSA, WXSB, B3) shall be "LOW". Valid DC0-3 or WRX/WXSA/WXSB operand(s) may be issued together with WCK2CK SYNC (WS_WR="H", WS_RD="L", WS_FS="L"). Valid B3 operand may be issued together with WCK2CK SYNC (WS_RD="H", WS_WR="L", WS_FS="L"). Refer to Table 8.
- 8) WRITE32 (WRITE with BL32) command is valid in a BG/16B mode only. WRITE command is WR in an 8B mode and WR16 in a BG/16B mode.
- 9) PD "HIGH" during SELF REFRESH ENTRY command indicates that both Self Refresh Entry and Power Down Entry will occur together. DSM "HIGH" during SELF REFRESH ENTRY command indicates that both Self Refresh Entry and Deep Sleep Mode Entry will occur together. Both PD "HIGH" and DSM "HIGH" are illegal.
- 10) READ command includes Read Burst Start Address (B3, B4) associated with Burst Length setting, B3 for BL16 in a BG/16B mode, or B3, C0 for BL32 in a BG/16B mode, or B3, B4 for BL32 in an 8B mode. Refer to MR1 Burst Sequence Table.
- 11) READ32 (READ with BL32) command is valid in a BG/16B mode only. READ command is RD in an 8B mode and RD16 in a BG/16B mode.
- 12) CS shall be compliant to Power Down Entry/Exit timing requirements. Refer to Power Down Function and Timing Description.
- 13) Refer to LPDDR5 low power function description, Data Copy function (DC0-DC3) and Write X function (WRX, WXSA, WXSB).
- 14) CA3 F1 edge is 'V' when RFM is not required, but becomes 'RFM' when read-only RFM (RFM Required): MR27 OP[0]=1_B and/or ARFM support: MR1 OP[1]=1_B.
- 15) Issuing the RFMpb or RFMab command allows the LPDDR5 to use the command period for additional Refresh Management.
- 16) CA4 F1 edge is 'V' when RFM is not required, or when RFMSBC is set to '1'. SB0 is always treated as 'V' for REFpb and REFab commands.

[Table 8] Allowable CAS command operand(s) combination

Operation Mode	CAS Command Operand(s)						
	WS_WR	WS_RD	WS_FS	DC0-3	WRX	WXSA	WXSBB3
WCK2CK SYNC (WR)	H	L	L	LLLL	L	L	L
WCK2CK SYNC (WR) + Write Data Copy	H	L	L	VVVV	L	L	L
WCK2CK SYNC (WR) + Write X	H	L	L	LLLL	H	V	V
WCK2CK SYNC (RD)	L	H	L	LLLL	L	L	V
WCK2CK FAST SYNC	L	L	H	LLLL	L	L	L
WCK2CK SYNC OFF	H	H	H	LLLL	L	L	L
Write Data Copy	L	L	L	VVVV	L	L	L
Write X	L	L	L	LLLL	H	V	V
Non-Zero Burst Start Address for RD	L	L	L	LLLL	L	L	H
WCK SUSPEND	H	L	H	LLLL	L	L	L
Illegal	Other Operands Combination						

NOTE :

1) "V" means "H" or "L" (a defined logic level).

2) All CAS command operands are non-sticky.

3) Write Data Copy function requires DC0-3 operands not to be all-zero.

4) CAS-WCK_SUSPEND command is valid only when MR0 OP[2]=1_B (Enhanced WCK Always On mode supported) and MR18 OP[4]= 1_B (WCK Always On mode enabled).

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8.0 MODE REGISTER DEFINITION

8.1 Mode Register Assignment and Definition in LPDDR5X SDRAM

Table 9 shows the mode registers for LPDDR5X SDRAM. Each register is denoted as "R" if it can be read but not written, "W" if it can be written but not read, and "R/W" if it can be read and written. A Mode Register Read command is used to read a mode register. A Mode register Write command is used to write a mode register. LPDDR5X device provides additional MRR capability to some write-only Mode Registers containing important memory system configuration and status info. Mode Registers 1, 3, 11, 16, 17, 37, and 40 can be read by issuing a Mode Register Read (MRR) command.

[Table 9] Mode Register Assignment in LPDDR5X SDRAM

MR#	MA [6:0]	Access	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]	
0	00 _H	R	Per-Pin DFE	Pre-Emphasis	Unified NT-ODT Behavior	DMI Output Behavior Mode	Optimized Refresh Mode	Enhanced WCK Always On Mode	Latency Mode	NT-ODT DQ/RDQS Timing Separation	
1	01 _H	R/W	WL				CK Mode	(RFU)	ARFM Support	CS ODT OP Support	
			WL				CK Mode				
			WL				CK Mode				
2	02 _H	W	nWR				RL and nRBTP				
			nWR				RL and nRBTP				
			nWR				RL and nRBTP				
3	03 _H	R/W	DBI-WR	DBI-RD	WLS	BK/BG ORG		PDDS			
			DBI-WR	DBI-RD	WLS	BK/BG ORG		PDDS			
			DBI-WR	DBI-RD	WLS	BK/BG ORG		PDDS			
4	04 _H	R	TUF	ZQ Initiator	ZQUF	Refresh Multiplier					
5	05 _H	R	LPDDR5 Manufacturer ID								
6	06 _H	R	Revision ID-1								
7	07 _H	R	Revision ID-2								
8	08 _H	R	IO Width			Density				Type	
9	09 _H	W	Vendor Specific Test Register								
10	0A _H	W	RDQS PST		RDQS PRE		WCK PST		RDQS Pre-2	RPST Mode	
			RDQS PST		RDQS PRE		WCK PST		RDQS Pre-2	RPST Mode	
			RDQS PST		RDQS PRE		WCK PST		RDQS Pre-2	RPST Mode	
11	0B _H	R/W	CS ODT OP	CA ODT			NT ODT	DQ ODT			
			CS ODT OP	CA ODT			NT ODT	DQ ODT			
			CS ODT OP	CA ODT			NT ODT	DQ ODT			
12	0C _H	R/W	VBS	VREF(CA)							
				VREF(CA)							
				VREF(CA)							
13	0D _H	W	Dual VDD2	CBT Mode	DMD	DMI I/O Control	(RFU)	VRO	Thermal Offset		
14	0E _H	R/W	VDLC	VREF(DQ[7:0])							
				VREF(DQ[7:0])							
				VREF(DQ[7:0])							

[Table 9] Mode Register Assignment in LPDDR5X SDRAM

MR#	MA [6:0]	Access	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]	
15	0F _H	R/W	OCC	VREF(DQ[15:8])							
				VREF(DQ[15:8])							
				VREF(DQ[15:8])							
16	10 _H	R/W	CBT-Phase	VRCG	CBT		FSP-OP		FSP-WR		
17	11 _H	R/W	x8 ODTD Upper	x8 ODTD Lower	ODTD-CA	ODTD-CS	ODTD-CK	SOC ODT			
					ODTD-CA	ODTD-CS	ODTD-CK	SOC ODT			
					ODTD-CA	ODTD-CS	ODTD-CK	SOC ODT			
18	12 _H	W	CKR	WCK2CK Leveling	(RFU)	WCK ON	WCK_FM	WCK ODT			
			CKR			WCK ON	WCK_FM	WCK ODT			
			CKR			WCK ON	WCK_FM	WCK ODT			
19	13 _H	R/W	CS ODT (CS termination)		WCK2DQ OSC FM support	WCK2DQ OSC FM	DVFSQ		DVFSC		
						WCK2DQ OSC FM	DVFSQ		DVFSC		
						WCK2DQ OSC FM	DVFSQ		DVFSC		
20	14 _H	W	RDC DQ Mode	RDC DMI Mode	MRWDU	MRWDL	WCK Mode		RDQS		
							WCK Mode		RDQS		
							WCK Mode		RDQS		
21	15 _H	R/W	WXS	WXFE	RDCFE	WDCFE	ODT-CSFS	WXFS	RDCFS	WDCFS	
22	16 _H	W	RECC		WECC		(RFU)				
			RECC		WECC						
			RECC		WECC						
23	17 _H	W	PASR Segment Mask								
24	18 _H	R/W	DFE Support	DFE Quantity for Upper Byte			Read DCA Support	DFE Quantity for Lower Byte			
				DFE Quantity for Upper Byte				DFE Quantity for Lower Byte			
				DFE Quantity for Upper Byte				DFE Quantity for Lower Byte			
25	19 _H	W	Optimized Refresh Enable	PARC	CA Inputs TERM	CK pair TERM	(RFU)				
26	1A _H	R/W	RDQSTFE	RDQSTFS	DCMU1	DCMU0	DCML1	DCML0	DCM Flip	DCM Start/Stop	
27	1B _H	R	RAAMULT		RAAIMT					RFM	
28	1C _H	W	(RFU)		ZQ Mode	(RFU)	ZQ interval		ZQ Stop	ZQ Reset	
29	1D _H	R	PPR Resource Bank7	PPR Resource Bank6	PPR Resource Bank5	PPR Resource Bank4	PPR Resource Bank3	PPR Resource Bank2	PPR Resource Bank1	PPR Resource Bank0	
30	1E _H	W	DCA for Upper byte				DCA for Lower byte				
			DCA for Upper byte				DCA for Lower byte				
			DCA for Upper byte				DCA for Lower byte				
31	1F _H	W	Lower-Byte per-bit control Register for DQ Calibration								
32	20 _H	W	Upper-Byte per-bit control Register for DQ Calibration								
33	21 _H	W	DQ Calibration Pattern "A"								
34	22 _H	W	DQ Calibration Pattern "B"								
35	23 _H	R	WCK2DQI Oscillator Count - LSB								
36	24 _H	R	WCK2DQI Oscillator Count - MSB								
37	25 _H	R/W	WCK2DQI interval timer run time setting								
38	26 _H	R	WCK2DQO Oscillator Count - LSB								
39	27 _H	R	WCK2DQO Oscillator Count - MSB								

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR HEADQUARTERS OF SAMSUNG ELECTRONICS.

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[Table 9] Mode Register Assignment in LPDDR5X SDRAM

MR#	MA [6:0]	Access	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
40	28 _H	R/W	WCK2DQO interval timer run time setting							
41	29 _H	R/W	NT DQ ODT			PPRE	E-DVFS ODT Option Support	DVFS/E-DVFS Sup- port		PDFEC
			NT DQ ODT							PDFEC
			NT DQ ODT							PDFEC
42	2A _H	W	PPR KEY Protection							
43	2B _H	R	DBE_flag	SBEC_Rul e	SBE_count					
44	2C _H	R	Data ECC Syndrome							
45	2D _H	R	Data ECC Syndrome	Error Byte Lane	DMI ECC Syndrome					
46	2E _H	W	(RFU)					FIFO RDQS Training	RDQS Toggle	Enhanced RDQS
47	2F _H	R	Serial ID-1							
48	30 _H	R	Serial ID-2							
49	31 _H	R	Serial ID-3							
50	32 _H	R	Serial ID-4							
51	33 _H	R	Serial ID-5							
52	34 _H	R	Serial ID-6							
53	35 _H	R	Serial ID-7							
54	36 _H	R	Serial ID-8							
55	37 _H	N/A	DNU (Do Not Use)							
56	38 _H	W	Valid 0 or 1							
57	39 _H	R/W	ARFM		RFMSBC		RFMSB		RAADEC	
58	3A _H	W	DQ Dn Emphasis UB		DQ Up Emphasis UB		DQ Dn Emphasis LB		DQ Up Emphasis LB	
			DQ Dn Emphasis UB		DQ Up Emphasis UB		DQ Dn Emphasis LB		DQ Up Emphasis LB	
			DQ Dn Emphasis UB		DQ Up Emphasis UB		DQ Dn Emphasis LB		DQ Up Emphasis LB	
59	3B _H	N/A	DNU (Do Not Use)							
60	3C _H	W	Valid 0 or 1							
61:63	3D _H :3F _H	N/A	DNU (Do Not Use)							
64:68	40 _H :44 _H	N/A	(RFU)							
69	45 _H	W	RDCAU				RDCAL			
70	46 _H	W	Offset DFE Quantity for DQ3 (DFEDQ3)		Offset DFE Quantity for DQ2 (DFEDQ2)		Offset DFE Quantity for DQ1 (DFEDQ1)		Offset DFE Quantity for DQ0 (DFEDQ0)	
71	47 _H	W	Offset DFE Quantity for DQ7 (DFEDQ7)		Offset DFE Quantity for DQ6 (DFEDQ6)		Offset DFE Quantity for DQ5 (DFEDQ5)		Offset DFE Quantity for DQ4 (DFEDQ4)	
72	48 _H	W	Offset DFE Quantity for DQ11 (DFEDQ11)		Offset DFE Quantity for DQ10 (DFEDQ10)		Offset DFE Quantity for DQ9 (DFEDQ9)		Offset DFE Quantity for DQ8 (DFEDQ8)	
73	49 _H	W	Offset DFE Quantity for DQ15 (DFEDQ15)		Offset DFE Quantity for DQ14 (DFEDQ14)		Offset DFE Quantity for DQ13 (DFEDQ13)		Offset DFE Quantity for DQ12 (DFEDQ12)	
74	4A _H	W	Offset DFE Quantity for RDQS1 (DFERDQS1)		Offset DFE Quantity for RDQS0 (DFERDQS0)		Offset DFE Quantity for DMI1 (DFEDMI1)		Offset DFE Quantity for DMI0 (DFEDMI0)	
75:119	4B _H :77 _H	N/A	(RFU)							
120:123	78 _H :7B _H	N/A	DNU (Do Not Use)							
124:127	7C _H :7F _H	N/A	(RFU)							

	Applied when FSP=0
	Applied when FSP=1
	Applied when FSP=2

NOTE :
1) RFU bits shall be set to '0' during writes.
2) All mode registers that are specified as RFU or write-only shall return undefined data when read.
3) All mode registers that are specified as RFU should not be written.
4) Writes to read-only registers shall have no impact on the functionality of the device.
5) In order to guarantee proper operation, all mode registers that can be written shall be set to specified data, except "Reserved" and "RFU".

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8.2 Mode Register Definition

MR0_Register Information (MA[7:0] = 00_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Per-pin DFE	Pre-Emphasis	Unified NT-ODT Behavior	DMI Output Behavior Mode	Optimized Refresh Mode	Enhanced WCK Always On Mode	Latency Mode	NT-ODT DQ/RDQS Timing Separation

Function	Register Type	Operand	Data	Notes
NT-ODT DQ/RDQS Timing separation	Read only	OP[0]	0 _B : Device supports same DQ/RDQS NT-ODT Timing 1 _B : Device supports separated DQ/RDQS NT-ODT Timing	2
Latency Mode		OP[1]	0 _B : Device supports X16 mode latency 1 _B : Device supports byte mode latency	1
Enhanced WCK Always On Mode		OP[2]	0 _B : Device does not support Enhanced WCK Always-ON mode. 1 _B : Device supports Enhanced WCK Always-ON mode.	3
Optimized Refresh Mode		OP[3]	0 _B : Device does not support Optimized Refresh mode 1 _B : Device supports Optimized Refresh mode.	4
DMI Output Behavior Mode		OP[4]	0 _B : Device only supports DMI Behavior mode1 1 _B : Device supports both DMI Behavior mode1, 2 and mode selection.	5
Unified NT-ODT Behavior		OP[5]	0 _B : The NT-ODT behavior does not follow the unified NT-ODT behavior. 1 _B : The NT-ODT behavior follows the unified NT-ODT behavior.	6
Pre-Emphasis		OP[6]	0 _B : Pre-Emphasis mode not supported 1 _B : Pre-Emphasis mode supported	
Per-pin DFE		OP[7]	0 _B : Per-pin DFE mode not supported 1 _B : Per-pin DFE mode supported	

NOTE :

- 1) Byte mode devices only support byte mode latency.
- 2) LPDDR5 SDRAM supports Separate DQ/RDQS NT-ODT timing as an optional feature. See "NT-ODT behavior for Read Operation" section.
- 3) Enhanced WCK Always On mode is an optional feature. See "Enhanced WCK Always On Mode" section.
- 4) Optimized Refresh is an optional feature for Self Refresh mode. See "Optimized Refresh" section.
- 5) DMI Behavior Mode 2 and DMI Behavior Mode selection at Read FIFO and Read DQ Calibration is an optional feature. DMI Behavior Mode can be selected by MR13 OP[4]. See "DMI Output Behavior Mode" section.
- 6) Unified NT-ODT behavior is an optional feature. See "NT-ODT behavior unification" section.

MR1_Register Information (MA[7:0] = 01_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WL				CK Mode	(RFU)	ARFM support	CS ODT OP Support

Function	Register Type	Operand	Data	Notes
CS ODT OP Support (CS ODT behavior option supported)	Read-Only	OP[0]	0 _B : CS ODT behavior option not supported 1 _B : CS ODT behavior option supported	
Adaptive Refresh Management (ARFM) support		OP[1]	0 _B : Adaptive Refresh Management not supported 1 _B : Adaptive Refresh Management supported	
CK mode	Read/Write	OP[3]	0 _B : Differential (Default) 1 _B : Single Ended	1,2,3
WL (Write Latency)		OP[7:4]	Refer to Table 10	2,3,4,5

[Table 10] WL (Write Latency) table

WL (x16/ch and x8/ch)			
OP[7:4]	WCK:CK	Set A	Set B
0000 _B	2:1	4	4
0001 _B		4	6
0010 _B		6	8
0011 _B		8	10
0100 _B		8	14
0101 _B		10	16
0000 _B (Default)	4:1	2	2
0001 _B		2	3
0010 _B		3	4
0011 _B		4	5
0100 _B		4	7
0101 _B		5	8
0110 _B		6	9
0111 _B		6	11
1000 _B		7	12
1001 _B		8	14
1010 _B		9	15
1011 _B		9	16
1100 _B		11	19
1101 _B		12	22

All others are RFU.

NOTE :

- When MR1 OP[3]= 1_B, CK_t is used as CK timing and CK_c is set to a valid logic state.
- There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- The Write Latency applies regardless of the following function setting: (Disable/Enable) Byte Mode, Write DBI, Write Data Copy, Link ECC and DVFS and Enhanced DVFS.
- Write Latency Set "A" and Set "B" is determined by MR3 OP[5]. When MR3 OP[5]=0_B, then Write Latency Set "A" should be used. When MR3 OP[5]=1_B, then Write Latency Set "B" should be used.

MR2_Register Information (MA[7:0] = 02_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
nWR				RL and nRBTP			

Function	Register Type	Operand	Data	Notes
RL (Read latency) and nRBTP (Read Burst end to Precharge delay)	Write-only	OP[3:0]	Refer to Table 11, Table 12 and Table 14	1,2,4,5
nWR (Write recovery)		OP[7:4]	Refer to Table 17	1,2,3

[Table 11] RL and nRBTP (Link ECC off case, DVFSC disabled, Enhanced DVFSC disable)

OP[3:0]	WCK:CK Ratio	RL Set 0	RL Set 1	RL Set 2	nRBTP
0000 _B (Default)	2:1	6	6	6	0
0001 _B		8	8	8	0
0010 _B		10	10	12	0
0011 _B		12	14	14	0
0100 _B		16	16	18	2
0101 _B		18	20	20	2
0000 _B	4:1	3	3	3	0
0001 _B		4	4	4	0
0010 _B		5	5	6	0
0011 _B		6	7	7	0
0100 _B		8	8	9	1
0101 _B		9	10	10	1
0110 _B		10	11	12	2
0111 _B		12	13	14	2
1000 _B		13	14	15	3
1001 _B		15	16	17	4
1010 _B		16	17	19	4
1011 _B		17	18	20	4
1100 _B		20	22	24	6
1101 _B		23	25	26	6

All others are RFU.

[Table 12] RL and nRBTP (Link ECC off case, DVFSC enable, Enhanced DVFSC disable)

OP[3:0]	WCK:CK Ratio	RL Set 0	RL Set 1	RL Set 2	nRBTP [nCK]
0000 _B	2:1	6	6	6	0
0001 _B		8	10	10	0
0010 _B		12	12	14	0
0000 _B	4:1	3	3	3	0
0001 _B		4	5	5	0
0010 _B		6	6	7	0

All others are RFU.

[Table 13] RL and nRBTP (Link ECC off case, DVFS disabled, Enhanced DVFS enabled)

OP[3:0]	WCK:CK Ratio	RL Set 0	RL Set 1	RL Set 2	nRBTP
0000 _B	2:1	6	6	6	0
0001 _B		9	10	10	0
0010 _B		13	13	14	0
0011 _B		16	16	20	2
0100 _B		20	20	24	3
0101 _B		24	24	28	4
0000 _B	4:1	3	3	3	0
0001 _B		5	5	5	0
0010 _B		7	7	7	0
0011 _B		8	8	10	1
0100 _B		10	10	12	2
0101 _B		12	12	14	2

All others are RFU.

[Table 14] RL and nRBTP (Link ECC on case, DVFS disabled, Enhanced DVFS disabled)

OP[3:0]	WCK:CK Ratio	RL Set 0	RL Set 1	nRBTP
0110 _B	4:1	12	13	2
0111 _B		13	14	2
1000 _B		15	16	3
1001 _B		17	18	4
1010 _B		18	20	4
1011 _B		19	21	4
1100 _B		23	24	6
1101 _B		26	28	6

All others are RFU.

NOTE :

- 1) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 3) The programmed value of nWR is the number of clock cycles the LPDDR5-SDRAM device uses to determine the starting point of an internal Pre-charge operation after a Write burst with AP (auto-pre-charge) enabled.
- 4) Some operating features can affect Read Latency: (Link ECC off case)

FEATURE	Description
1	Byte Mode
2	Read DBI and/or Read Data Copy

RL Set 0 applies when no features are enabled.

RL Set 1 applies when one feature is enabled (1 or 2).

RL Set 2 applies when two features are enabled.

- 5) RL Set 1 applies when the device is byte-mode. (Link ECC on case)

- 6) Mixed package with byte-mode and non byte-mode configuration should follow the latency settings for byte-mode device, specifically RL.

[Table 15] nWR Latency

MR2 OP[7:4]	WCK:CK Ratio	Data Rate Range [Mbps]		CK Frequency Range [MHz]		nWR								Unit
		Lower Limit (>)	Upper Limit (≤)	Lower Limit (>)	Upper Limit (≤)	x16 w/o DVFSC w/o Write Link ECC	x8 w/o DVFSC w/o Write Link ECC	x16 w/o DVFSC w Write Link ECC	x8 w/o DVFSC w Write Link ECC	x16 w DVFSC w/o Write Link ECC	x8 w DVFSC w/o Write Link ECC	x16 w Enhance d DVFSC w/o Write Link ECC	x8 w Enhance d DVFSC w/o Write Link ECC	
0000 _B	2:1	40	533	10	133	5	5	N/A	N/A	6	6	6	6	nCK
0001 _B	2:1	533	1067	133	267	10	10	N/A	N/A	11	12	11	12	nCK
0010 _B	2:1	1067	1600	267	400	14	15	N/A	N/A	17	18	17	18	nCK
0011 _B	2:1	1600	2133	400	533	19	20	N/A	N/A	N/A	N/A	22	23	nCK
0100 _B	2:1	2133	2750	533	688	24	25	N/A	N/A	N/A	N/A	29	30	nCK
0101 _B	2:1	2750	3200	688	800	28	29	N/A	N/A	N/A	N/A	33	35	nCK
0000 _B	4:1	40	533	5	67	3	3	N/A	N/A	3	3	3	3	nCK
0001 _B	4:1	533	1067	67	133	5	5	N/A	N/A	6	6	6	6	nCK
0010 _B	4:1	1067	1600	133	200	7	8	N/A	N/A	9	9	9	9	nCK
0011 _B	4:1	1600	2133	200	267	10	10	N/A	N/A	N/A	N/A	11	12	nCK
0100 _B	4:1	2133	2750	267	344	12	13	N/A	N/A	N/A	N/A	15	15	nCK
0101 _B	4:1	2750	3200	344	400	14	15	N/A	N/A	N/A	N/A	17	18	nCK
0110 _B	4:1	3200	3733	400	467	16	17	18	19	N/A	N/A	N/A	N/A	nCK
0111 _B	4:1	3733	4267	467	533	19	20	21	22	N/A	N/A	N/A	N/A	nCK
1000 _B	4:1	4267	4800	533	600	21	22	23	24	N/A	N/A	N/A	N/A	nCK
1001 _B	4:1	4800	5500	600	688	24	25	27	28	N/A	N/A	N/A	N/A	nCK
1010 _B	4:1	5500	6000	688	750	26	28	29	31	N/A	N/A	N/A	N/A	nCK
1011 _B	4:1	6000	6400	750	800	28	29	31	32	N/A	N/A	N/A	N/A	nCK
1100 _B	4:1	6400	7500	800	937.5	32	34	36	38	N/A	N/A	N/A	N/A	nCK

MR3_Register Information (MA[7:0] = 03_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DBI-WR	DBI-RD	WLS	BK/BG ORG		PDDS		

Function	Register Type	Operand	Data	Notes
PDDS (Pull-Down Drive Strength)	Read/Write	OP[2:0]	000_B : RFU 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 (Default) 111_B : Reserved	1,2,3
BK/BG ORG (Bank/Bank Group Organization)		OP[4:3]	00_B : BG Mode 01_B : 8B Mode 10_B : 16B Mode 11_B : Reserved	2,3,5,6,7
WLS (Write Latency Set)		OP[5]	0_B : Write Latency Set "A" (Default) 1_B : Write Latency Set "B"	2,3
DBI-RD (DBI-Read select)		OP[6]	0_B : Disabled (Default) 1_B : Read DBI-DC Enabled	2,3
DBI-WR (DBI-Write select)		OP[7]	0_B : Disabled (Default) 1_B : Write DBI-DC Enabled	2,3,4

NOTE :

- 1) All values are "typical". The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re-calibration may be required as voltage and temperature vary.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 4) Masked Write must use DBI DC only. For more details concerning masked writes with DBI enabled, refer to Data Mask (DM) and Data Bus Inversion (DBI-DC) Function section for information on data bus inversion.
- 5) 16B Mode with DVFS enabled can support up to 1600Mbps(≤1600Mbps). Please refer to Table 8.
- 6) The supported operation data rate for each Bank/Bank Group Organization is as follows.

Device Type (MR8 OP[1:0])	Bank Organization Support
00 _B : LPDDR5	- BG Mode for more than 3200Mbps (>3200Mbps). - 8B Mode for all data rate range. - 16B Mode for equal or less than 3200Mbps (≤3200Mbps).
01 _B : LPDDR5X	- BG Mode for more than 3200Mbps (>3200Mbps). - 16B Mode for equal or less than 3200Mbps (≤3200Mbps).

- 7) The LPDDR5X SDRAM does not support 8B Mode. If the MR8 OP[1:0] is 01B:LPDDR5X SDRAM, MR3 OP[4:3] shall be set to 00B: BG Mode or 10B: 16B Mode.

MR4_Register Information (MA[7:0] = 04_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
TUF	ZQ Initiator	ZQUF	Refresh Multiplier (RM)				

Function	Register Type	Operand	Data	Notes
Refresh Multiplier (RM)	Read-only	OP[4:0]	00000_B : SDRAM Low temperature operating limit exceeded 00001_B : 8x 00010_B : 6x 00011_B : 4x 00100_B : 3.3x 00101_B : 2.5x 00110_B : 2.0x 00111_B : 1.7x 01000_B : 1.3x 01001_B : 1x 01010_B : 0.7x 01011_B : 0.5x 01100_B : 0.25x, no de-rating 01101_B : 0.25x, with de-rating 01110_B : 0.125x, no de-rating 01111_B : 0.125x, with de-rating 11111_B : SDRAM High temperature operating limit exceeded All others are reserved.	1,2,3,4,6,7
ZQUF (ZQ Update Flag)		OP[5]	0_B : No change in calibration code since previous ZQ latch command 1_B : Calibration code has changed since previous ZQ Latch command	8
ZQ Initiator		OP[6]	0_B : Not a Initiator die 1_B : Initiator die for ZQ Calibration purposes	9,10
TUF (Temperature Update Flag)		OP[7]	0_B : No change in OP[4:0] since last MR4 read 1_B : Change in OP[4:0] since last MR4 read (Default)	5,6,7

- NOTE :**
- 1) The refresh rate for each MR4-OP[4:0] setting applies to t_{REFI} , t_{REFIPB} and t_{REFW} . OP[4:0]=01001_B corresponds to a device temperature of 85°C. Other values require either a longer (1.3x, 8x) refresh interval at lower temperatures, or a shorter (0.7x, 0.125x) refresh interval at higher temperatures. If OP[4:0] is from 01010_B to 11111_B, the device temperature is greater than 85°C.
 - 2) At higher temperatures (>85°C), AC timing derating may be required. If derating is required the LPDDR5-SDRAM will set OP[4:0]= 01101_B or 01111_B. See derating timing requirements in Temperature Derating AC Timing.
 - 3) DRAM vendors may or may not report all of the possible settings over the operating temperature range of the device. Each vendor guarantees that their device will work at any temperature within the range using the refresh interval requested by their device.
 - 4) The device may not operate properly when OP[4:0]=00000_B or 11111_B.
 - 5) When OP[7]=1_B, the refresh rate reported in OP[4:0] has changed since the last MR4 read. A mode register read from MR4 will reset OP[7] to '0'.
 - 6) OP[4:0] bits are indicated the latest Refresh Rate whenever OP[7]=1_B.
 - 7) See "Temperature Sensor" section for information on the recommended frequency of reading MR4.
 - 8) After Power up initialization and reset sequence have been completed ZQUF MR4 OP[5] indicate 0_B.
 - 9) In command-based calibration mode, ZQCal Start commands only need to be issued to the ZQ Initiator die or dice to maintain accurate calibration. ZQCal Start commands received by non-ZQ Initiator die will be ignored. All die which share ZQ resources with a ZQ Initiator die that receives a valid ZQCal Start command will be calibrated. ZQCal Latch commands may be issued to each of these die after t_{ZQCAL4} , t_{ZQCAL8} or $t_{ZQCAL16}$ has been met.
 - 10) LPDDR5 packages with more than one ZQ pin may include more than one ZQ Initiator die.

MR5_Register Information (MA[7:0] = 05_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
LPDDR5 Manufacturer ID							

Function	Register Type	Operand	Data	Notes
LPDDR5 Manufacturer ID	Read-only	OP[7:0]	0000 0001 _B : Samsung	

MR6_Register Information (MA[7:0] = 06_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Revision ID-1							

Function	Register Type	Operand	Data	Notes
LPDDR5 Revision ID-1	Read-only	OP[7:0]	0000 1001 _B : J-version	1

NOTE :

1) MR6 is vendor specific.

MR7_Register Information (MA[7:0] = 07_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Revision ID-2							

Function	Register Type	Operand	Data	Notes
LPDDR5 Revision ID-2	Read-only	OP[7:0]	0000 0000 _B : A-version	1

NOTE :

1) MR7 is vendor specific.

MR8_Register Information (MA[7:0] = 08_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
IO Width		Density				Type	

Function	Register Type	Operand	Data	Notes
Type	Read-only	OP[1:0]	00 _B : S32 (32n pre-fetch/8Banks), S16 SDRAM (16n pre-fetch/4Banks 4Bank Groups), S16 SDRAM (16n pre-fetch /16 Banks) 01 _B : LPDDR5X SDRAM All Others: Reserved	
Density		OP[5:2]	0000 _B : 2Gb 0001 _B : 3Gb 0010 _B : 4Gb 0011 _B : 6Gb 0100 _B : 8Gb 0101 _B : 12Gb 0110 _B : 16Gb 0111 _B : 24Gb 1000 _B : 32Gb All Others: Reserved	
IO Width		OP[7:6]	00 _B : x16 01 _B : x8 All Others: Reserved	

MR9_Register Information (MA[7:0] = 09_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Vendor Specific Test Register							

Function	Register Type	Operand	Data	Notes
Vendor Specific Test Register	Write-only	OP[7:0]	Vendor Specific	1

NOTE :

1) Only 00_H should be written to this register.

MR10_Register Information (MA[7:0] = 0A_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RDQS PST		RDQS PRE-1		WCK PST		RDQS PRE-2	RPST Mode

Function	Register Type	Operand	Data	Notes
RDQS Post-amble mode	Write-only	OP[0]	0 _B : Toggle Mode (Default) 1 _B : Static Mode	1,2,3
RDQS PRE-2 (RD Pre-amble Length)		OP[1]	0 _B : Follow MR10 OP[5:4] (Default) 1 _B : Ignore MR10 OP[5:4], and Static : tRDQS_PRE+2*tWCK, Toggle : 2*tWCK	1,2,3,6,8
WCK PST (WCK Post-amble Length)		OP[3:2]	00 _B : 2.5*tWCK (Default) 01 _B : 4.5*tWCK 10 _B : 6.5*tWCK 11 _B : Reserved	1,2,3,4,5
RDQS PRE-1 (RD Pre-amble Length)		OP[5:4]	00 _B : Static:4*tWCK, Toggle:0 (Default) 01 _B : Static:2*tWCK, Toggle:2*tWCK 10 _B : Static:0, Toggle:4*tWCK 11 _B : Static:tRDQS_PRE, Toggle:4*tWCK	1,2,3,6,7
RDQS PST (RD Post-amble Length)		OP[7:6]	00 _B : 0.5*tWCK (Default) 01 _B : 2.5*tWCK 10 _B : 4.5*tWCK 11 _B : Reserved	1,2,3,4

NOTE :

- 1) All units are t_{WCK}.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 4) t_{WCKPST} length should be larger than t_{RPST} length.
- 5) WCK PST OP[3:2] applies to both read and write operation timing as same setting.
- 6) OP[5:4]=11_B can be supported over 3200Mbps operation and t_{RDQS_PRE} is Min 2*tWCK and Max 4*tWCK.
- 7) OP[5:4]=00_B/01_B/10_B can be supported over all frequency range.
- 8) OP[1]=1_B can be supported on LPDDR5X, and over 3200Mbps operation. t_{RDQS_PRE} is Min 2*tWCK and Max 4*tWCK.

MR11_Register Information (MA[7:0] = 0B_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CS ODT OP	CA ODT			NT-ODT Enable	DQ ODT		

Function	Register Type	Operand	Data	Notes
DQ ODT (DQ Bus Receiver On-Die-Termination)	Read/Write	OP[2:0]	000_B : Disable (Default) 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 111_B : RFU	1,2,3
NT-ODT Enable (Non Target ODT Enable)		OP[3]	0_B : Non-Target ODT is disabled (Default) 1_B : Non-Target ODT is enabled	2,3
CA ODT (CA Bus Receiver On-Die-Termination)		OP[6:4]	000_B : Disable (Default) 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 111_B : RFU	1,2,3
CS ODT OP (CS ODT behavior option)	Write-only	OP[7]	0_B : Basic CS ODT behavior (Default) 1_B : CS ODT behavior option	2,3,4

NOTE :

- 1) All values are "typical". The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re-calibration may be required as voltage and temperature vary.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 4) Optional function of CS ODT OP is for LPDDR5X MR8 OP[1:0]=01_B only. Therefore 0_B is required to be written for LPDDR5 SDRAM: MR8 OP[1:0]=00_B.

MR12_Register Information (MA[7:0] = 0C_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
VBS	V _{REF(CA)}						

Function	Register Type	Operand	Data	Notes
V _{REF(CA)} (V _{REF(CA)} Setting)	Read/ Write	OP[6:0]	0000000_B : -- Thru -- 1111111_B : See Table 16	1,2,3,4, 5,7
VBS (V _{REF(CA)} Byte Select)	Write	OP[7]	0_B : Write the V _{REF(CA)} values to OP[6:0] for x16 device, and Byte mode device which is assigned lower byte: DQ[7:0]. 1_B : Write the V _{REF(CA)} values to OP[6:0] for Byte mode device which is assigned upper byte: DQ[15:8].	6,7

NOTE :

- 1) This register controls the V_{REF(CA)} levels for Frequency-Set-Point[2:0].
- 2) A read (MRR) to this register places the contents of OP[6:0] on DQ[6:0]. DQ[7] will read 0_B. See Mode Register Read section for more information on MRR Operation.
- 3) A write to MR12 OP[6:0] sets the internal V_{REF(CA)} level for FSP[0] when MR16 OP[1:0]=00_B, sets FSP[1] when MR16 OP[1:0]=01_B or sets FSP[2] when MR16 OP[1:0]=10_B. The time required for V_{REF(CA)} to reach the set level depends on the step size from the current level to the new level. See CA VREF Training section for more information on VREF(CA) training.
- 4) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
- 5) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 6) Even if when the MRW command issues to the x16 device, writing "0" to OP[7] which is not a sticky bit is required.
- 7) Byte Mode device output each individual MR12 OP[6:0] from DQ[7:0] for lower byte device and from DQ[15:8] for upper byte device by MRR command. OP[7] controls only the MRW operation.

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[Table 16] MR12 VREF(CA) Settings

Function	Operand	V _{REF} Values (% of VDDQ)								Notes
V _{REF} Settings for MR12	OP [6:0]	0000000 _B :	10.0 ⁺⁵	0100000 _B :	26.0	1000000 _B :	42.0	1100000 _B :	58.0	1,2,3,4,5
		0000001 _B :	10.5 ⁺⁵	0100001 _B :	26.5	1000001 _B :	42.5	1100001 _B :	58.5	
		0000010 _B :	11.0 ⁺⁵	0100010 _B :	27.0	1000010 _B :	43.0	1100010 _B :	59.0	
		0000011 _B :	11.5 ⁺⁵	0100011 _B :	27.5	1000011 _B :	43.5	1100011 _B :	59.5	
		0000100 _B :	12.0 ⁺⁵	0100100 _B :	28.0	1000100 _B :	44.0	1100100 _B :	60.0	
		0000101 _B :	12.5 ⁺⁵	0100101 _B :	28.5	1000101 _B :	44.5	1100101 _B :	60.5	
		0000110 _B :	13.0 ⁺⁵	0100110 _B :	29.0	1000110 _B :	45.0	1100110 _B :	61.0	
		0000111 _B :	13.5 ⁺⁵	0100111 _B :	29.5	1000111 _B :	45.5	1100111 _B :	61.5	
		0001000 _B :	14.0 ⁺⁵	0101000 _B :	30.0	1001000 _B :	46.0	1101000 _B :	62.0	
		0001001 _B :	14.5 ⁺⁵	0101001 _B :	30.5	1001001 _B :	46.5	1101001 _B :	62.5	
		0001010 _B :	15.0	0101010 _B :	31.0	1001010 _B :	47.0	1101010 _B :	63.0	
		0001011 _B :	15.5	0101011 _B :	31.5	1001011 _B :	47.5	1101011 _B :	63.5	
		0001100 _B :	16.0	0101100 _B :	32.0	1001100 _B :	48.0	1101100 _B :	64.0	
		0001101 _B :	16.5	0101101 _B :	32.5	1001101 _B :	48.5	1101101 _B :	64.5	
		0001110 _B :	17.0	0101110 _B :	33.0	1001110 _B :	49.0	1101110 _B :	65.0	
		0001111 _B :	17.5	0101111 _B :	33.5	1001111 _B :	49.5	1101111 _B :	65.5	
		0010000 _B :	18.0	0110000 _B :	34.0	1010000 _B : (Default)	50.0	1110000 _B :	66.0	
		0010001 _B :	18.5	0110001 _B :	34.5	1010001 _B :	50.5	1110001 _B :	66.5	
		0010010 _B :	19.0	0110010 _B :	35.0	1010010 _B :	51.0	1110010 _B :	67.0	
		0010011 _B :	19.5	0110011 _B :	35.5	1010011 _B :	51.5	1110011 _B :	67.5	
		0010100 _B :	20.0	0110100 _B :	36.0	1010100 _B :	52.0	1110100 _B :	68.0	
		0010101 _B :	20.5	0110101 _B :	36.5	1010101 _B :	52.5	1110101 _B :	68.5	
		0010110 _B :	21.0	0110110 _B :	37.0	1010110 _B :	53.0	1110110 _B :	69.0	
		0010111 _B :	21.5	0110111 _B :	37.5	1010111 _B :	53.5	1110111 _B :	69.5	
		0011000 _B :	22.0	0111000 _B :	38.0	1011000 _B :	54.0	1111000 _B :	70.0	
		0011001 _B :	22.5	0111001 _B :	38.5	1011001 _B :	54.5	1111001 _B :	70.5	
		0011010 _B :	23.0	0111010 _B :	39.0	1011010 _B :	55.0	1111010 _B :	71.0	
		0011011 _B :	23.5	0111011 _B :	39.5	1011011 _B :	55.5	1111011 _B :	71.5	
		0011100 _B :	24.0	0111100 _B :	40.0	1011100 _B :	56.0	1111100 _B :	72.0	
		0011101 _B :	24.5	0111101 _B :	40.5	1011101 _B :	56.5	1111101 _B :	72.5	
		0011110 _B :	25.0	0111110 _B :	41.0	1011110 _B :	57.0	1111110 _B :	73.0	
		0011111 _B :	25.5	0111111 _B :	41.5	1011111 _B :	57.5	1111111 _B :	73.5	

NOTE :

- 1) These values may be used for MR12 OP[6:0] to set the V_{REF(CA)} levels in the LPDDR5-SDRAM.
- 2) The MR12 registers represents either FSP[0], FSP[1] or FSP[2]. Three frequency-set-points each for CA and CK are provided to allow for faster switching between terminated and un-terminated operation, or between different high-frequency setting which may use different terminations values.
- 3) Absolute CA V_{REF} low level (%code * V_{DDQ}) must be higher than or equal to 75mV for normal operation. Vref error is not included this calculation.
- 4) Absolute CA Vref high level (%code * V_{DDQ}) must be lower than or equal to 350mV. Vref error is not included this calculation.
- 5) Vref codes from 0000000_B to 0001001_B are able to be used only for testing purpose not for normal operation. Vref accuracy are not guaranteed from 0000000_B to 0001001_B.

MR13_Register Information (MA[7:0] = 0D_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Dual VDD2	CBT Mode	DMD	DMI I/O Control	(RFU)	VRO	Thermal Offset	

Function	Register Type	Operand	Data	Notes
Thermal Offset	Write-only	OP[1:0]	00_B : No offset, 0~5°C gradient (Default) 01_B : 5°C offset, 5~10°C gradient 10_B : 10°C offset, 10~15°C gradient 11_B : Reserved	
VRO (VREF Output)		OP[2]	0_B : Normal operation (Default) 1_B : Output the VREF(CA) and VREF(DQ) values on DQ bits	1
DMI I/O Control (DMI input/output behavior control Mode)		OP[4]	0_B : DMI Output behavior follows MR setting Read DBI, Read Link ECC and Read Data Copy. (Default) 1_B : DMI outputs a valid data, if Read FIFO Command and Read DQ Calibration Command is issued when Data Mask and/or Write DBI are enabled even though Read DBI, Read Link ECC and Read Data Copy are disabled.	6
DMD (Data Mask Disable)		OP[5]	0_B : Data Mask Operation Enabled (Default) 1_B : Data Mask Operation Disabled	5
CBT Mode		OP[6]	0_B : CBT Training Mode 1 (Default) 1_B : CBT Training Mode 2	
Dual VDD2		OP[7]	0_B : Dual VDD2 rail (1.05V & 0.9V) used (Default) 1_B : Single 1.05V VDD2 rail used	2,3,4

- NOTE :**
- When set, the LPDDR5-SDRAM will output the V_{REF(CA)} and V_{REF(DQ)} voltages on DQ pins. Only the "active" frequency-set-point, as defined by MR16 OP[3:2], will be output on the DQ pins. This function allows an external test system to measure the internal VREF levels and its accuracy is guaranteed at or less than 25°C. The DQ pins used for VREF output are vendor specific.
 - LPDDR5 SDRAM may be powered up / initialized/ reset using either a single or dual VDD2 configuration regardless of the OP[7] setting. OP[7] shall be set based on the VDD2 configuration during initialization before normal operation. It is illegal to change OP[7] setting during normal operation. See "Voltage Ramp and Device Initialization" for additional information.
 - In the single VDD2 configuration (1.05V), the same voltage shall be supplied to all VDD2L and VDD2H balls.
 - When enabled (OP[7]=1_B), MR19 OP[1:0] setting is ignored and DRAM operates in High Speed mode.
 - When enabled (OP[5]=0_B) data masking is enabled for the device. When disabled (OP[5]=1_B), masked write command is illegal. See Data Mask(DM) and Data Bus Inversion (DBI-DC) Function.
 - DMI I/O Control: MR13 OP[4]=1_B is optional. Refer to vendor's data sheet. The feature of DMI input/output behavior control Mode is defined at Read DQ Calibration and WCK - DQ (FIFO) training section.

MR14_Register Information (MA[7:0] = 0E_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
VDLC	VREF(DQ[7:0])						

Function	Register Type	Operand	Data	Notes
VREF(DQ[7:0]) (VREF(DQ[7:0]) Setting)	Read/Write	OP[6:0]	0000000_B : -- Thru -- 1111111_B : See Table 17	1,2,3,4,5
VDLC (VREF DQ Lower byte copy)		OP[7]	X16 device only 0_B : VREF (DQ[15:8]) follow MR15 OP[6:0] code (Default) 1_B : VREF (DQ[15:8]) follow MR14 OP[6:0] code Byte Mode device ignores OP[7]	6,7

- NOTE :**
- This register controls the VREF(DQ[7:0]) levels for Frequency-Set-Point[2:0].
 - A read (MRR) to this register places the contents of OP[7:0] on DQ[7:0]. Any RFU bits and unused DQ's shall be set to'0'. See Mode Register Read section for more information on MRR Operation.
 - A write to OP[6:0] sets the internal VREF(DQ[7:0]) level for FSP[0] when MR16 OP[1:0]=00_B, sets FSP[1] when MR16 OP[1:0]=01_B or sets FSP[2] when MR16 OP[1:0]=10_B. The time required for VREF(DQ[7:0]) to reach the set level depends on the step size from the current level to the new level. See DQ VREF Training for more information on VREF(DQ) training.
 - There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address or read from with an MRR command to this address.
 - There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
 - When OP[7] is 1_B, MR14 MRR will return VREF(DQ[15:8]) and MR15 MRR value is undefined. To verify the VREF code, please refer to Table 17 for MR14 OP[6:0].
 - Byte mode device doesn't support VDLC function. SOC need to set MR14 and MR15 to individual devices.

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR HEADQUARTERS OF SAMSUNG ELECTRONICS.

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[Table 17] MR14 VREF(DQ[7:0]) Settings

Function	Operand	V _{REF} Values (% of V _{DDQ})								Notes
V _{REF} Settings for MR14	OP [6:0]	0000000 _B	10.0 ⁺⁵	0100000 _B	26.0	1000000 _B	42.0	1100000 _B	58.0	1,2,3,4,5,6
		0000001 _B	10.5 ⁺⁵	0100001 _B	26.5	1000001 _B	42.5	1100001 _B	58.5	
		0000010 _B	11.0 ⁺⁵	0100010 _B	27.0	1000010 _B	43.0	1100010 _B	59.0	
		0000011 _B	11.5 ⁺⁵	0100011 _B	27.5	1000011 _B	43.5	1100011 _B	59.5	
		0000100 _B	12.0 ⁺⁵	0100100 _B	28.0	1000100 _B	44.0	1100100 _B	60.0	
		0000101 _B	12.5 ⁺⁵	0100101 _B	28.5	1000101 _B	44.5	1100101 _B	60.5	
		0000110 _B	13.0 ⁺⁵	0100110 _B	29.0	1000110 _B	45.0	1100110 _B	61.0	
		0000111 _B	13.5 ⁺⁵	0100111 _B	29.5	1000111 _B	45.5	1100111 _B	61.5	
		0001000 _B	14.0 ⁺⁵	0101000 _B	30.0	1001000 _B	46.0	1101000 _B	62.0	
		0001001 _B	14.5 ⁺⁵	0101001 _B	30.5	1001001 _B	46.5	1101001 _B	62.5	
		0001010 _B	15.0	0101010 _B	31.0	1001010 _B	47.0	1101010 _B	63.0	
		0001011 _B	15.5	0101011 _B	31.5	1001011 _B	47.5	1101011 _B	63.5	
		0001100 _B	16.0	0101100 _B	32.0	1001100 _B	48.0	1101100 _B	64.0	
		0001101 _B	16.5	0101101 _B	32.5	1001101 _B	48.5	1101101 _B	64.5	
		0001110 _B	17.0	0101110 _B	33.0	1001110 _B	49.0	1101110 _B	65.0	
		0001111 _B	17.5	0101111 _B	33.5	1001111 _B	49.5	1101111 _B	65.5	
		0010000 _B	18.0	0110000 _B	34.0	1010000 _B (Default)	50.0	1110000 _B	66.0	
		0010001 _B	18.5	0110001 _B	34.5	1010001 _B	50.5	1110001 _B	66.5	
		0010010 _B	19.0	0110010 _B	35.0	1010010 _B	51.0	1110010 _B	67.0	
		0010011 _B	19.5	0110011 _B	35.5	1010011 _B	51.5	1110011 _B	67.5	
		0010100 _B	20.0	0110100 _B	36.0	1010100 _B	52.0	1110100 _B	68.0	
		0010101 _B	20.5	0110101 _B	36.5	1010101 _B	52.5	1110101 _B	68.5	
		0010110 _B	21.0	0110110 _B	37.0	1010110 _B	53.0	1110110 _B	69.0	
		0010111 _B	21.5	0110111 _B	37.5	1010111 _B	53.5	1110111 _B	69.5	
		0011000 _B	22.0	0111000 _B	38.0	1011000 _B	54.0	1111000 _B	70.0	
		0011001 _B	22.5	0111001 _B	38.5	1011001 _B	54.5	1111001 _B	70.5	
		0011010 _B	23.0	0111010 _B	39.0	1011010 _B	55.0	1111010 _B	71.0	
		0011011 _B	23.5	0111011 _B	39.5	1011011 _B	55.5	1111011 _B	71.5	
		0011100 _B	24.0	0111100 _B	40.0	1011100 _B	56.0	1111100 _B	72.0	
		0011101 _B	24.5	0111101 _B	40.5	1011101 _B	56.5	1111101 _B	72.5	
		0011110 _B	25.0	0111110 _B	41.0	1011110 _B	57.0	1111110 _B	73.0	
		0011111 _B	25.5	0111111 _B	41.5	1011111 _B	57.5	1111111 _B	73.5	

NOTE :

- 1) These values may be used for MR14 OP[6:0] to set the V_{REF}(DQ[7:0]) levels in the LPDDR5-SDRAM.
- 2) The MR14 registers represents either FSP[0], FSP[1] or FSP[2]. Three frequency-set-points each for DQ are provided to allow for faster switching between terminated and un-terminated operation, or between different high-frequency setting which may use different terminations values.
- 3) Absolute DQ[7:0] Vref low level (%code * V_{DDQ}) must be higher than or equal to 75mV for normal operation. Vref error is not include this calculation.
- 4) Absolute DQ[7:0] Vref high level (%code * V_{DDQ}) must be lower than or equal to 350mV when WCK is less than or equal to 1600MHz. Absolute DQ[7:0] Vref high level (%code * V_{DDQ}) must be lower than or equal to 225mV when WCK is higher than 1600MHz and less than or equal to 3200MHz. Absolute DQ[7:0] Vref high level (%code * V_{DDQ}) must be lower than or equal to 180mV when WCK is higher than 3200MHz. V_{REF} error including a receiver offset and a training error is not included this calculation. Higher VREF level code can be used for testing and training purposes.
- 5) Absolute DQ[7:0] Vref high level (%code * V_{DDQ}) must be lower than or equal to 300mV regardless of WCK frequency when Enhanced DVFS is enabled.
- 6) VREF codes from 0000000_B to 0001001_B are able to be used only for testing purpose not for normal operation. VREF accuracy are not guaranteed from 0000000_B to 0001001_B.

MR15_Register Information (MA[7:0] = 0F_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
OCC	VREF(DQ[15:8])						

Function	Register Type	Operand	Data	Notes
V _{REF} (DQ[15:8]) (V _{REF} (DQ[15:8]) Setting)	Read/ Write	OP[6:0]	0000000_B : -- Thru -- 1111111_B : See Table 18 below	1,2,3, 4,5,6
Offset Calibration Control (OCC)	Write-Only	OP[7]	0_B : Offset calibration disable (Default) 1_B : Offset calibration enable	7

NOTE :

- 1) This register controls the V_{REF}(DQ[15:8]) levels for Frequency-Set-Point[2:0].
- 2) A read (MRR) to this register places the contents of OP[7:0] on DQ[15:8]. Any RFU bits and unused DQ's shall be set to'0'. See Mode Register Read section for more information on MRR Operation.
- 3) A write to OP[6:0] sets the internal VREF(DQ[15:8]) level for FSP[0] when MR16 OP[1:0]=00_B, sets FSP[1] when MR16 OP[1:0]=01_B or sets FSP[2] when MR16 OP[1:0]=10_B. The time required for V_{REF}(DQ[15:8]) to reach the set level depends on the step size from the current level to the new level. See DQ VREF Training section for more information on VREF(DQ) training.
- 4) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address or read from with an MRR command to this address.
- 5) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 6) MR15 MRR value is undefined when MR14 OP[7] is 1B. When MR14 OP[7] is 1_B, VREF(DQ[15:8]) can be read from MR14.
- 7) This Offset Calibration Control (OCC) bit is valid only when MR8 OP[1:0]=01B (LPDDR5X SDRAM).

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[Table 18] MR15 OP[6:0] VREF(DQ[15:8]) Settings

Function	Operand	V _{REF} Values (% of V _{DDQ})								Notes
V _{REF} Settings for MR15	OP [6:0]	0000000 _B	10.0 ⁺⁵	0100000 _B	26.0	1000000 _B	42.0	1100000 _B	58.0	1,2,3,4,5,6
		0000001 _B	10.5 ⁺⁵	0100001 _B	26.5	1000001 _B	42.5	1100001 _B	58.5	
		0000010 _B	11.0 ⁺⁵	0100010 _B	27.0	1000010 _B	43.0	1100010 _B	59.0	
		0000011 _B	11.5 ⁺⁵	0100011 _B	27.5	1000011 _B	43.5	1100011 _B	59.5	
		0000100 _B	12.0 ⁺⁵	0100100 _B	28.0	1000100 _B	44.0	1100100 _B	60.0	
		0000101 _B	12.5 ⁺⁵	0100101 _B	28.5	1000101 _B	44.5	1100101 _B	60.5	
		0000110 _B	13.0 ⁺⁵	0100110 _B	29.0	1000110 _B	45.0	1100110 _B	61.0	
		0000111 _B	13.5 ⁺⁵	0100111 _B	29.5	1000111 _B	45.5	1100111 _B	61.5	
		0001000 _B	14.0 ⁺⁵	0101000 _B	30.0	1001000 _B	46.0	1101000 _B	62.0	
		0001001 _B	14.5 ⁺⁵	0101001 _B	30.5	1001001 _B	46.5	1101001 _B	62.5	
		0001010 _B	15.0	0101010 _B	31.0	1001010 _B	47.0	1101010 _B	63.0	
		0001011 _B	15.5	0101011 _B	31.5	1001011 _B	47.5	1101011 _B	63.5	
		0001100 _B	16.0	0101100 _B	32.0	1001100 _B	48.0	1101100 _B	64.0	
		0001101 _B	16.5	0101101 _B	32.5	1001101 _B	48.5	1101101 _B	64.5	
		0001110 _B	17.0	0101110 _B	33.0	1001110 _B	49.0	1101110 _B	65.0	
		0001111 _B	17.5	0101111 _B	33.5	1001111 _B	49.5	1101111 _B	65.5	
		0010000 _B	18.0	0110000 _B	34.0	1010000 _B (Default)	50.0	1110000 _B	66.0	
		0010001 _B	18.5	0110001 _B	34.5	1010001 _B	50.5	1110001 _B	66.5	
		0010010 _B	19.0	0110010 _B	35.0	1010010 _B	51.0	1110010 _B	67.0	
		0010011 _B	19.5	0110011 _B	35.5	1010011 _B	51.5	1110011 _B	67.5	
		0010100 _B	20.0	0110100 _B	36.0	1010100 _B	52.0	1110100 _B	68.0	
		0010101 _B	20.5	0110101 _B	36.5	1010101 _B	52.5	1110101 _B	68.5	
		0010110 _B	21.0	0110110 _B	37.0	1010110 _B	53.0	1110110 _B	69.0	
		0010111 _B	21.5	0110111 _B	37.5	1010111 _B	53.5	1110111 _B	69.5	
		0011000 _B	22.0	0111000 _B	38.0	1011000 _B	54.0	1111000 _B	70.0	
		0011001 _B	22.5	0111001 _B	38.5	1011001 _B	54.5	1111001 _B	70.5	
		0011010 _B	23.0	0111010 _B	39.0	1011010 _B	55.0	1111010 _B	71.0	
		0011011 _B	23.5	0111011 _B	39.5	1011011 _B	55.5	1111011 _B	71.5	
		0011100 _B	24.0	0111100 _B	40.0	1011100 _B	56.0	1111100 _B	72.0	
		0011101 _B	24.5	0111101 _B	40.5	1011101 _B	56.5	1111101 _B	72.5	
		0011110 _B	25.0	0111110 _B	41.0	1011110 _B	57.0	1111110 _B	73.0	
		0011111 _B	25.5	0111111 _B	41.5	1011111 _B	57.5	1111111 _B	73.5	

- NOTE :**
- 1) These values may be used for MR15 OP[6:0] to set the V_{REF}(DQ[15:8]) levels in the LPDDR5 SDRAM.
 - 2) The MR15 registers represents either FSP[0], FSP[1] or FSP[2]. Three frequency-set-points each for DQ are provided to allow for faster switching between terminated and un-terminated operation, or between different high-frequency setting which may use different terminations values.
 - 3) Absolute DQ[15:8] V_{REF} low level (%code * V_{DDQ}) must be higher than or equal to 75mV for normal operation. V_{REF} error is not include this calculation.
 - 4) Absolute DQ[15:8] V_{REF} high level (%code * V_{DDQ}) must be lower than or equal to 350mV when WCK is less than or equal to 1600MHz. Absolute DQ[15:8] V_{REF} high level (%code * V_{DDQ}) must be lower than or equal to 225mV when WCK is higher than 1600MHz and less than or equal to 3200MHz. Absolute DQ[15:8] Vref high level (%code * VDDQ) must be lower than or equal to 180mV when WCK is higher than 3200MHz. V_{REF} error including a receiver offset and a training error is not included this calculation. Higher VREF level code can be used for testing and training purposes.
 - 5) Absolute DQ[15:8] Vref high level (%code * VDDQ) must be lower than or equal to 300mV regardless of WCK frequency when Enhanced DVFS is enabled.
 - 6) V_{REF} codes from 0000000_B to 0001001_B are able to be used only for testing purpose not for normal operation. V_{REF} accuracy are not guaranteed from 0000000_B to 0001001_B.

MR16_Register Information (MA[7:0] = 10_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CBT-Phase	VRCG	CBT		FSP-OP		FSP-WR	

Function	Register Type	Operand	Data	Notes
FSP-WR (Frequency Set Point Write Enable)	Write/ Read	OP[1:0]	00_B : Frequency-Set-Point [0] (Default) 01_B : Frequency-Set-Point [1] 10_B : Frequency-Set-Point [2] 11_B : Reserved	1
FSP-OP (Frequency Set Point Operation Mode)		OP[3:2]	00_B : Frequency-Set-Point [0] (Default) 01_B : Frequency-Set-Point [1] 10_B : Frequency-Set-Point [2] 11_B : Reserved	2
CBT (Command Bus Training)		OP[5:4]	00_B : Normal Operation (Default) 01_B : Command Bus Training Mode Enabled FSP0 10_B : Command Bus Training Mode Enabled FSP1 11_B : Command Bus Training Mode Enabled FSP2	3
VRCG (VREF Current Generator)		OP[6]	0_B : Normal Operation (Default) 1_B : VREF Fast Response (high current) mode	4
CBT-Phase		OP[7]	0_B : DQ outputs CA pattern latched by CK rising edge (Default) 1_B : DQ outputs CA pattern latched by CK falling edge	

NOTE :

- 1) FSP-WR determines which frequency-set-point registers are accessed with MRW commands for the following functions such as V_{REF(CA)} Setting, V_{REF(DQ)} Setting. For more information, refer to Frequency Set Point section.
- 2) FSP-OP determines which frequency-set-point register values are currently used to specify device operation for the following functions such as V_{REF(CA)} Setting, V_{REF(DQ)} Setting. For more information, refer to Frequency Set Point section.
- 3) A write to set OP[5:4]= 01_B, 10_B or 11_B causes the LPDDR5-SDRAM to enter the Command Bus Training mode. When OP[5:4]= 01_B, 10_B or 11_B, commands are ignored and the contents of CA[6:0] are mapped to the DQ bus. See the Command Bus Training section for more information.
- 4) When OP[6]=1_B, the VREF circuit uses a high-current mode to improve VREF settling time.

MR17_Register Information (MA[7:0] = 11_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
x8ODTD Upper	x8ODTD Lower	ODTD-CA	ODTD-CS	ODTD-CK	SOC ODT		

Function	Register Type	Operand	Data	Notes
SoC ODT (Controller ODT Value for VOH calibration)	Read/Write	OP[2:0]	000_B : Disable (Default) 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 111_B : RFU	1,2,3,6
ODTD-CK (CK ODT termination)		OP[3]	0_B : ODTD-CK Enable 1_B : ODTD-CK Disable (Default)	2,3,4
ODTD-CS (CS ODT termination)		OP[4]	0_B : ODTD-CS Enable RZQ/3 1_B : ODTD-CS Disable (Default)	2,3,5,7,8
ODTD-CA (CA ODT termination)		OP[5]	0_B : ODTD-CA Enable 1_B : ODTD-CA Disable (Default)	2,3,4
X8 ODTD Lower (CA/CS/CK ODT termination disable, Lower Byte select)		OP[6]	x8 per ch. only, Lower Byte selected Device 0_B : ODTD-CA/CS/CK follows MR17 OP[5], OP[4], OP[3] & MR11 OP[6:4] (Default) 1_B : ODTD-CA/CS/CK Disabled	7
X8 ODTD Upper (CA/CS/CK ODT termination disable, Upper Byte select)		OP[7]	x8 per ch. only, Upper Byte selected Device 0_B : ODTD-CA/CS/CK follows MR17 OP[5], OP[4], OP[3] & MR11 OP[6:4] (Default) 1_B : ODTD-CA/CS/CK Disabled	7

- NOTE :**
- 1) All values are "typical".
 - 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
 - 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
 - 4) To ensure proper operation in a multi-rank configuration, when CA, CK ODT is enabled via MR11 OP[6:4], MR17 OP[3] and OP[5], the rank providing ODT will continue to terminate the command bus in all DRAM states including Active Self-refresh, Self-refresh Power-down, Active Power-down and Idle Power-down.
 - 5) To ensure proper operation in a multi-rank configuration, when CS is enabled via MR17 OP[4], the rank providing ODT will continue to terminate in all DRAM states except Idle Power-down, Active Power-down, Self-refresh Power-down and Deep Sleep Mode. CS ODT state goes OFF ignoring MRS ODT state after Power-Down Entry is issued and returns to MRS ODT state with Power-Down Exit.
 - 6) DRAM Pull-up driver strength is controlled by OP[2:0] SOC ODT setting when DQ termination is disabled.
 - 7) DRAM will ignore all CS ODT settings, and CS will remain unterminated at all times, when MR21 OP[3] = 0b.
 - 8) In LPDDR5, CS ODT value is RZQ/3. In LPDDR5X, CS ODT value is determined by MR19 OP[7:6].

MR18_Register Information (MA[7:0] = 12_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CKR	WCK2CK Leveling	(RFU)	WCK ON	WCK_FM	WCK ODT		

Function	Register Type	Operand	Data	Notes
WCK ODT	Write-only	OP[2:0]	000_B : ODT disable (Default) 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 111_B : RFU	1,2,4
WCK_FM (WCK Frequency Mode)		OP[3]	0_B : Low frequency mode (Default) 1_B : High frequency mode	1,2,5
WCK ON (WCK always ON mode)		OP[4]	0_B : WCK Always On Mode Disabled (Default) 1_B : WCK Always On Mode Enabled	1,2
WCK2CK Leveling		OP[6]	0_B : WCK2CK Leveling Mode Disable (Default) 1_B : WCK2CK Leveling Mode Enable	
CKR (WCK to CK frequency ratio)		OP[7]	0_B : 4:1 ratio 1_B : 2:1 ratio (Default)	1,2,3

NOTE :

- 1) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The SDRAM will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the SDRAM and may be changed without affecting SDRAM operation.
- 3) CKR 2:1 can support up to 3200Mbps.
- 4) The SDRAM will continue to terminate WCK in all states, if termination is enabled by MR18 OP[2:0].
- 5) t_{WCK2DQ} AC parameters can be changed by MR18 OP[3]. Refer to t_{WCK2DQ} AC parameter table. WCK single-ended mode (MR20 OP[3:2]) can be allowed during Low frequency mode only.

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MR19_Register Information (MA[7:0] = 13_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CS ODT		WCK2DQ OSC FM support	WCK2DQ OSC FM	DVFSQ		DVFSC	

Function	Register Type	Operand	Data	Notes
DVFSC (V _{DD2} Dynamic Voltage and Frequency Scaling Core)	Write-only	OP[1:0]	00_B : High Speed mode (only V _{DD2H} : 1.05V rail) (Default) 01_B : Low speed mode: DVFSC mode (use V _{DD2L} : 0.9V rail) 10_B : Low Speed mode: Enhanced DVFSC mode (use VDD2L: 0.9V rail) 11_B : RFU	1,2,7,8,9
DVFSQ (V _{DDQ} Dynamic Voltage and Frequency Scaling V _{DDQ})		OP[3:2]	00_B : V _{DDQ} = 0.5V (Default) 01_B : V _{DDQ} = 0.3V All others : reserved	1,2,3
WCK2DQ OSC FM		OP[4]	0_B : WCK2DQ oscillator for WCK low frequency mode (Default) 1_B : WCK2DQ oscillator for WCK high frequency mode	1,2,4,5
WCK2DQ OSC FM support	Read-only	OP[5]	0_B : WCK2DQ OSC FM not supported 1_B : WCK2DQ OSC FM supported	
CS ODT (CS termination)	Read/Write	OP[7:6]	00_B : RZQ/3 (Default) 01_B : RZQ/2 10_B : RZQ/1 11_B : RFU	1,2,6

- NOTE :**
- 1) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address or read from with an MRR command to this address.
 - 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
 - 3) When DVFSQ OP[3:2] is 01_B, DRAM will turn off all ODT DQ, CS, CA, CK and WCK as well as DQ NT-ODT will turn off regardless of MR setting about ODT and NT-ODT.
 - 4) In the case of MR19 OP[5]=0_B (LPDDR5 WCK2DQ OSC FM not supported), MR19 OP[4] is "don't care" and WCK2DQ OSC follows MR18 OP[3] WCK Frequency Mode.
 - 5) When operating in WCK low frequency mode (MR18 OP[3]=0_B), WCK2DQ Oscillator for WCK high frequency mode (MR19 OP[4]=1_B) can be enabled and vice versa.
 - 6) OP[7:6] can be supported on LPDDR5X, and all values are "typical".
 - 7) MR19 OP[1:0]=00_B or 01_B is allowed when MR41 OP[2:1]=00_B.
 - 8) MR19 OP[1:0]=00_B or 10_B is allowed when MR41 OP[2:1]=01_B.
 - 9) MR19 OP[1:0]=00_B, 01_B or 10_B is allowed when MR41 OP[2:1]=10_B.

MR20_Register Information (MA[7:0] = 14_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RDC DQ mode	RDC DMI mode	MRWDU	MRWDL	WCK mode		RDQS	

Function	Register Type	Operand	Data	Notes
RDQS (Read DQS)	Write-only	OP[1:0]	00_B : RDQS_t and RDQS_c disabled 01_B : RDQS_t enabled and RDQS_c disabled (Default) 10_B : RDQS_t and RDQS_c enabled 11_B : RDQS_t disabled and RDQS_c enabled	1,2,3, 6,7,9
WCK mode		OP[3:2]	00_B : differential (Default) 01_B : single-ended from WCK_t 10_B : single-ended from WCK_c 11_B : reserved	1,2,4, 5,8
MRWDL (Mode Register Write Disable Lower byte)		OP[4]	0_B : Lower Byte MRW is enabled (Default) 1_B : Lower Byte MRW is disabled	10
MRWDU (Mode Register Write Disable Upper byte)		OP[5]	0_B : Upper Byte MRW is enabled (Default) 1_B : Upper Byte MRW is disabled	10
RDC DMI mode		OP[6]	In Read DQ Calibration, DMI output pattern is controlled as: 0_B : DMI pattern will be decided by MR33/34 (Default) 1_B : DMI pattern will be low-fixed	
RDC DQ mode		OP[7]	In Read DQ Calibration, DQ output pattern is controlled by MR33/34 (pattern) and MR31/32 (per-bit control), where, MR31/32 function is defined as: 0_B : MR31/32 decides whether "invert" or not. (Default) 1_B : MR31/32 decides whether "low-fix" or not.	

NOTE :

- 1) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 3) WCK clocking generates RDQS_t and RDQS_c.
- 4) When MR20 OP[3:2] = 01_B, WCK_t is used as WCK timing, and WCK_c should be maintained at a valid logic level.
- 5) When MR20 OP[3:2] = 10_B, WCK_c is used as WCK timing, and WCK_t should be maintained at a valid logic level.
- 6) When MR20 OP[1:0] = 01_B, RDQS_t is used as RDQS timing, and RDQS_c should be Hi-Z state.
- 7) When MR20 OP[1:0] = 11_B, RDQS_c is used as RDQS timing, and RDQS_t should be Hi-Z state.
- 8) When MR20 OP[3:2] = 01_B, WCK_t polarity is the same as WCK_t in MR20 OP[3:2] = 00_B, and when MR20 OP[3:2] = 10_B, WCK_c polarity is the same as WCK_c in MR20 OP[3:2] = 00_B.
- 9) When MR20 OP[1:0] = 01_B, RDQS_t polarity is the same as RDQS_t in MR20 OP[1:0] = 10_B, and when MR20 OP[1:0] = 11_B, RDQS_c polarity is the same as RDQS_c in MR20 OP[1:0] = 10_B.
- 10) This MR setting is not valid to a standard X16 die.

MR21_Register Information (MA[7:0] = 15_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WXS	WXFE	RDCFE	WDCFE	ODTD-CSFS	WXFS	RDCFS	WDCFS

Function	Register Type	Operand	Data	Notes
WDCFS (WRITE Data Copy Function Support)	Read-only	OP[0]	0 _B : LPDDR5 WRITE data copy function not supported 1 _B : LPDDR5 WRITE data copy function supported	4
RDCFS (READ Data Copy Function Support)		OP[1]	0 _B : LPDDR5 READ data copy function not supported 1 _B : LPDDR5 READ data copy function supported	4
WXFS (Write X Function Support)		OP[2]	0 _B : LPDDR5 Write-X function not supported 1 _B : LPDDR5 Write-X function supported	4
ODTD-CSFS (ODTD-CS Function Support)		OP[3]	0 _B : LPDDR5 ODTD-CS function not supported 1 _B : LPDDR5 ODTD-CS function supported	4
WDCFE (WRITE Data Copy Function Enable)	Write-only	OP[4]	0 _B : LPDDR5 WRITE data copy function disable (Default) 1 _B : LPDDR5 WRITE data copy function enable	1
RDCFE (READ Data Copy Function Enable)		OP[5]	0 _B : LPDDR5 READ data copy function disable (Default) 1 _B : LPDDR5 READ data copy function enable	2
WXFE (Write X Function Enable)		OP[6]	0 _B : LPDDR5 write-X function disable (Default) 1 _B : LPDDR5 write-X function enable	3
WXS (Selection function of data to be written by Write X)	Read-only	OP[7]	0 _B : Data to be written is "0" only. 1 _B : Data to be written can be selected with "0" and "1" and Byte control.	4

NOTE :

- 1) MR21 OP[4] is "don't care" in case of MR21 OP[0]=0_B (LPDDR5 WRITE data copy function not supported).
- 2) MR21 OP[5] is "don't care" in case of MR21 OP[1]=0_B (LPDDR5 READ data copy function not supported).
- 3) MR21 OP[6] is "don't care" in case of MR21 OP[2]=0_B (LPDDR5 write-X function not supported).
- 4) Data Copy function, Write X and ODTD-CS function are optional. Refer to vendor's data sheet.

MR22_Register Information (MA[7:0] = 16_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RECC		WECC		(RFU)			

Function	Register Type	Operand	Data	Notes
WECC (Write link ECC Control)	Write-only	OP[5:4]	00 _B : Write link ECC disable (Default) 01 _B : Write link ECC enable 10 _B : Reserved 11 _B : Reserved	1,2
RECC (Read link ECC Control)		OP[7:6]	00 _B : Read link ECC disable (Default) 01 _B : Read link ECC enable 10 _B : Reserved 11 _B : Reserved	1,2

NOTE :

- 1) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 2) Refer to Link ECC for LPDDR5 link ECC descriptions.

MR23_Register Information (MA[7:0] = 17_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PASR Segment Mask							

Function	Register Type	Operand	Data	Notes
PASR Segment Mask	Write Only	[7:0]	0_B : Segment Refresh Enable (Default) 1_B : Segment Refresh Disable	1,2

[Table 19] Row Address of Masked Segment for x16 Mode

Segment	OP[n]	Segment Mask	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb
			R12:R10	R13:R11	R13:R11	R14:R12	R14:R12	R15:R13	R15:R13	R16:R14	R16:R14
0	0	xxxxxx1	000 _B								
1	1	xxxxxx1x	001 _B								
2	2	xxxxx1xx	010 _B								
3	3	xxxx1xxx	011 _B								
4	4	xxx1xxxx	100 _B								
5	5	xx1xxxxx	101 _B								
6	6	x1xxxxxx	110 _B	Not Allowed	110 _B	Not Allowed	110 _B	Not Allowed	110 _B	Not Allowed	110 _B
7	7	1xxxxxxx	111 _B		111 _B		111 _B		111 _B		111 _B

NOTE :

- 1) This table indicates the range of row addresses in each masked segment. "X" is don't care for a particular segment.
- 2) For 3Gb, 6Gb, 12Gb, and 24Gb densities, OP[7:6] must always be LOW (=00_B).

[Table 20] Row Address of Masked Segment for x8 Mode

Segment	OP[n]	Segment Mask	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb
			R13:R11	R14:R12	R14:R12	R15:R13	R15:R13	R16:R14	R16:R14	R17:R15	R17:R15
0	0	xxxxxx1	000 _B								
1	1	xxxxxx1x	001 _B								
2	2	xxxxx1xx	010 _B								
3	3	xxxx1xxx	011 _B								
4	4	xxx1xxxx	100 _B								
5	5	xx1xxxxx	101 _B								
6	6	x1xxxxxx	110 _B	Not Allowed	110 _B	Not Allowed	110 _B	Not Allowed	110 _B	Not Allowed	110 _B
7	7	1xxxxxxx	111 _B		111 _B		111 _B		111 _B		111 _B

NOTE :

- 1) This table indicates the range of row addresses in each masked segment. "X" is don't care for a particular segment.
- 2) For 3Gb, 6Gb, 12Gb, and 24Gb densities, OP[7:6] must always be LOW (=00_B).

MR24_Register Information (MA[7:0] = 18_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DFES	DFE Quantity for Upper Byte (DFEQU)			Read DCA support	DFE Quantity for Lower Byte (DFEQL)		

Function	Register Type	Operand	Data	Notes
DFE Quantity for Lower Byte (DFEQL)	Write-Only	OP[2:0]	000_B : DFE disabled (Default) 001_B : Minimum negative feedback quantity : : 111_B : Maximum negative feedback quantity	1,2,3,4,5,6
Read DCA support	Read-Only	OP[3]	0_B : Read DCA not supported 1_B : Read DCA supported	
DFE Quantity for Upper Byte (DFEQU)	Write-Only	OP[6:4]	000_B : DFE disabled (Default) 001_B : Minimum negative feedback quantity : : 111_B : Maximum negative feedback quantity	1,2,3,4,5,6
DFE support (DFES)	Read-Only	OP[7]	0_B : DFE is not supported 1_B : DFE is supported	

- NOTE :**
- 1) DFE quantity is in DFE Quantity when the device supports 3 step DFE table and DFE Quantity when the device supports 7 step DFE table.
 - 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address or read from with an MRR command to this address.
 - 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
 - 4) 000_B means DFE disabled when per-pin DFE is disabled, and zero negative feedback quantity when per-pin DFE is enabled.
 - 5) LPDDR5(MR8 OP[1:0]=00_B) supports 3 step DFE only. LPDDR5X(MR8 OP[1:0]=01_B) supports 3 step DFE or 7 step DFE. DFE quantities from 100_B to 111_B are valid for 7 step DFE only.
 - 6) LPDDR5X SDRAM (MR8 OP[1:0]=01_B) supports either 3-step or 7-step DFE quantity. Refer to vendor's data sheet for detail.

MR25_Register Information (MA[7:0] = 19_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Optimized Refresh mode	PARC	CA Inputs TERM	CK Pair TERM	(RFU)			

Function	Register Type	Operand	Data	Notes
CK Pair TERM (Other Shared dies' CK ODT Info)	Write-only	OP[4]	0_B : All ranks sharing CK Pair are un-terminated (Default) 1_B : One of ranks sharing CK Pair are terminated	1,2
CA Inputs TERM (Other Shared dies' CA ODT Info)		OP[5]	0_B : All ranks sharing CA Inputs are un-terminated (Default) 1_B : One of ranks sharing CA Inputs are terminated	1,2
PARC (Partial Array Refresh Control)		OP[6]	0_B : PARC disable (Default) 1_B : PARC enable	3,4
Optimized Refresh mode		OP[7]	0_B : Optimized Refresh mode disabled 1_B : Optimized Refresh mode enabled (Default)	5,6

- NOTE :**
- 1) MR25 OP[5]/[4] is set to notify CA/CK ODT status of other shared dies.
 - 2) When CK and CA ODT status is different from each other (ex. CK termination, CA un-termination) and MR25 OP[5] is disabled, the un-terminated CA input buffer uses the fixed level reference voltage (TBD).
 - 3) MR23 PASR Segment Mask is applied to PARC (Partial Array Refresh Control) operation if PARC enabled.
 - 4) Refer to Partial Array Self Refresh (PASR) for LPDDR5 PASR descriptions.
 - 5) When LPDDR5 SDRAM does not support Optimized Refresh mode (MR0 OP[3]=0B), MR25 OP[7] is don't care. In case of MR0 OP[3]=0B or MR25 OP[7]=0B, Refresh operation needs to follow the refresh requirement which is defined in Refresh Requirement.
 - 6) SoC which does not support the optimized refresh mode can use SDRAM without changing MR OP[7] to 0B, if the SoC follows the defines in Refresh Requirement.

MR26_Register Information (MA[7:0] = 1A_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RDQST FE	RDQST FS	DCMU1	DCMU0	DCML1	DCML0	DCM Flip	DCM Start/ Stop

Function	Register Type	Operand	Data	Notes
DCM_Start/Stop	Write-Only	OP[0]	0_B : Stop (Default) 1_B : Start	
Flip inputs to cancel offset (DCM_Flip)		OP[1]	0_B : No flip (Default) 1_B : Flip	
Duty cycle result for lower byte when DCM Flip=0 (DCML0)	Read-Only	OP[2]	0_B : High duty cycle < 50% for lower byte 1_B : High duty cycle ≥ 50% for lower byte	
Duty cycle result for lower byte when DCM Flip=1 (DCML1)		OP[3]	0_B : High duty cycle < 50% for lower byte 1_B : High duty cycle ≥ 50% for lower byte	
Duty cycle result for upper byte when DCM Flip=0 (DCMU0)		OP[4]	0_B : High duty cycle < 50% for upper byte 1_B : High duty cycle ≥ 50% for upper byte	
Duty cycle result for upper byte when DCM Flip=1 (DCMU1)		OP[5]	0_B : High duty cycle < 50% for upper byte 1_B : High duty cycle ≥ 50% for upper byte	
RD/WR-based WCK-RDQS_t Training mode support (RDQSTFS)		OP[6]	0_B : SDRAM does not support RD/WR-based WCK-RDQS_t Training mode 1_B : SDRAM supports RD/WR-based WCK-RDQS_t Training mode	
RD/WR-based WCK-RDQS_t Training mode Enable (RDQSTFE)	Write-Only	OP[7]	0_B : SDRAM does not support RD/WR-based WCK-RDQS_t Training Disable (Default) 1_B : SDRAM supports RD/WR-based WCK-RDQS_t Training Enable	

MR27_Register Information (MA[7:0] = 1B_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RAAMULT		RAAIMT					RFM

Function	Register Type	Operand	Data	Notes
RFM (RFM Required)	Read-Only	OP[0]	0 _B : RFM not required 1 _B : RFM required	1
RAAIMT (Rolling Accumulated ACT Initial Management Threshold)		OP[5:1]	00000 _B : Invalid 00001 _B : 8 00010 _B : 16 (step +8) 11110 _B : 240 11111 _B : 248	1
RAAMULT (Rolling Accumulated ACT Multiplier)		OP[7:6]	00 _B : 2X 01 _B : 4X 10 _B : 6X 11 _B : 8X	1,2

NOTE :

- 1) Vendor programmed, no need to periodical read.
- 2) RAAMMT = RAAMULT * RAAIMT

MR28_Register Information (MA[7:0] = 1C_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
(RFU)		ZQ mode	(RFU)	ZQ Interval		ZQ Stop	ZQ Reset

Function	Register Type	Operand	Data	Notes
ZQ Reset	Write-only	OP[0]	0 _B : Normal Operation (Default) 1 _B : ZQ Reset	1,2
ZQ Stop		OP[1]	0 _B : Normal Operation (Default) 1 _B : Background ZQ Calibration is halted after t _{ZQSTOP}	3,4
ZQ Interval		OP[3:2]	00 _B : Background Cal Interval ≤32ms 01 _B : Background Cal Interval ≤64ms (Default) 10 _B : Background Cal Interval ≤128ms 11 _B : Background Cal Interval ≤256ms	5
ZQ Mode		OP[5]	0 _B : Background ZQ Calibration (Default) 1 _B : Command-Based ZQ Calibration	5

NOTE :

- 1) See ZQCalibration Timing Parameters table for calibration latency and timing.
- 2) Asserting ZQ Reset will set the calibration values to their default setting.
- 3) When ZQ stop is enabled, the ZQ resource is available for use by other devices. See ZQ Stop Functionality section.
- 4) In Command-Based Calibration mode ZQCal Start commands are ignored when MR28 OP[1]=1_B.
- 5) ZQ Interval and ZQ Mode MR settings are only applicable to ZQ Initiator die. These settings will be ignored by ZQ Target die.

MR29_Register Information (MA[7:0] = 1D_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PPR Resource Bank 7	PPR Resource Bank 6	PPR Resource Bank 5	PPR Resource Bank 4	PPR Resource Bank 3	PPR Resource Bank 2	PPR Resource Bank 1	PPR Resource Bank 0

Function	Register Type	Operand	Data	Notes
PPR Resource Bank 0	Read-only	OP[0]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 1		OP[1]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 2		OP[2]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 3		OP[3]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 4		OP[4]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 5		OP[5]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 6		OP[6]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1
PPR Resource Bank 7		OP[7]	0 _B : PPR Resource is not available 1 _B : PPR Resource is available	1

NOTE :
1) PPR Resources describe as follows:

PPR Resource		MR29 OP[0]	MR29 OP[1]	MR29 OP[2]	MR29 OP[3]	MR29 OP[4]	MR29 OP[5]	MR29 OP[6]	MR29 OP[7]
8Bank	Bank	0	1	2	3	4	5	6	7
16Bank	Bank	0, 8	1, 9	2, 10	3, 11	4, 12	5, 13	6, 14	7, 15
BG	BG	0, 2	0, 2	0, 2	0, 2	1, 3	1, 3	1, 3	1, 3
	Bank	0	1	2	3	0	1	2	3

MR30_Register Information (MA[7:0] = 1E_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DCA for Upper Byte (DCAU)				DCA for Lower Byte (DCAL)			

Function	Register Type	Operand	Data	Notes
DCA for Lower Byte (DCAL)	Write-only	OP[3:0]	0000_B : 0 Steps (Default no adjustment) 0001_B : -1 Steps 0010_B : -2 Steps 0011_B : -3 Steps 0100_B : -4 Steps 0101_B : -5 Steps 0110_B : -6 Steps 0111_B : -7 Step 1000_B : RFU 1001_B : +1 Step 1010_B : +2 Steps 1011_B : +3 Steps 1100_B : +4 Steps 1101_B : +5 Steps 1110_B : +6 Steps 1111_B : +7 Steps	1,2,3,4,5
DCA for Upper Byte (DCAU)		OP[7:4]	0000_B : 0 Steps (Default no adjustment) 0001_B : -1 Steps 0010_B : -2 Steps 0011_B : -3 Steps 0100_B : -4 Steps 0101_B : -5 Steps 0110_B : -6 Steps 0111_B : -7 Step 1000_B : RFU 1001_B : +1 Step 1010_B : +2 Steps 1011_B : +3 Steps 1100_B : +4 Steps 1101_B : +5 Steps 1110_B : +6 Steps 1111_B : +7 Steps	1,2,3,4,5

NOTE :

- 1) 0001_B to 0111_B of bit sets will decrease the internal WCK duty cycle.
- 2) 1001_B to 1111_B of bit sets will increase the internal WCK duty cycle.
- 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 4) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The SDRAM will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the SDRAM and may be changed without affecting SDRAM operation.
- 5) With byte mode device, OP[3:0] applies lower byte device and OP[7:4] applies upper byte device. Unused operands (OP[7:4] for lower byte device and OP[3:0] for upper byte device) should be set 0_B or 1_B by which are ignored SDRAM though.

MR31_Register Information (MA[7:0] = 1F_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Lower-byte per-bit control Register for DQ Calibration							

Function	Register Type	Operand	Data	Notes
Lower-byte per-bit control for DQ Calibration	Write-only	OP[7:0]	The following values may be written for any operand OP[7:0], and will be applied to the corresponding DQ locations DQ[7:0] within a byte lane: In DQ inversion mode (MR20[7] = 0), 0_B: Do not invert 1_B: Invert the DQ Calibration patterns in MR33 and MR34 In DQ output fix0 mode (MR20[7] = 1), 0_B: Do not apply low-fix 1_B: Data pattern is low-fixed Default value for OP[7:0]=55_H	1,2,3,4

NOTE :

- 1) This register will invert or apply low-fix for the DQ Calibration pattern found in MR33 and MR34 for any single DQ, or any combination of DQ's.
 Example: In case of DQ inversion mode, If MR31 OP[7:0]=00010101B, then the DQ Calibration patterns transmitted on DQ[7,6,5,3,1] will not be inverted, but the DQ Calibration patterns transmitted on DQ[4,2,0] will be inverted.
 In case of DQ output fix0 mode, If MR31 OP[7:0]=00010101B, then the DQ Calibration patterns transmitted on DQ[7,6,5,3,1] will not be low-fixed, but the DQ Calibration patterns transmitted on DQ[4,2,0] will be low-fixed.
- 2) DMI[0] is not inverted. DMI pattern is decided by MR33/34, where MR20[6] defines whether DMI output low-fix or not.
- 3) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[7:6].
- 4) In case of byte mode, MR31 is valid only for lower byte selected device.

[Table 21] MR31 Invert Register Pin Mapping

PIN	DQ0	DQ1	DQ2	DQ3	DMI0	DQ4	DQ5	DQ6	DQ7
MR31	OP0	OP1	OP2	OP3	NO-Invert	OP4	OP5	OP6	OP7

MR32_Register Information (MA[7:0] = 20_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Upper-byte per-bit control Register for DQ Calibration							

Function	Register Type	Operand	Data	Notes
Upper-byte per-bit control for DQ Calibration	Write-only	OP[7:0]	The following values may be written for any operand OP[7:0], and will be applied to the corresponding DQ locations DQ[15:8] within a byte lane: In DQ inversion mode (MR20[7] = 0), 0_B: Do not invert 1_B: Invert the DQ Calibration patterns in MR33 and MR34 In DQ output fix0 mode (MR20[7] = 1), 0_B: Do not apply low-fix 1_B: Data pattern is low-fixed Default value for OP[7:0] = 55_H	1,2,3,4

NOTE :

- 1) This register will invert or apply low-fix for the DQ Calibration pattern found in MR33 and MR34 for any single DQ, or any combination of DQ's.
 Example: In case of DQ inversion mode, If MR32OP[7:0]=00010101B, then the DQ Calibration patterns transmitted on DQ[15,14,13,11,9] will not be inverted, but the DQ Calibration patterns transmitted on DQ[12,10,8] will be inverted.
 In case of DQ output fix0 mode, If MR31 OP[7:0]=00010101B, then the DQ Calibration patterns transmitted on DQ[15,14,13,11,9] will not be low-fixed, but the DQ Calibration patterns transmitted on DQ[12,10,8] will be low-fixed.
- 2) DMI[1] is not inverted. DMI pattern is decided by MR33/34, where MR20[6] defines whether DMI output low-fix or not.
- 3) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[7:6].
- 4) In case of byte mode, MR32 is valid only for upper byte selected device.

[Table 22] MR32 Invert Register Pin Mapping

PIN	DQ8	DQ9	DQ10	DQ11	DMI1	DQ12	DQ13	DQ14	DQ15
MR32	OP0	OP1	OP2	OP3	NO-Invert	OP4	OP5	OP6	OP7

MR33_Register Information (MA[7:0] = 21_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ Calibration Pattern "A"							

Function	Register Type	Operand	Data	Notes
Return DQ Calibration Pattern MR33 + MR34	Write-only	OP[7:0]	X_B : DQ Calibration Pattern A "5A _H " (Default)	1,2,3,4

NOTE :

- 1) Read DQ Calibration command (RDC) causes the device to return the DQ Calibration Pattern contained in this register followed by the contents of MR34. The pattern contained in MR33 is transmitted on DQ[15:0] and DMI[1:0] when DQ Read Calibration is issued. The pattern is transmitted serially on each data lane, is organized as "big endian" such that the low-order bit in a byte is transmitted first. If the data pattern in MR33 is 27H, then the first bit transmitted will be a '1', followed by '1', '0', '0', '1', '0', and '0'. The bit stream will be 11100100B. A default pattern "5AH" is loaded at power-up or RESET, or the pattern may be overwritten with an MRW to this register. The contents of MR31 and MR32 will invert the data pattern for a given DQ. See MR31 definition for more information.
- 2) MR31 and MR32 may be used to invert the MR33/MR34 data patterns on the DQ pins. See MR31 and MR32 for more information. Data is never inverted on the DMI[1:0] pins.
- 3) The data pattern is not transmitted on the DMI[1:0] pins if DBI-RD is disabled via MR3-OP[6:5].
- 4) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[6:5].

MR34_Register Information (MA[7:0] = 22_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ Calibration Pattern "B"							

Function	Register Type	Operand	Data	Notes
Return DQ Calibration Pattern MR33 + MR34	Write-only	OP[7:0]	X_B : DQ Calibration Pattern B "3C _H " (Default)	1,2,3,4

NOTE :

- 1) Read DQ Calibration command (RDC) causes the device to return the DQ Calibration Pattern contained in MR33 and followed by the contents of this register. The pattern contained in MR34 is concatenated to the end of MR33 and transmitted on DQ[15:0] and DMI[1:0] when DQ Read Calibration is issued. The pattern is transmitted serially on each data lane, organized "big endian" such that the low-order bit in a byte is transmitted first. If the data pattern in MR33 is 27H, then the first bit transmitted will be a '1', followed by '1', '0', '0', '1', '0', and '0'. The bit stream will be 11100100B. A default pattern "3CH" is loaded at power-up or RESET, or the pattern may be overwritten with an MRW to this register. See MR33 for more information.
- 2) MR31 and MR32 may be used to invert the MR33/MR34 data patterns on the DQ pins. See MR31 and MR32 for more information. Data is never inverted on the DMI[1:0] pins.
- 3) The data pattern is not transmitted on the DMI[1:0] pins if DBI-RD is disabled via MR3-OP[6:5].
- 4) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[6].

MR35_Register Information (MA[7:0] = 23_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WCK2DQI Oscillator Count - LSB							

Function	Register Type	Operand	Data	Notes
WCK2DQI Oscillator (DQ input Training WCK Oscillator)	Read-only	OP[7:0]	0 - 255 LSB DRAM WCK2DQI Oscillator Count	1,2,3,4

NOTE :

- 1) MR35 reports the LSB bits of the DRAM WCK2DQI Oscillator count. The DRAM WCK2DQI Oscillator count value is used to train WCK to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of WCK relative to DQ.
- 2) Both MR35 and MR36 must be read (MRR) and combined to get the value of the WCK2DQI Oscillator count.
- 3) A new MPC [Start WCK2DQI Oscillator] could be issued to reset the contents of MR35/MR36.
- 4) WCKDQI & WCKDQO cannot be operated simultaneously.

MR36_Register Information (MA[7:0] = 24_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WCK2DQI Oscillator Count - MSB							

Function	Register Type	Operand	Data	Notes
WCK2DQI Oscillator (DQ input Training WCK Oscillator)	Read-only	OP[7:0]	0 - 255 MSB DRAM WCK2DQI Oscillator Count	1,2,3,4

NOTE :
 1) MR36 reports the MSB bits of the DRAM WCK2DQI Oscillator count. The DRAM WCK2DQI Oscillator count value is used to train WCK to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of WCK relative to DQ.
 2) Both MR35 and MR36 must be read (MRR) and combined to get the value of the WCK2DQI Oscillator count.
 3) A new MPC [Start WCK2DQI Oscillator] could be issued to reset the contents of MR35/MR36.
 4) WCKDQI & WCKDQO cannot be operated simultaneously.

MR37_Register Information (MA[7:0] = 25_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WCK2DQI interval timer run time setting							

Function	Register Type	Operand	Data	Notes
WCK2DQI interval timer run time	Write/Read	OP[7:0]	00000000_B : WCK2DQI interval timer stop via MPC Command (Default) 00000001_B : WCK2DQI timer stops automatically at 16th clocks after timer start 00000010_B : WCK2DQI timer stops automatically at 32nd clocks after timer start 00000011_B : WCK2DQI timer stops automatically at 48th clocks after timer start 00000100_B : WCK2DQI timer stops automatically at 64th clocks after timer start ----- Thru ----- 00111111_B : WCK2DQI timer stops automatically at (63X16)th clocks after timer start 01XXXXXX_B : WCK2DQI timer stops automatically at 2048th clocks after timer start 10XXXXXX_B : WCK2DQI timer stops automatically at 4096th clocks after timer start 11XXXXXX_B : WCK2DQI timer stops automatically at 8192nd clocks after timer start	1,2

NOTE :
 1) MPC command with OP[6:0]=10000010_B (Stop WCK2DQI Interval Oscillator) stops WCK2DQI interval timer in case of MR37 OP[7:0] = 00000000_B.
 2) MPC command with OP[6:0]=10000010_B (Stop WCK2DQI Interval Oscillator) is illegal with non-zero values in MR37 OP[7:0].

MR38_Register Information (MA[7:0] = 26_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WCK2DQO Oscillator Count - LSB							

Function	Register Type	Operand	Data	Notes
WCK2DQO Oscillator (DQ output Training WCK Oscillator)	Read-Only	OP[7:0]	0 - 255 LSB DRAM WCK2DQO Oscillator Count	1,2,3

NOTE :
 1) MR38 reports the LSB bits of the DRAM WCK2DQO Oscillator count. The DRAM WCK2DQO Oscillator count value is used to train WCK to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQ output relative to WCK.
 2) Both MR38 and MR39 must be read (MRR) and combined to get the value of the WCK2DQO Oscillator count.
 3) A new MPC [Start WCK2DQO Oscillator] can be issued at any time before sending MPC [Stop WCK2DQO Oscillator]. A new MPC [Start WCK2DQO Oscillator] resets the contents of MR38/MR39.

MR39_Register Information (MA[7:0] = 27_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WCK2DQO Oscillator Count - MSB							

Function	Register Type	Operand	Data	Notes
WCK2DQO Oscillator (DQ output Training WCK Oscillator)	Read-Only	OP[7:0]	0 - 255 MSB DRAM WCK2DQO Oscillator Count	1,2,3

NOTE :
 1) MR39 reports the MSB bits of the DRAM WCK2DQO Oscillator count. The DRAM WCK2DQO Oscillator count value is used to train WCK to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQ output relative to WCK.
 2) Both MR38 and MR39 must be read (MRR) and combined to get the value of the WCK2DQO Oscillator count.
 3) A new MPC [Start WCK2DQO Oscillator] can be issued at any time before sending MPC [Stop WCK2DQO Oscillator]. A new MPC [Start WCK2DQO Oscillator] resets the contents of MR38/MR39.

MR40_Register Information (MA[7:0] = 28_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
WCK2DQO interval timer run time setting							

Function	Register Type	Operand	Data	Notes
WCK2DQO interval timer run time	Write/Read	OP[7:0]	00000000_B : WCK2DQO interval timer stop via MPC Command (Default) 00000001_B : WCK2DQO timer stops automatically at 16th clocks after timer start 00000010_B : WCK2DQO timer stops automatically at 32nd clocks after timer start 00000011_B : WCK2DQO timer stops automatically at 48th clocks after timer start 00000100_B : WCK2DQO timer stops automatically at 64th clocks after timer start ----- Thru ----- 00111111_B : WCK2DQO timer stops automatically at (63X16)th clocks after timer start 01XXXXXX_B : WCK2DQO timer stops automatically at 2048th clocks after timer start 10XXXXXX_B : WCK2DQO timer stops automatically at 4096th clocks after timer start 11XXXXXX_B : WCK2DQO timer stops automatically at 8192nd clocks after timer start	1,2

NOTE :
 1) MPC command with OP[7:0]= 10000100_B (Stop WCK2DQO Interval Oscillator) stops WCK2DQO interval timer in case of MR40 OP[7:0] = 00000000_B.
 2) MPC command with OP[7:0]= 10000100_B (Stop WCK2DQO Interval Oscillator) is illegal with non-zero values in MR40 OP[7:0].

MR41_Register Information (MA[7:0] = 29_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
NT DQ ODT			PPRE	E-DVFS ODT Option Support	DVFS/E-DVFS Support		PDFEC

Function	Register Type	Operand	Data	Notes
Per-pin DFE Control (PDFEC)	Write-only	OP[0]	0 _B : Per-pin DFE disable (Default) 1 _B : Per-pin DFE enable	2,3,4
DVFS/E-DVFS Support (DVFS/Enhanced-DVFS Support)	Read-Only	OP[2:1]	00 _B : Only DVFS Mode supported 01 _B : Only Enhanced DVFS Mode supported 10 _B : Both DVFS and Enhanced DVFS Mode supported 11 _B : RFU	5,6
E-DVFS ODT Option Support (Enhanced DVFS ODT Option Support)	Read-Only	OP[3]	0 _B : Enhanced DVFS ODT option not supported 1 _B : Enhanced DVFS ODT option supported	7
PPRE	Write-Only	OP[4]	0 _B : PPR Disable (Default) 1 _B : PPR Enable	
NT DQ ODT (Non-Target DQ Bus Receiver On-Die-Termination)		OP[7:5]	000 _B : Disable 001 _B : RZQ/1 010 _B : RZQ/2 011 _B : RZQ/3 (Default) 100 _B : RZQ/4 101 _B : RZQ/5 110 _B : RZQ/6 111 _B : RFU	1,2,3

NOTE :

- 1) All values are "typical". The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re-calibration may be required as voltage and temperature vary.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address or read from with an MRR command to this address.
- 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1 and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device and may be changed without affecting device operation.
- 4) PDFEC bit is valid only when MR8 OP[1:0]=01_B (LPDDR5X SDRAM) and MR0 OP[7]=1_B (Per-pin DFE mode supported). Otherwise, MR41 OP[0] shall be set 0_B.
- 5) LPDDR5 SDRAM (MR8 OP[1:0]=00_B) supports DVFS mode as mandatory feature.
- 6) MR41 OP[2:1] is only valid for LPDDR5X SDRAM (MR8 OP[1:0]=01_B). LPDDR5 SDRAM (MR8 OP[1:0]=00_B) shall return undefined data when MR41 OP[2:1] is read.
- 7) LPDDR5X SDRAM (MR8 OP[1:0]=01_B) supports ODT for CA/CS/CK/WCK/DQ and NT-ODT under E-DVFS mode (MR19 OP[1:0]=10_B) only when MR41 OP[3]=1_B. If MR41 OP[3]=0_B, it is illegal to set 0_B for CK ODT(MR17 OP[3]) / CS ODT (MR17 OP[4]) / CA ODT(MR17 OP[5]), 1_B for NT ODT(MR11 OP[3]) and DQ ODT(MR11 OP[0:2]) / WCK ODT(MR18 OP[0:2]) should be set 000_B during an E-DVFS mode.

MR42_Register Information (MA[7:0] = 2A_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PPR Key Protection							

Function	Register Type	Operand	Data	Notes
PPR Key Protection	Write-only	OP[7:0]	PPR Key Protection code	1

NOTE :

- 1) PPR entry and exit sequence details are described in Post Package Repair section.

MR43_Register Information (MA[7:0] = 2B_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DBE_flag	SBEC_Rule	SBE_count					

Function	Register Type	Operand	Data	Notes
SBE_count	Read-only	OP[5:0]	The number of times a single-bit error or errors have occurred on the DRAM interface	1,2,3,4,5
SBEC_Rule		OP[6]	0_B : Simultaneous SBE on each DQ byte and DMI are counted as one error 1_B : Simultaneous SBE on each DQ byte and DMI are independently counted	3
DBE_flag		OP[7]	0_B : No double-bit errors have occurred on the DRAM interface 1_B : One or more double-bit errors have occurred on the DRAM interface	1,2,4,6

- NOTE :**
- These bits are cleared on power-up, DRAM reset, any Power-Down Exit, and on each read of this mode register. They are also held in the cleared state whenever MR22 OP[5:4]=00b. Because time is required for the DRAM to perform the clear on reads, the delay after reading this mode register should be tMRR + tMRW (rather than simply tMRR as required on most registers).
 - The DRAM should disable detection & recording of ECC errors during Write-FIFO commands, as these are used for training purposes and some errors are expected during training.
 - In a x16 DRAM device, up to 4 single-bit errors could potentially be detected in a single BL16 burst (byte 0 data SBE, byte 0 DMI SBE, byte 1 data SBE, and byte 1 DMI SBE). SBEC_Rule 0_B : Any combination of these 4 simultaneous errors is considered a single SBE occurrence and would only increment the SBE_count by 1. A BL32 burst can have up to 2 such SBE occurrences and could increment the SBE_count by up to 2. In a x8 DRAM device, any simultaneous combination of data SBE and DMI SBE is considered a single SBE occurrence, and would only increment the SBE_count by 1 (up to 2 in a BL32 burst). SBEC_Rule 1_B : Each of the maximum 4 simultaneous errors is considered as an independent error and would increment the SBE count up to 4. A BL32 burst can have up to 8 such SBE occurrences and could increment the SBE_count by up to 8. In a x8 DRAM device, any simultaneous combination of data SBE and DMI SBE is considered an independent SBE occurrence and would only increment the SBE_count by up to 2 (up to 4 in a BL32 burst).
 - In x16 mode, errors from either interface byte are stored in a single register per DRAM (SBE_count would be the number of SBE occurrences on BOTH bytes & DBE_flag would indicate a DBE on EITHER byte). In x8 mode, errors are stored for a single interface byte and MRR returns both copies of this register, each on its corresponding byte lane.
 - SBE_count should be a saturating counter.
 - Once set, the DBE_flag bit remains set until explicitly cleared by one of the conditions described in note 1 above.

MR44_Register Information (MA[7:0] = 2C_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Data ECC Syndrome[7:0]							
S[7]	S[6]	S[5]	S[4]	S[3]	S[2]	S[1]	S[0]

Function	Register Type	Operand	Data	Notes
Data ECC Syndrome[7:0]	Read-only	OP[7:0]	Bits 7:0 of the data ECC syndrome from the most recent single-bit error	1,2,3,4,5,6

- NOTE :**
- These bits are cleared on power-up, DRAM reset, any Power-Down Exit, and on each read of this mode register. They are also held in the cleared state whenever MR22 OP[5:4]=00b. Because time is required for the DRAM to perform the clear on reads, the delay after reading this mode register should be tMRR + tMRW (rather than simply tMRR as required on most registers).
 - The DRAM should disable detection & recording of ECC errors during Write-FIFO commands, as these are used for training purposes and some errors are expected during training.
 - In x16 mode, error syndromes from either interface byte are stored in a single register per DRAM. In x8 mode, error syndromes are stored for a single interface byte and MRR returns both copies of this register, each on its corresponding byte lane.
 - In x16 mode, it is possible for errors to occur on both bytes simultaneously. In this rare case, only the syndrome from DQ[7:0] should be stored.
 - Regardless of detecting a data ECC error or a DMI ECC error, both syndromes should be stored at either error occurrence.
 - The error syndrome is not affected by a DBE and retains its previous value.

MR45_Register Information (MA[7:0] = 2D_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Data ECC Syndrome [8]	Error Byte Lane	DMI ECC Syndrome					
S[8]	Error Byte Lane	DS[5]	DS[4]	DS[3]	DS[2]	DS[1]	DS[0]

Function	Register Type	Operand	Data	Notes
DMI ECC Syndrome (DS[5:0])	Read-only	OP[5:0]	The DMI ECC syndrome from the most recent single-bit error	1,2,3,4,5,7
Error Byte Lane		OP[6]	0_B : The most recent single bit error occurred on DQ[7:0] or DMI0 1_B : The most recent single bit error occurred on DQ[15:8] or DMI1	1,2,3,4,5,6,8
Data ECC Syndrome[8] (S[8])		OP[7]	Bit 8 of the data ECC syndrome from the most recent single-bit error	1,2,3,4,5,7

NOTE :

- These bits are cleared on power-up, DRAM reset, any Power-Down Exit, and on each read of this mode register. They are also held in the cleared state whenever MR22 OP[5:4]=00_B. Because time is required for the DRAM to perform the clear on reads, the delay after reading this mode register should be tMRR + tMRW (rather than simply tMRR as required on most registers).
- The DRAM should disable detection & recording of ECC errors during Write-FIFO commands, as these are used for training purposes and some errors are expected during training.
- In x16 mode, error syndromes from either interface byte are stored in a single register per DRAM. In x8 mode, error syndromes are stored for a single interface byte and MRR returns both copies of this register, each on its corresponding byte lane.
- In x16 mode, it is possible for single-bit errors to occur on both bytes simultaneously. In this rare case, only the error syndromes and error byte lane from byte 0 (DQ[7:0] & DMI0) should be stored.
- Regardless of detecting a data ECC error or a DMI ECC error, both syndromes should be stored at either error occurrence.
- In x8 mode, this bit is unused and shall always read back as 0.
- These syndrome fields are not affected by a DBE and retains their previous values.
- The Error Byte Lane is only updated by a SBE.

Table 23 is provided to help clarify when and how each field is updated.

[Table 23] Updating ECC Syndromes and Error Byte Lane

Errors at Byte 0	Error at Byte 1	Syndromes (MR44 and MR45 OP[7] & OP[5:0])	Error Byte Lane (MR45 OP[6])
None	None	No Change	No Change
SBE	None	Byte 0	Byte 0
DBE	None	No Change	No Change
None	SBE	Byte 1	Byte 1
SBE	SBE	Byte 0	Byte 0
DBE	SBE	Byte 1	Byte 1
None	DBE	No Change	No Change
SBE	DBE	Byte 0	Byte 0
DBE	DBE	No Change	No Change

MR46_Register Information (MA[7:0] = 2E_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
(RFU)					FIFO RDQS Training	RDQS Toggle	Enhanced RDQS

Function	Register Type	Operand	Data	Notes
Enhanced RDQS (Enhanced RDQS training mode)	Write-only	OP[0]	0 _B : Disabled (Default) 1 _B : Enhanced RDQS training mode Enabled	
RDQS Toggle (RDQS toggle mode)		OP[1]	0 _B : Disabled (Default) 1 _B : RDQS toggle mode Enabled	
FIFO RDQS Training (WCK-RDQS_t/Parity Training)		OP[2]	0 _B : Disabled (Default) 1 _B : WCK-RDQS_t/Parity Training Enabled	1

NOTE :
1) When MR46 OP[2] = 1_B, Write FIFO command will allow data to be written through the RDQS_t pin. The data written via the RDQS_t pin can then be read back via the DMI by a Read FIFO command.

MR47_Register Information (MA[7:0] = 2F_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-1							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-1	Read-only	OP[7:0]	Serial ID-1	1

NOTE :
1) MR47 is vendor specific.

MR48_Register Information (MA[7:0] = 30_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-2							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-1	Read-only	OP[7:0]	Serial ID-2	1

NOTE :
1) MR48 is vendor specific.

MR49_Register Information (MA[7:0] = 31_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-3							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-3	Read-only	OP[7:0]	Serial ID-3	1

NOTE :
1) MR49 is vendor specific.

MR50_Register Information (MA[7:0] = 32_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-4							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-4	Read-only	OP[7:0]	Serial ID-4	1

NOTE :
1) MR50 is vendor specific.

MR51_Register Information (MA[7:0] = 33_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-5							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-5	Read-only	OP[7:0]	Serial ID-5	1

NOTE :

1) MR51 is vendor specific.

MR52_Register Information (MA[7:0] = 34_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-6							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-6	Read-only	OP[7:0]	Serial ID-6	1

NOTE :

1) MR52 is vendor specific.

MR53_Register Information (MA[7:0] = 35_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-7							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-7	Read-only	OP[7:0]	Serial ID-7	1

NOTE :

1) MR53 is vendor specific.

MR54_Register Information (MA[7:0] = 36_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial ID-8							

Function	Register Type	Operand	Data	Notes
LPDDR5 Serial ID-8	Read-only	OP[7:0]	Serial ID-8	1

NOTE :

1) MR54 is vendor specific.

MR55_DNU (Do Not Use) (MA[7:0] = 37_H) :

MR56_Register Information (MA[7:0] = 38_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Valid 0 or 1							

Function	Register Type	Operand	Data	Notes
SDRAM will ignore	Write-only	OP[7:0]	Don't Care	1

NOTE :

1) This register is reserved for testing purpose. The logical data values written to OP[7:0] shall have no effect on SDRAM operation, however timings need to be observed as for any other MR access command.

MR57_Register Information (MA[7:0] = 39_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
ARFM		RFMSBC		RFMSB		RAADEC	

Function	Register Type	Operand	Data	Notes
RAADEC (RAA Count Decrement per RFM Command)	Read-only	OP[1:0]	00 _B : RAAIMT 01 _B : RAAIMT*1.5 10 _B : RAAIMT*2 11 _B : RAAIMT*4	
RFMSB (RFM Allowable Single- Banks)		OP[3:2]	00 _B : Not support Single-bank mode 01 _B : Support Single-bank mode 10 _B : RFU 11 _B : RFU	
RFMSBC (RFM Single-Bank Counters Implemented)	Write-only	OP[5:4]	00 _B : One RAA counter per two banks (1 of 8) (Default) 01 _B : One RAA counter per one bank (1 of 16) 10 _B : RFU 11 _B : RFU	
ARFM (Adaptive RFM)		OP[7:6]	00 _B : Default 01 _B : Level A 10 _B : Level B 11 _B : Level C	

NOTE :

- 1) If LPDDR5 SDRAM has MR57 OP[3:2]=00_B, it is not allowed to set MR57 OP[5:4]=01_B.
- 2) MR57 OP[5:4]=01_B is not supported for 8Bank mode.

MR58_Register Information (MA[7:0] = 3A_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQ Dn Emphasis UB		DQ Up Emphasis UB		DQ Dn Emphasis LB		DQ Up Emphasis LB	

Function	Register Type	Operand	Data	Notes
DQ Up Emphasis LB (DQ pull-up pre-emphasis lower Byte)	Write-only	OP[1:0]	00 _B : Disabled (Default) 01 _B : Weak strength 10 _B : Middle strength 11 _B : Strong strength	1,2,3,4,5,6
DQ Dn Emphasis LB (DQ pull-down pre-emphasis lower Byte)		OP[3:2]	00 _B : Disabled (Default) 01 _B : Weak strength 10 _B : Middle strength 11 _B : Strong strength	1,2,3,4,5,6
DQ Up Emphasis UB (DQ pull-up pre-emphasis upper Byte)		OP[5:4]	00 _B : Disabled (Default) 01 _B : Weak strength 10 _B : Middle strength 11 _B : Strong strength	1,2,3,4,5,6
DQ Dn Emphasis UB (DQ pull-down pre-emphasis upper Byte)		OP[7:6]	00 _B : Disabled (Default) 01 _B : Weak strength 10 _B : Middle strength 11 _B : Strong strength	1,2,3,4,5,6

NOTE :

- 1) The Pre-Emphasis function apply to RDQS_t/c and DMI too, if RDQS_t/c and/or DMI output is enabled.
- 2) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1, and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
- 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1, and set point 2. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 4) Active MR bit which is selected by FSP-OP: MR16 OP[3:2] cannot be changed directly by MRW command except Reset and Initialization procedure.
- 5) Each MR bit can be enabled only by FSP procedure.
- 6) The pre-emphasis applies more than 6400Mbps.

MR59_DNU (Do Not Use) (MA[7:0] = 3B_H) :**MR60_Register Information (MA[7:0] = 3C_H) :**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Valid 0 or 1							

Function	Register Type	Operand	Data	Notes
SDRAM will ignore	Write-only	OP[7:0]	Don't Care	1

NOTE :

- 1) This register is reserved for testing purpose. The logical data values written to OP[7:0] shall have no effect on SDRAM operation, however timings need to be observed as for any other MR access command.

MR69_Register Information (MA[7:0] = 45_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RDCAU				RDCAL			

Function	Register Type	Operand	Data	Notes
Read DCA for Lower Byte (RDCAL)	Write-only	OP[3:0]	0000_B : 0 Steps (default no adjustment) 0001_B : -1 Steps 0010_B : -2 Steps 0011_B : -3 Steps 0100_B : -4 Steps 0101_B : -5 Steps 0110_B : -6 Steps 0111_B : -7 Step 1000_B : RFU 1001_B : +1 Step 1010_B : +2 Steps 1011_B : +3 Steps 1100_B : +4 Steps 1101_B : +5 Steps 1110_B : +6 Steps 1111_B : +7 Steps	1,2,3,4,5
Read DCA for Upper Byte (RDCAU)		OP[7:4]	0000_B : 0 Steps (default no adjustment) 0001_B : -1 Steps 0010_B : -2 Steps 0011_B : -3 Steps 0100_B : -4 Steps 0101_B : -5 Steps 0110_B : -6 Steps 0111_B : -7 Step 1000_B : RFU 1001_B : +1 Step 1010_B : +2 Steps 1011_B : +3 Steps 1100_B : +4 Steps 1101_B : +5 Steps 1110_B : +6 Steps 1111_B : +7 Steps	1,2,3,4,5

- NOTE :**
- 1) 0001_B to 0111_B of bit sets will decrease the internal WCK duty cycle.
 - 2) 1001_B to 1111_B of bit sets will increase the internal WCK duty cycle.
 - 3) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1, and set point 2. Only the registers for the set point determined by the state of the FSP-WR bit (MR16 OP[1:0]) will be written to with an MRW command to this MR address.
 - 4) There are three physical registers assigned to each bit of this MR parameter, designated set point 0, set point 1, and set point 2. The SDRAM will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR16 OP[3:2]). The values in the registers for the inactive set point will be ignored by the SDRAM and may be changed without affecting SDRAM operation.
 - 5) With byte mode device, OP[3:0] applies lower byte device and OP[7:4] applies upper byte device. Unused operands (OP[7:4] for lower byte device and OP[3:0] for upper byte device) should be set 0_B or 1_B by which are ignored SDRAM though.

MR70_Register Information (MA[7:0] = 46_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Offset DFE Quantity for DQ3 (DFEDQ3)		Offset DFE Quantity for DQ2 (DFEDQ2)		Offset DFE Quantity for DQ1 (DFEDQ1)		Offset DFE Quantity for DQ0 (DFEDQ0)	

Function	Register Type	Operand	Data	Notes
Offset DFE Quantity for DQ0 (DFEDQ0)	Write-only	OP[1:0]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ1 (DFEDQ1)		OP[3:2]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ2 (DFEDQ2)		OP[5:4]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ3 (DFEDQ3)		OP[7:6]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2

NOTE :

1) DFE quantity is vender specific. Please refer to each vender specification.

2) Offset values are valid only when MR0 OP[7]=1_B(Per-Pin DFE mode supported) and MR41 OP[0]=1_B (Perpin DFE enable)

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MR71_Register Information (MA[7:0] = 47_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Offset DFE Quantity for DQ7 (DFEDQ7)		Offset DFE Quantity for DQ6 (DFEDQ6)		Offset DFE Quantity for DQ5 (DFEDQ5)		Offset DFE Quantity for DQ4 (DFEDQ4)	

Function	Register Type	Operand	Data	Notes
Offset DFE Quantity for DQ4 (DFEDQ4)	Write-only	OP[1:0]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ5 (DFEDQ5)		OP[3:2]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ6 (DFEDQ6)		OP[5:4]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ7 (DFEDQ7)		OP[7:6]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2

NOTE :

1) DFE quantity is vender specific. Please refer to each vender specification.

2) Offset values are valid only when MR0 OP[7]=1_B(Per-Pin DFE mode supported) and MR41 OP[0]=1_B (Perpin DFE enable)

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MR72_Register Information (MA[7:0] = 48_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Offset DFE Quantity for DQ11(DFEDQ11)		Offset DFE Quantity for DQ10 (DFEDQ10)		Offset DFE Quantity for DQ9 (DFEDQ9)		Offset DFE Quantity for DQ8 (DFEDQ8)	

Function	Register Type	Operand	Data	Notes
Offset DFE Quantity for DQ8 (DFEDQ8)	Write-only	OP[1:0]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ9 (DFEDQ9)		OP[3:2]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ10 (DFEDQ10)		OP[5:4]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ11(DFEDQ11)		OP[7:6]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2

NOTE :

1) DFE quantity is vender specific. Please refer to each vender specification.

2) Offset values are valid only when MR0 OP[7]=1_B(Per-Pin DFE mode supported) and MR41 OP[0]=1_B (Perpin DFE enable)

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MR73_Register Information (MA[7:0] = 49_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Offset DFE Quantity for DQ15(DFEDQ15)		Offset DFE Quantity for DQ14 (DFEDQ14)		Offset DFE Quantity for DQ13 (DFEDQ13)		Offset DFE Quantity for DQ12 (DFEDQ12)	

Function	Register Type	Operand	Data	Notes
Offset DFE Quantity for DQ12 (DFEDQ12)	Write-only	OP[1:0]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ13 (DFEDQ13)		OP[3:2]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ14 (DFEDQ14)		OP[5:4]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DQ15(DFEDQ15)		OP[7:6]	00 _B : No offset negative feedback quantity (Default) 01 _B : Add 1-step offset negative feedback quantity 10 _B : Add 2-step offset negative feedback quantity 11 _B : Add 3-step offset negative feedback quantity	1,2

NOTE :

1) DFE quantity is vender specific. Please refer to each vender specification.

2) Offset values are valid only when MR0 OP[7]=1_B(Per-Pin DFE mode supported) and MR41 OP[0]=1_B (Per-pin DFE enable)

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MR74_Register Information (MA[7:0] = 4A_H) :

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Offset DFE Quantity for RDQS1 (DFERDQS1)		Offset DFE Quantity for RDQS0 (DFERDQS0)		Offset DFE Quantity for DMI1 (DFEDMI1)		Offset DFE Quantity for DMI0 (DFEDMI0)	

Function	Register Type	Operand	Data	Notes
Offset DFE Quantity for DMI0 (DFEDMI0)	Write-only	OP[1:0]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for DMI1 (DFEDMI1)		OP[3:2]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for RDQS0 (DFERDQS0)		OP[5:4]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2
Offset DFE Quantity for RDQS1 (DFERDQS1)		OP[7:6]	00_B : No offset negative feedback quantity (Default) 01_B : Add 1-step offset negative feedback quantity 10_B : Add 2-step offset negative feedback quantity 11_B : Add 3-step offset negative feedback quantity	1,2

NOTE :

1) DFE quantity is vender specific. Please refer to each vender specification.

2) Offset values are valid only when MR0 OP[7]=1_B(Per-Pin DFE mode supported) and MR41 OP[0]=1_B (Per pin DFE enable)

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9.0 ABSOLUTE MAXIMUM DC RATINGS

Stresses greater than those listed may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

[Table 24] Absolute Maximum DC Ratings

Parameter	Symbol	Min	Max	Unit	Note
VDD1 Supply Voltage relative to VSS	VDD1	-0.4	2.1	V	1
VDD2H Supply Voltage relative to VSS	VDD2H	-0.4	1.4	V	1
VDD2L Supply Voltage relative to VSS	VDD2L	-0.4	1.4	V	1
VDDQ Supply Voltage relative to VSS	VDDQ	-0.4	1.4	V	1
Voltage on Any Ball Except VDD1 relative to VSS	VIN, VOUT	-0.4	1.4	V	
Storage Temperature	TSTG	-55	125	°C	2

NOTE :

1) See "Power-Up, Initialization" in Operation and Timing datasheet for relationships between power supplies.

2) Storage Temperature is the case surface temperature on the center/top side of the LPDDR5 device. For the measurement conditions, please refer to JESD51-2A

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10.0 AC AND DC OPERATING CONDITIONS

10.1 Recommended DC Operating Conditions

[Table 25] Recommended Voltage Operating Conditions

DRAM	Symbol		Low Freq Voltage Spec Freq: DC to 2 MHz				Z(f) Spec Freq: 2 MHz to 10 MHz		Z(f) Spec Freq: 20 MHz		Notes
			Min	Typ	Max	Unit	Zmax	Unit	Zmax	Unit	
Core 1 Power	VDD1		1.70	1.80	1.95	V	100	mOhm	170	mOhm	1,2,9
Core 2 Power/Input Buffer Power	VDD2H		1.01	1.05	1.12	V	40	mOhm	80	mOhm	1,2,9
	VDD2L	Single Core Power Rail (MR13 OP[7]=1 _B)	1.01	1.05	1.12	V	120	mOhm	190	mOhm	1,2,9
		Dual Core Power Rail (MR13 OP[7]=0 _B)	0.87	0.9	0.97		120		190		
I/O Buffer Power	VDDQ	SPEC Range 1	0.47	0.5	0.57	V	40	mOhm	80	mOhm	2,3,6,9
		SPEC Range 2	0.27	0.3	0.37	V					2,4,9
		Allowable Range	0.27	N/A	0.57	V	N/A		N/A		5,6,7,8

NOTE :

- VDD1 uses significantly less current than VDD2H and VDD2L.
- DC to 2MHz voltage range includes all noise at DRAM ball, both DC and AC ripple fluctuations. This noise is included in the aperture mask defined by VdIVW. Refer to Figure 4.
- SPEC Range 1 is intended for IO operation with both ODT enabled and disabled.
- SPEC Range 2 is intended for IO operation with ODT disabled.
- IO operation at VDDQ levels between outside SPEC Range 1 or SPEC Range 2 is allowed with ODT disabled.
- Allowable range is valid only when DVFSQ enabled.
- 100mV tolerance(-30mV/+70mV) is applied to VDDQ allowable ranges.
- Vendors may support LPDDR5 (MR8 OP [1:0] = 00B) 0.6V and LPDDR5X (MR8 OP [1:0] = 01B) 0.45V VDDQ(typ) as an option. Because ZQ calibration is optimized at VDDQ=0.5V, the output drive strength may not be guaranteed at LPDDR5 0.6V and LPDDR5X 0.45V. Refer to a vendor's data sheet.
- Z(f) is defined for all pins per voltage domain per channel. Z(f) does not include the DRAM package and silicon die.

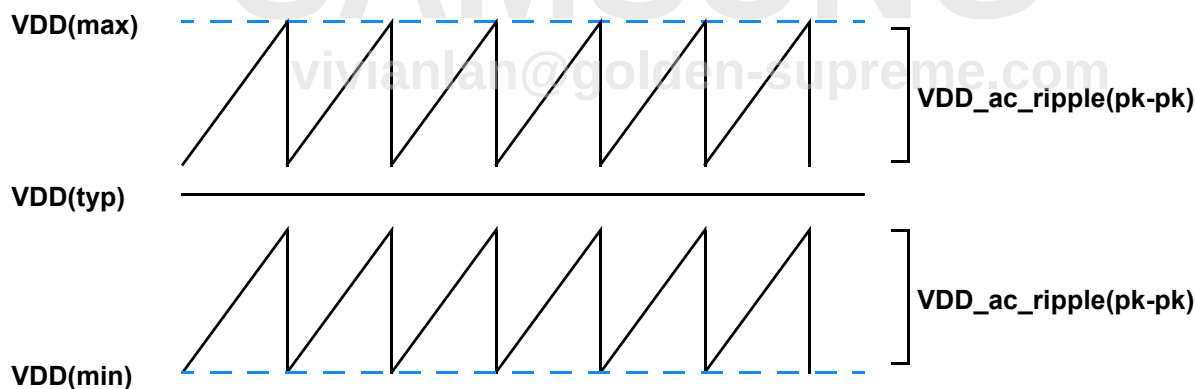


Figure 4. DC Voltage Range

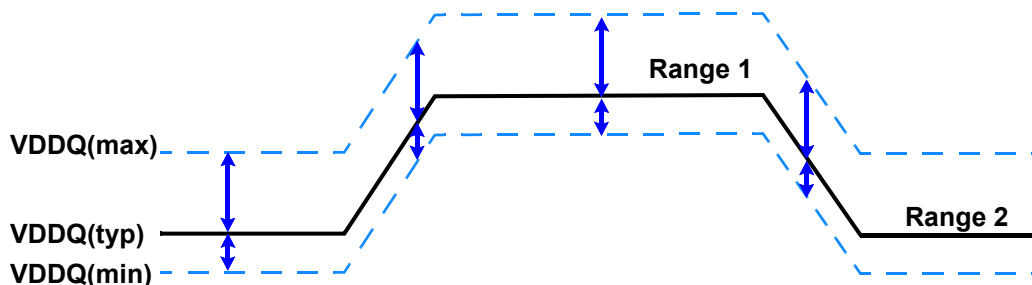


Figure 5. VDDQ Tolerance Definition in Allowable Range

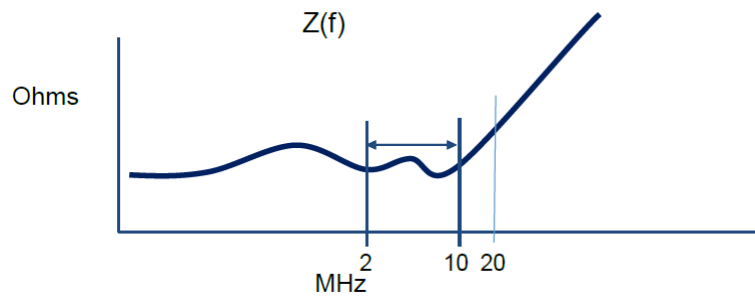


Figure 6. Zprofile $Z(f)$ of the system at the DRAM package solder ball (without the DRAM component)

A simplified electrical system load model for $Z(f)$ with the general frequency response is shown in Figure 7. The system resistance and inductance can be scaled to generalize the spec response to the DRAM pins.

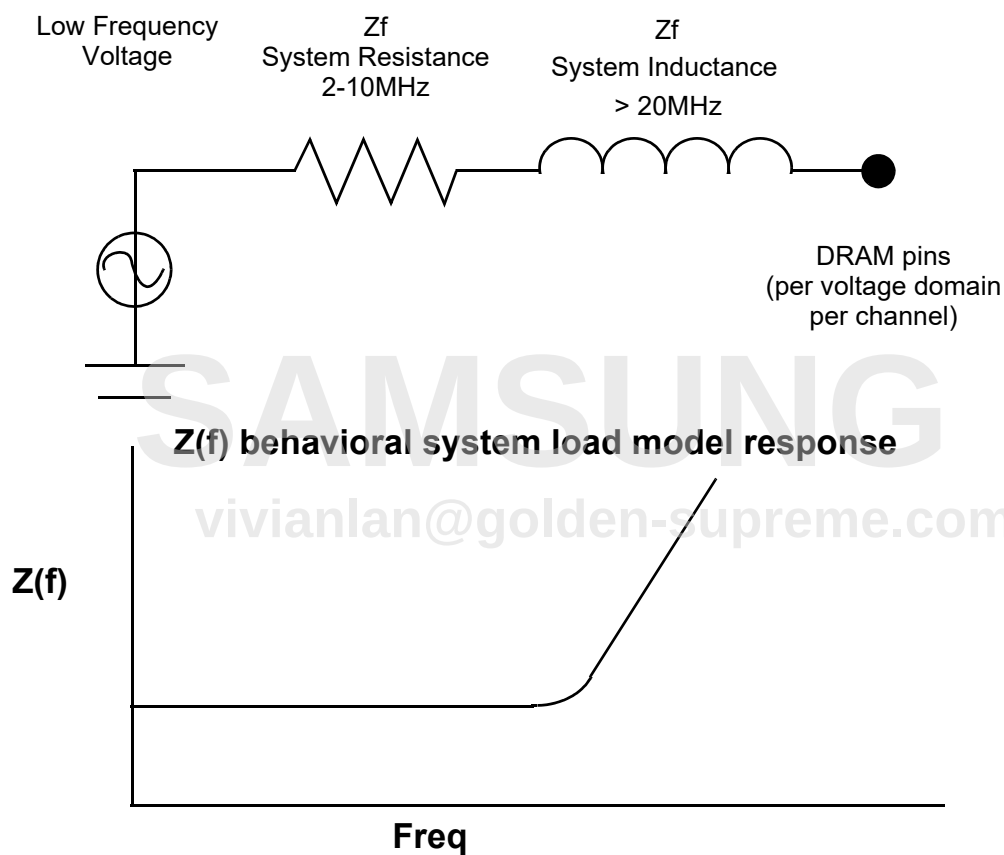


Figure 7. A simplified $Z(f)$ system electrical model and frequency response of the behavioral PDN electrical load model without the DRAM component per voltage domain per channel.

10.2 Input Leakage Current

[Table 26] Input Leakage Current

Parameter/Condition	Symbol	Min	Max	Unit	Notes
Input Leakage Current	I_L	-8	8	μA	1,2,3,6
CS Leakage Current	I_{LCS}	-16	16	μA	1,2,4,5,6

NOTE :

- 1) For CK_t, CK_c, WCK_t, WCK_c, and CA. Any input $0V \leq V_{IN} \leq V_{DDQ}$ (All other pins not under test = 0V).
- 2) For CS and RESET_n. Any input $0V \leq V_{IN} \leq V_{DD2H}$ (All other pins not under test = 0V).
- 3) CA ODT is disabled for CA, CK ODT is disabled for CK_t and CK_c, and WCK ODT is disabled for WCK_t and WCK_c.
- 4) CS ODT is disabled for CS if optional CS ODT is supported.
- 5) ILCS is applied when CS ODT is supported. If CS ODT is not supported, the input leakage current for CS shall meet IL.
- 6) The leakage current per a mono die.

10.3 Input/Output Leakage Current

[Table 27] Input/Output Leakage Current

Parameter/Condition	Symbol	Min	Max	Unit	Notes
Input/Output Leakage Current	I_{OZ}	-10	10	μA	1,2,3

NOTE :

- 1) For DQ, RDQS_t, RDQS_c and DMI. Any I/O $0V \leq V_{OUT} \leq V_{DDQ}$.
- 2) I/Os status are disabled: High Impedance and ODT Off.
- 3) The leakage current per a mono die.

10.4 Operating Temperature Range

[Table 28] Operating Temperature Range

Parameter / Condition	Symbol	Min	Max	Unit	Notes
Standard	$T_{oper_standard}$	-25	85	$^{\circ}C$	1,2

NOTE :

- 1) Operating Temperature is the case surface temperature on the center-top side of the LPDDR5 device. For the measurement conditions, please refer to JESD51-2A.
- 2) Either the device case temperature rating or the temperature sensor may be used to set an appropriate refresh rate, determine the need for AC timing de-rating and/or monitor the operating temperature. When using the temperature sensor, the actual device case temperature may be higher than the TOPER rating that applies for the Standard or Elevated Temperature Ranges. For example, TCASE may be above 85 $^{\circ}C$ when the temperature sensor indicates a temperature of less than 85 $^{\circ}C$.

10.5 Electrostatic Discharge Sensitivity Characteristics

[Table 29] Electrostatic Discharge Sensitivity Characteristics

Parameter	Symbol	Min	Max	Unit	Notes
Human body model (HBM)	ESD_{HBM}	1000		V	1
Charged-device model (CDM)	ESD_{CDM}	250		V	2

NOTE :

- 1) Refer to ESDA/ JEDEC Joint Standard JS-001 for measurement procedures.
- 2) Refer JS-002 for measurement procedures.

11.0 AC AND DC INPUT/OUTPUT MEASUREMENT LEVELS

11.1 1.05V High speed LVCMOS

11.1.1 Standard specifications

All voltages are referenced to ground except where noted.

11.1.2 Input Level for Reset_n

LPDDR5 uses CMOS with VDD2H Reset_n signaling to ensure stable LPDDR5 reset operation.

[Table 30] Reset input level specification

Parameter	Symbol	Min/Max	Value	Unit	Note
Reset_n ViH	V _{IH_RS}	Min	$0.8 \times V_{DD2H}$	V	
		Max	$V_{DD2H} + 0.2$	V	1
Reset_n ViL	V _{IL_RS}	Min	-0.2	V	1
		Max	$0.2 \times V_{DD2H}$	V	

NOTE :

1) Refer to 11.1.3 AC Overshoot / Undershoot.

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11.1.3 AC Overshoot / Undershoot

[Table 31] AC Overshoot / Undershoot

Parameter	Specification
Maximum Peak Amplitude allowed for overshoot area	0.35V
Maximum Peak Amplitude allowed for undershoot area	0.35V
Maximum overshoot area above V_{DD2H}/V_{DDQ}	0.8V-ns
Maximum undershoot area above V_{SS}	0.8V-ns

11.1.4 AC Overshoot / Undershoot for LVSTL

[Table 32] LPDDR5 AC Overshoot/Undershoot for LVSTL

Parameter	Min/Max	Data Rate				Units
		1600	3200	5500	6400	
Maximum peak amplitude allowed for overshoot area	Max	0.3	0.3	0.3	0.3	V
Maximum peak amplitude allowed for undershoot area	Max	0.3	0.3	0.3	0.3	V
Maximum overshoot area above V_{DD2H}/V_{DDQ}	Max	0.1	0.1	0.1	0.1	V-ns
Maximum undershoot area above V_{SS}	Max	0.1	0.1	0.1	0.1	V-ns

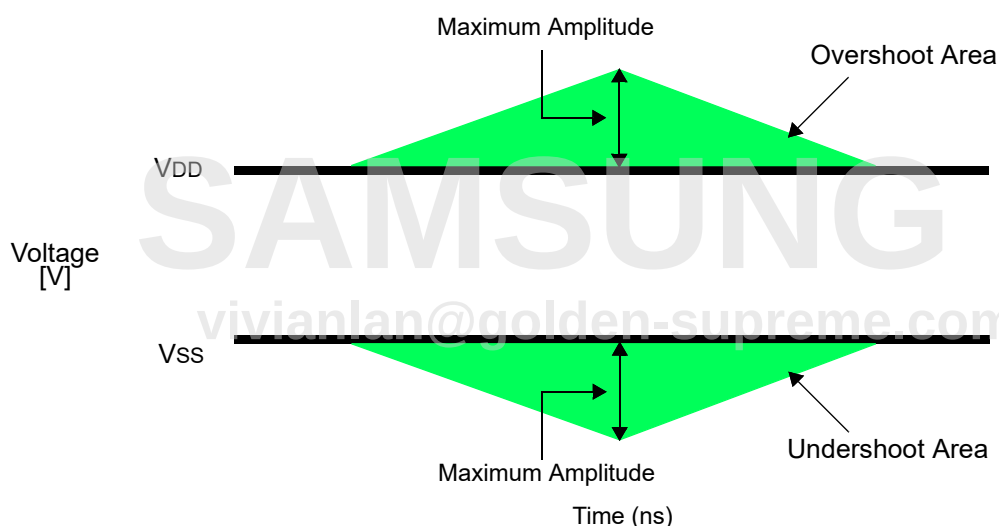


Figure 8. Overshoot and Undershoot Definition

NOTE :

- 1) V_{DD} is V_{DD2H} for CS and RESET_n. V_{DD} is V_{DDQ} for CA[6:0], CK_t/c, DQ, DMI, RDQS_t, WCK_t/c.
- 2) Maximum peak amplitude values are referenced from actual V_{DD} and V_{SS} values.
- 3) Maximum area values are referenced from maximum operating V_{DD} and V_{SS} values.

11.2 Differential Input Voltage

11.2.1 Differential Input Voltage for CK

The minimum input voltage need to satisfy both V_{indiff_CK} and $V_{indiff_CK}/2$ specification at input receiver and their measurement period is $1t_{CK}$. V_{indiff_CK} is the peak to peak voltage centered on 0 volts differential and $V_{indiff_CK}/2$ is max and min peak voltage from 0V.

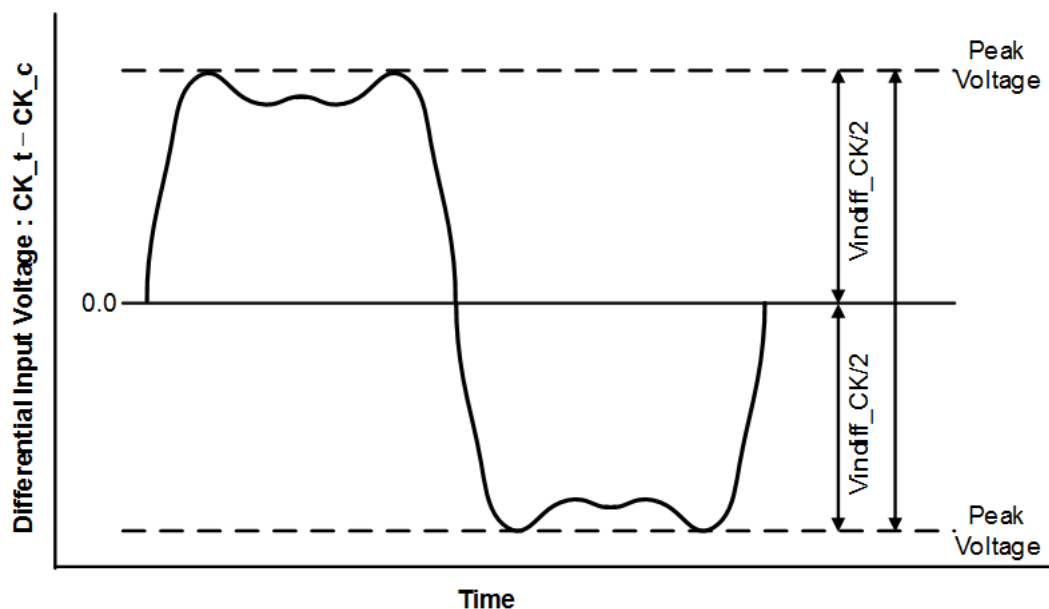


Figure 9. CK Differential Input Voltage

[Table 33] CK Differential Input Voltage

Parameter	Symbol	Clock Rate								Unit	Note
		266MHz		533MHz		800MHz		937Mhz			
		Min	Max	Min	Max	Min	Max	Min	Max		
CK Differential Input Voltage	Vindiff_CK	350	-	350	-	350	-	350	-	mV	1,2

NOTE :

- 1) Refer to the latency table to match the clock rate to data rate.
- 2) The peak voltage of Differential CK signals is calculated in a following equation.
 $V_{indiff_CK} = (\text{Max Peak Voltage}) - (\text{Min Peak Voltage})$
 $\text{Max Peak Voltage} = \text{Max}(f(t))$
 $\text{Min Peak Voltage} = \text{Min}(f(t))$
 $f(t) = V_{CK_t} - V_{CK_c}$

11.2.1.1 Peak Voltage Calculation Method

The peak voltage of Differential Clock signals are calculated in a following equation.

$$V_{IH.DIFF.Peak Voltage} = \text{Max}(f(t))$$

$$V_{IL.DIFF.Peak Voltage} = \text{Min}(f(t))$$

$$f(t) = V_{CK_t} - V_{CK_c}$$

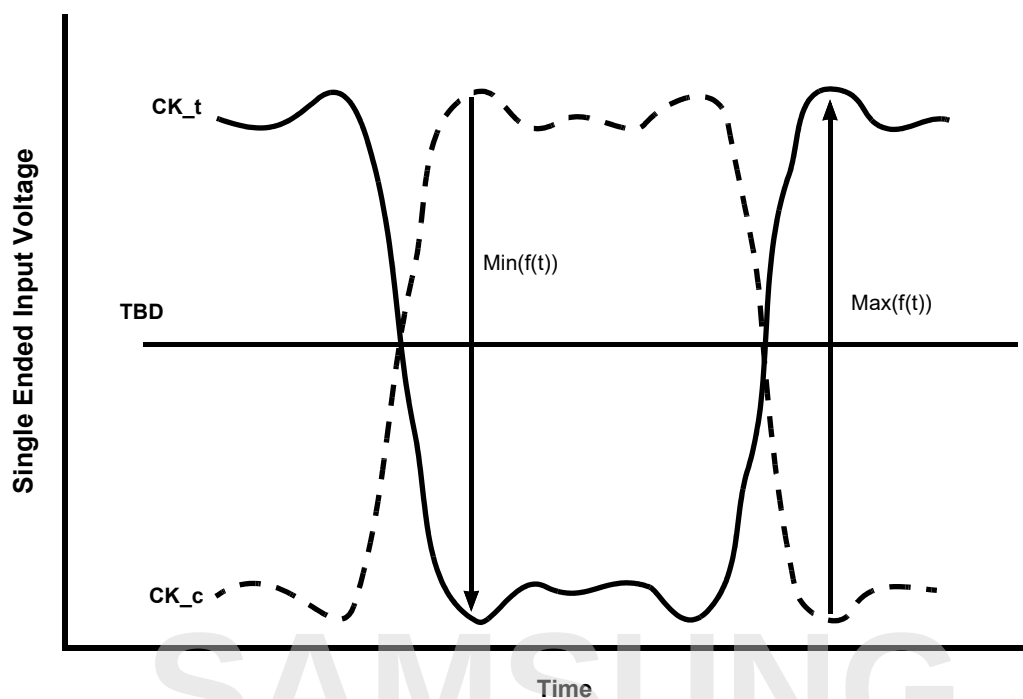


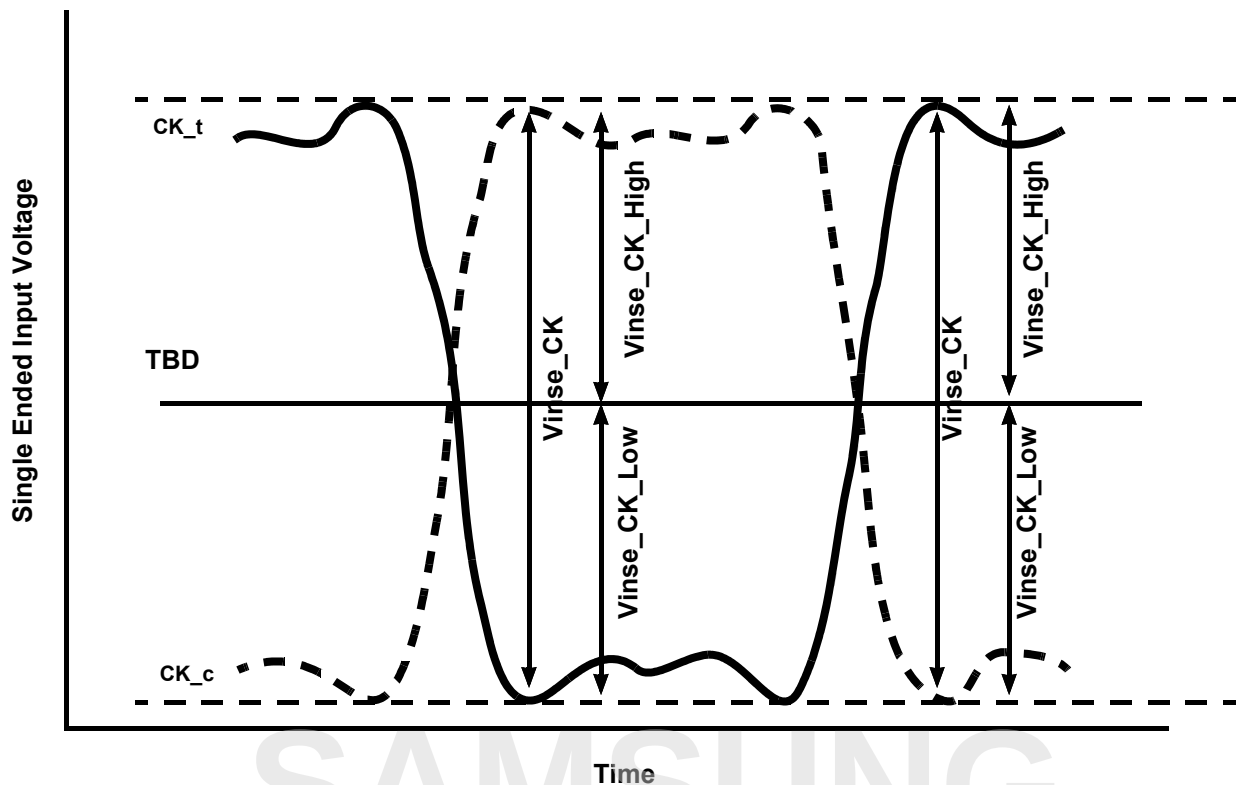
Figure 10. Definition of differential Clock Peak Voltage

NOTE :

1) TBD is LPDDR5 SDRAM internal setting value by Vref training.

11.2.1.2 Single ended Input Voltage for CK

The minimum input voltage need to satisfy both Vinse_CK, Vinse_CK_High/Low specification at input receiver.



NOTE :
1) TBD is LPDDR5 SDRAM internal setting value by Vref training.

[Table 34] Clock Single-Ended Input Voltage

Parameter	Symbol	Clock Rate								Unit
		266MHz		533MHz		800MHz		937Mhz		
		Min	Max	Min	Max	Min	Max	Min	Max	
Clock Single-ended input Voltage	Vinse_CK	175	-	175	-	175	-	175	-	mV
Clock Single-ended input Voltage High from TBD	Vinse_CK_High	87.5	-	87.5	-	87.5	-	87.5	-	mV
Clock Single-ended input Voltage Low from TBD	Vinse_CK_Low	87.5	-	87.5	-	87.5	-	87.5	-	mV

11.2.1.3 Differential Input Slew Rate Definition for CK

Input slew rate for differential signals (CK_t, CK_c) are defined and measured as shown in Figure 12. and the following Tables.

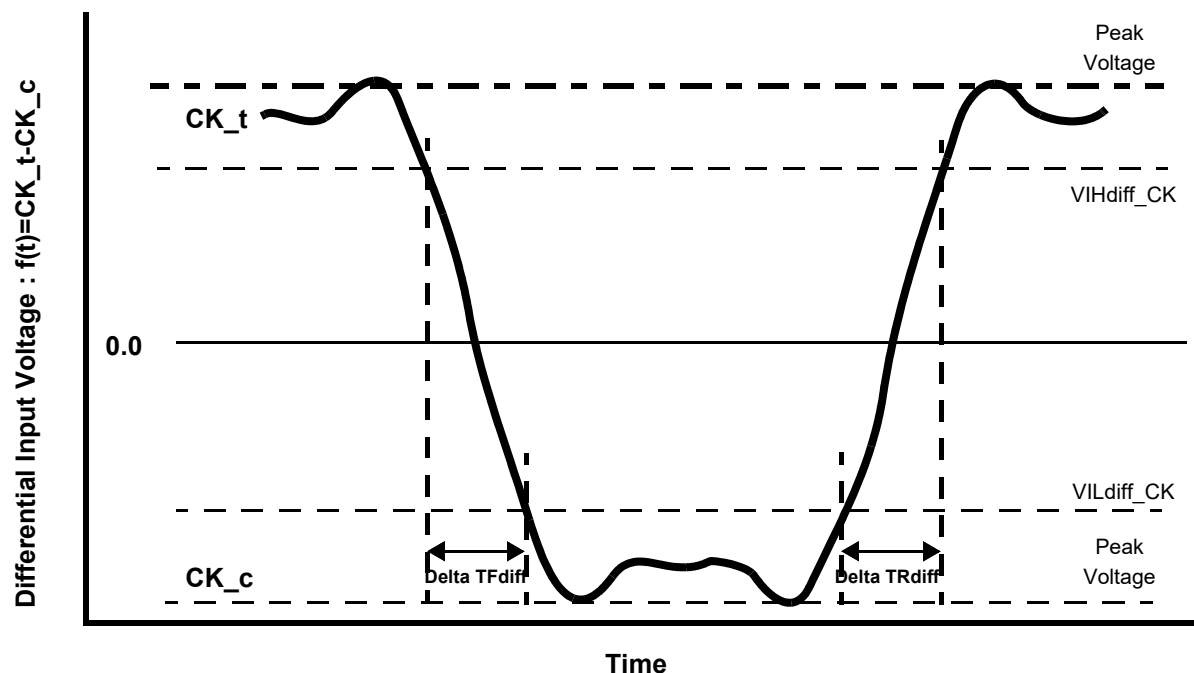


Figure 12. Differential Input Slew Rate Definition for CK_t, CK_c

NOTE :

- 1) Differential signal rising edge from VILdiff_CK to VIHdiff_CK must be monotonic slope.
- 2) Differential signal falling edge from VIHdiff_CK to VILdiff_CK must be monotonic slope.

[Table 35] Differential Input Slew Rate Definition for CK_t, CK_c

Description	From	To	Defined by
Differential input slew rate for rising edge (CK_t - CK_c)	VILdiff_CK	VIHdiff_CK	$ VILdiff_CK - VIHdiff_CK / \Delta TRdiff$
Differential input slew rate for falling edge (CK_t - CK_c)	VIHdiff_CK	VILdiff_CK	$ VILdiff_CK - VIHdiff_CK / \Delta TFdiff$

[Table 36] Differential Input Level for CK_t, CK_c

Parameter	Symbol	Clock Rate								Unit
		266MHz		533MHz		800MHz		937Mhz		
		Min	Max	Min	Max	Min	Max	Min	Max	
Differential Input High	VIHdiff_CK	145	-	145	-	145	-	145	-	mV
Differential Input Low	VILdiff_CK	-	145	-	145	-	145	-	145	mV

[Table 37] Differential Input Slew Rate for CK_t, CK_c

Parameter	Symbol	Clock Rate								Unit
		266MHz		533MHz		800MHz		937Mhz		
		Min	Max	Min	Max	Min	Max	Min	Max	
Differential Input Slew Rate for CK	SRIdiff_CK	2	14	2	14	2	14	2	14	V/ns

11.2.1.4 Differential Input Cross Point Voltage for CK

The cross point voltage of differential input signals (CK_t, CK_c) must meet the requirements in Table 38. The differential input cross point voltage VIX is measured from the actual cross point of true and complement signals to the mid level that is TBD.

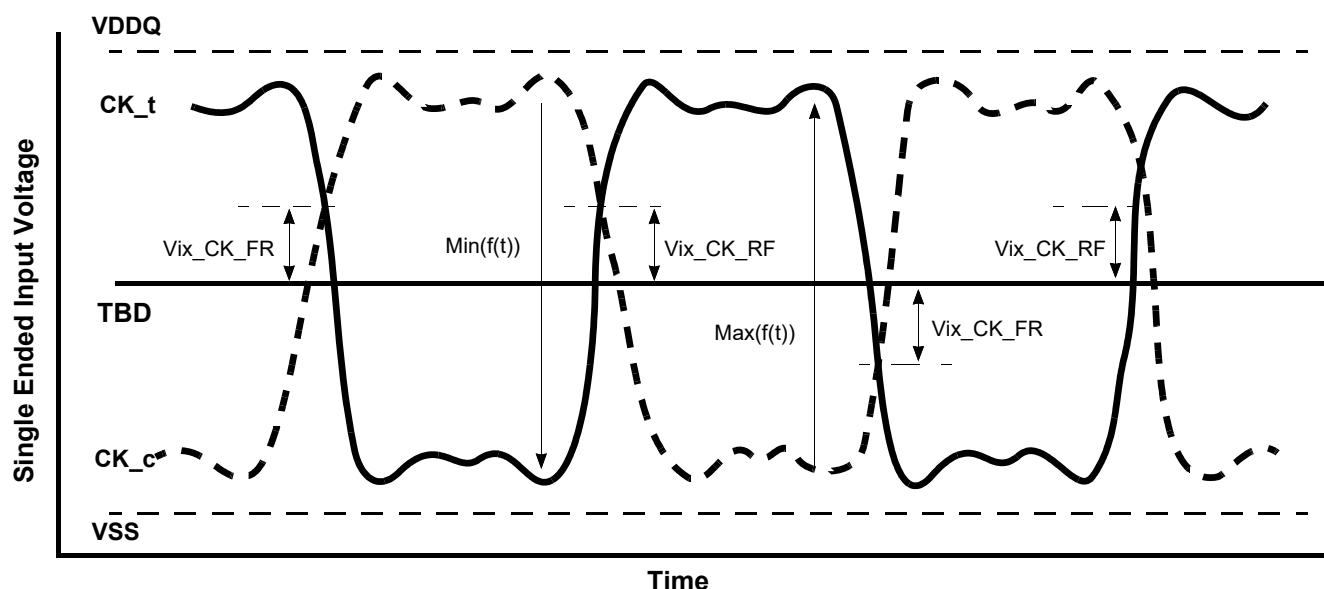


Figure 13. Differential Input Slew Rate Definition for CK_t, CK_c

NOTE :

1) The base level of Vix_CK_FR/RF is TBD that is LPDDR5 SDRAM internal setting value by Vref training.

[Table 38] Cross point voltage for differential input signals (Clock)

Parameter	Symbol	Clock Rate								Unit	Note
		266MHz		533MHz		800MHz		937Mhz			
		Min	Max	Min	Max	Min	Max	Min	Max		
Clock Differential input crosspoint voltage ratio	Vix_CK_ratio	-	25	-	25	-	25	-	25	%	1,2

NOTE :

1) Vix_CK_Ratio is defined by this equation: $Vix_CK_Ratio = Vix_CK_FR / |Min(f(t))|$

2) Vix_CK_Ratio is defined by this equation: $Vix_CK_Ratio = Vix_CK_RF / Max(f(t))$

11.2.2 Differential Input Voltage for WCK

The minimum input voltage need to satisfy both V_{indiff_WCK} and $V_{indiff_WCK}/2$ specification at input receiver and their measurement period is $1t_{WCK}$. V_{indiff_WCK} is the peak to peak voltage centered on 0 volts differential and $V_{indiff_WCK}/2$ is max and min peak voltage from 0V.

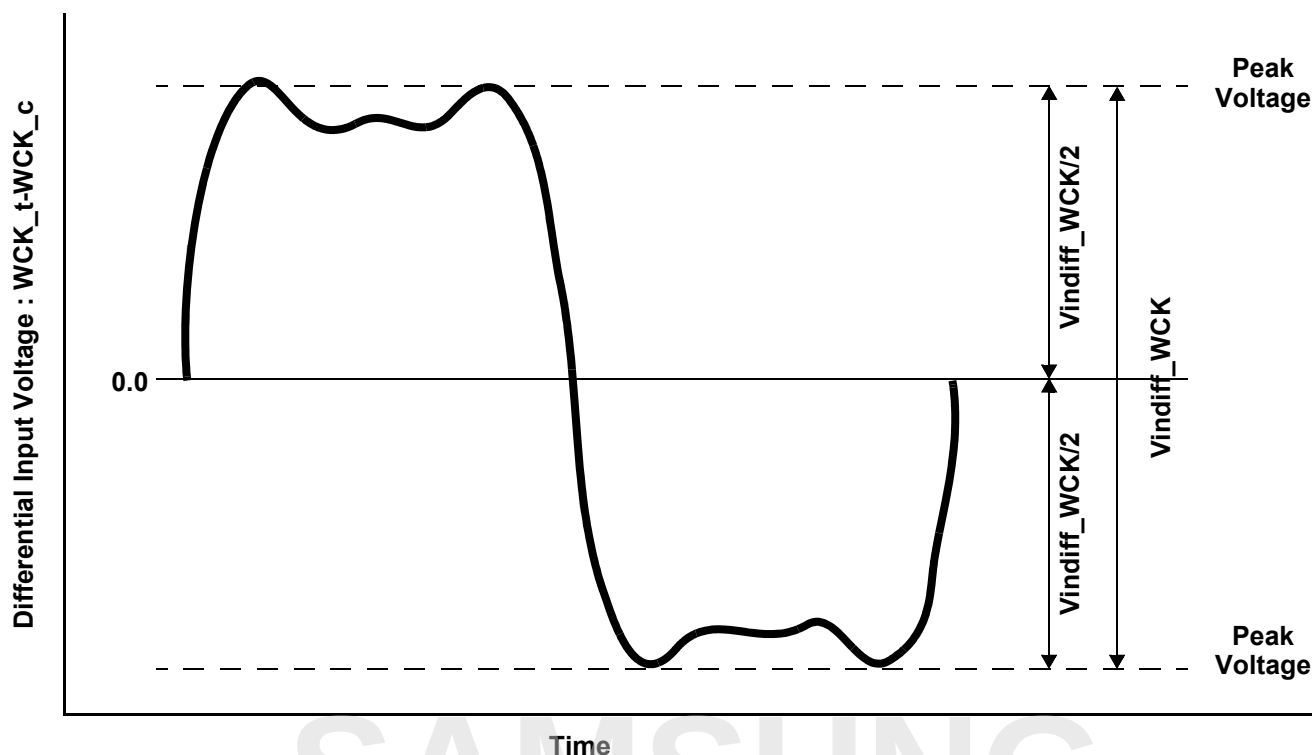


Figure 14. WCK Differential Input Voltage

[Table 39] WCK differential input voltage

Parameter	Symbol	WCK Rate [MHz]														Unit	Note
		800		1066		1600		2133		2750		3200		3750			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
WCK Differential Input Voltage	Vindiff_WCK	300	-	300	-	300	-	280	-	280	-	280	-	280	-	mV	1,2

NOTE :

- 1) Refer to the latency table to match the WCK rate to data rate.
- 2) The peak voltage of Differential WCK signals is calculated in a following equation.
 $V_{indiff_WCK} = (\text{Max Peak Voltage}) - (\text{Min Peak Voltage})$
 $\text{Max Peak Voltage} = \text{Max}(f(t))$
 $\text{Min Peak Voltage} = \text{Min}(f(t))$
 $f(t) = VWCK_t - VWCK_c$

11.2.2.1 Peak Voltage Calculation Method

The peak voltage of Differential WCK signals are calculated in a following equation.

$$V_{IH.DIFF.PEAK} \text{ Voltage} = \text{Max}(f(t))$$

$$V_{IL.DIFF.PEAK} \text{ Voltage} = \text{Min}(f(t))$$

$$f(t) = V_{WCK_t} - V_{WCK_c}$$

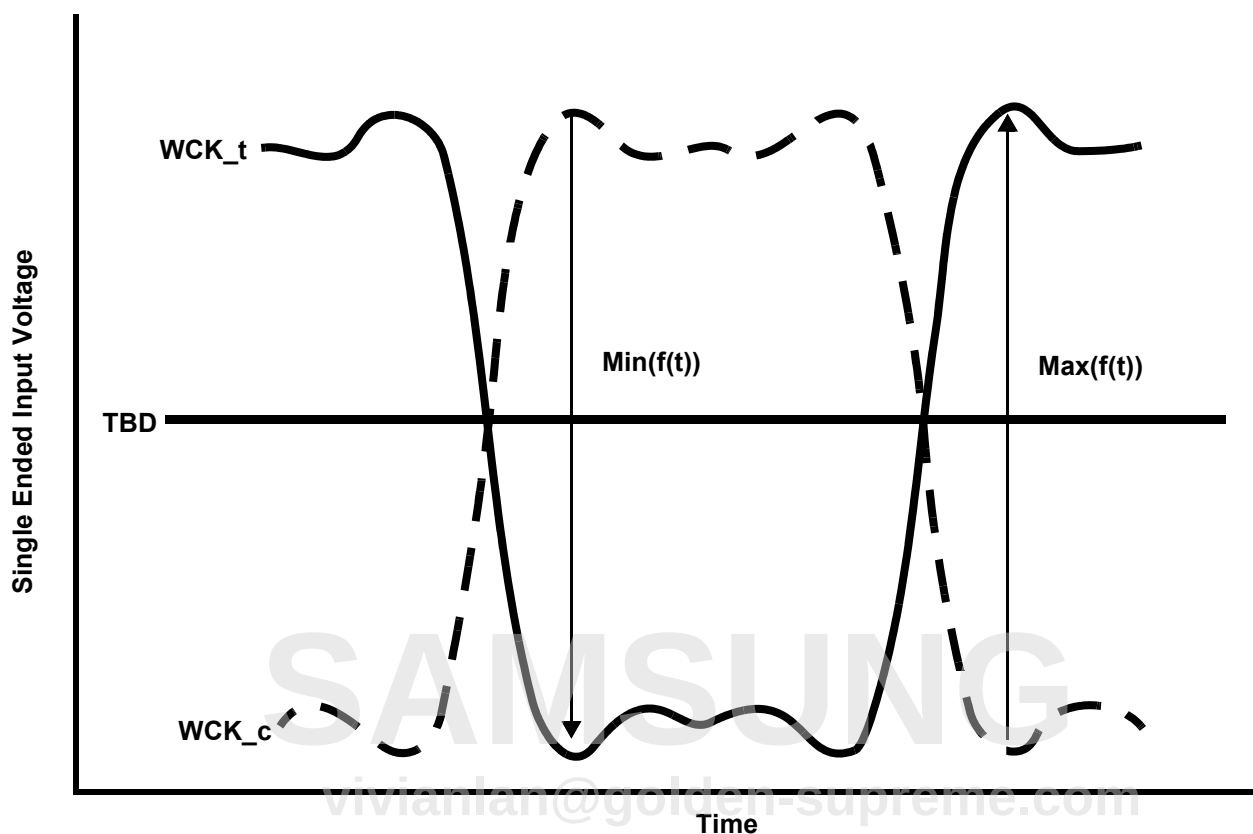


Figure 15. Definition of differential WCK Peak Voltage

NOTE :

1) TBD is LPDDR5 SDRAM internal setting value by Vref training.

11.2.2.2 Single ended Input Voltage for WCK

The minimum input voltage need to satisfy both Vinse_WCK, Vinse_WCK_High/Low specification at input receiver.

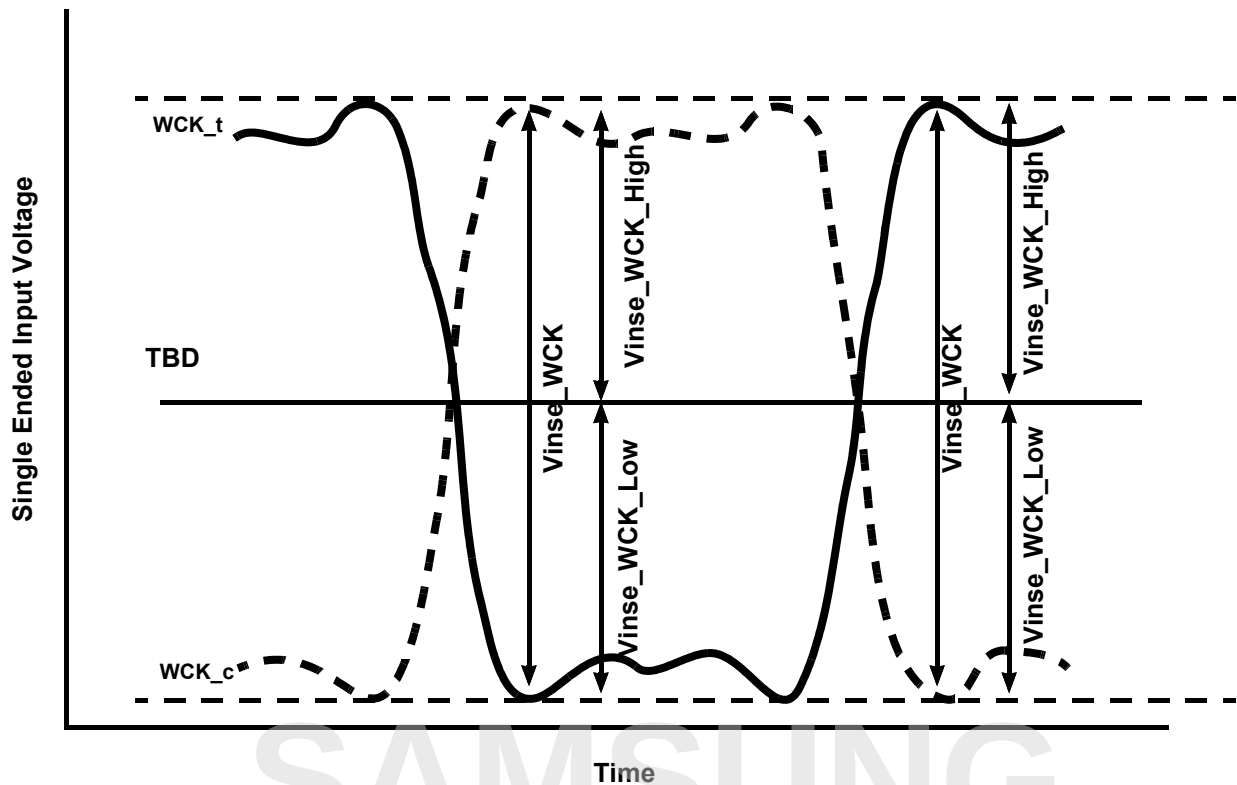


Figure 16. WCK Single ended Input Voltage

NOTE :
1) TBD is LPDDR5 SDRAM internal setting value by Vref training.

[Table 40] WCK Single-Ended Input Voltage

Parameter	Symbol	WCK Rate [MHz]														Unit
		800		1066		1600		2133		2750		3200		3750		
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
WCK Single-ended input Voltage	Vinse_WCK	150	-	150	-	150	-	140	-	140	-	140	-	140	-	mV
WCK Single-ended input Voltage High from TBD	Vinse_WCK_High	75	-	75	-	75	-	75	-	75	-	75	-	75	-	mV
WCK Single-ended input Voltage Low from TBD	Vinse_WCK_Low	75	-	75	-	75	-	75	-	75	-	75	-	75	-	mV

11.2.2.3 Differential Input Slew Rate Definition for WCK

Input slew rate for differential signals (WCK_t, WCK_c) are defined and measured as shown in Figure 17. and the following Tables.

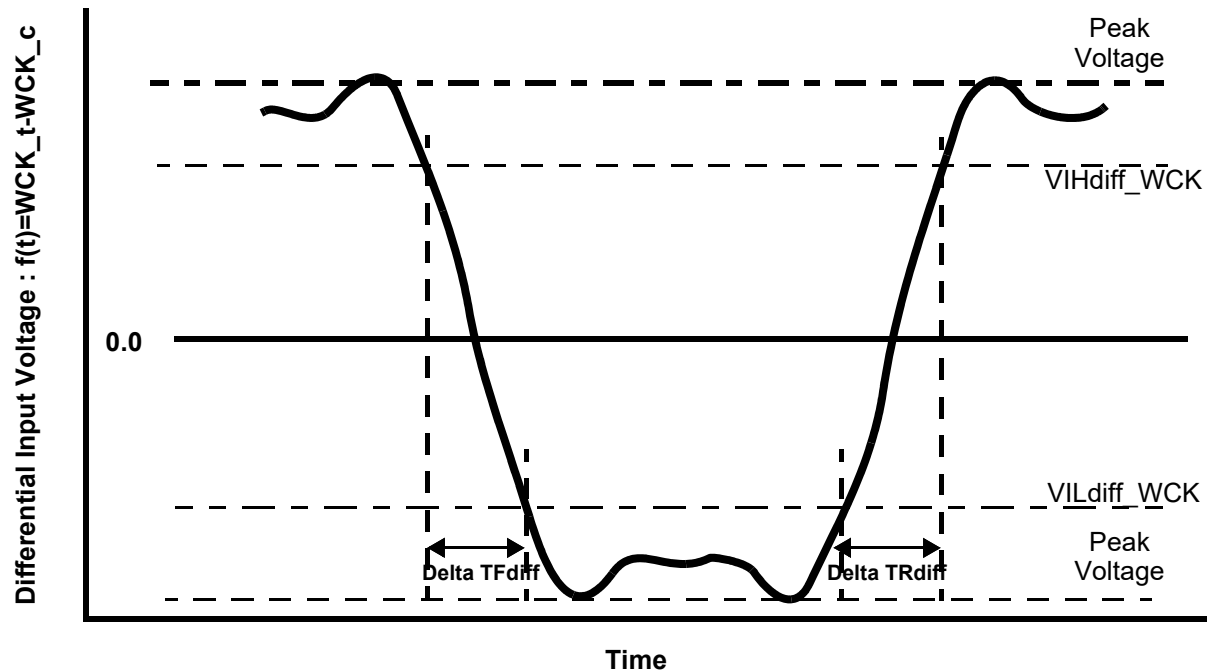


Figure 17. Differential Input Slew Rate Definition for WCK_t, WCK_c

NOTE :

- 1) Differential signal rising edge from VILdiff_WCK to VIHdiff_WCK must be monotonic slope.
- 2) Differential signal falling edge from VIHdiff_WCK to VILdiff_WCK must be monotonic slope.

[Table 41] Differential Input Slew Rate Definition for WCK_t, WCK_c

Description	From	To	Defined by
Differential input slew rate for rising edge (WCK_t - WCK_c)	VILdiff_WCK	VIHdiff_WCK	$ VILdiff_WCK - VIHdiff_WCK / \Delta TRdiff$
Differential input slew rate for falling edge (WCK_t - WCK_c)	VIHdiff_WCK	VILdiff_WCK	$ VILdiff_WCK - VIHdiff_WCK / \Delta TFdiff$

[Table 42] Differential Input Level for WCK_t, WCK_c

Parameter	Symbol	WCK Rate [MHz]														Unit
		800		1066		1600		2133		2750		3200		3750		
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Differential Input High	VIHdiff_WCK	120	-	120	-	120	-	100	-	100	-	100	-	100	-	mV
Differential Input Low	VILdiff_WCK	-	120	-	120	-	120	-	100	-	100	-	100	-	100	mV

[Table 43] Differential Input Slew Rate for WCK_t, WCK_c

Parameter	Symbol	WCK Rate [MHz]														Unit
		800		1066		1600		2133		2750		3200		3750		
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Differential Input Slew Rate for WCK	SRIdiff_WCK	2	14	2	14	2	14	2	14	2	14	2	14	3	14	V/ns

11.2.2.4 Differential Input Cross Point Voltage for WCK

The cross point voltage of differential input signals (WCK_t, WCK_c) must meet the requirements in Table 43. The differential input cross point voltage VIX is measured from the actual cross point of true and complement signals to the mid level that is TBD.

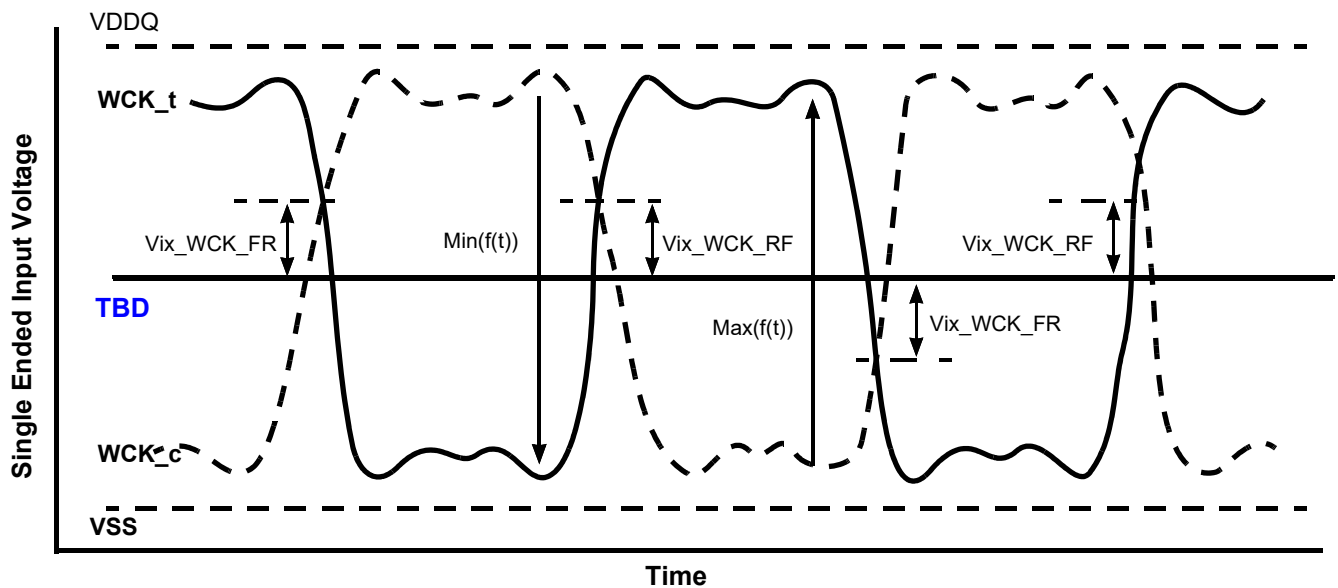


Figure 18. Vix Definition (WCK)

NOTE :

1) The base level of Vix_WCK_FR/RF is TBD that is LPDDR5 SDRAM internal setting value by Vref training.

[Table 44] Cross point voltage for differential input signals (WCK)

Parameter	Symbol	WCK Rate [MHz]														Unit	Note
		800		1066		1600		2133		2750		3200		3750			
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
WCK Differential input crosspoint voltage ratio	Vix_WCK_ratio	-	20	-	20	-	20	-	20	-	20	-	20	-	20	%	1,2

NOTE :

1) Vix_WCK_Ratio is defined by this equation: $Vix_WCK_Ratio = Vix_WCK_FR / |Min(f(t))|$

2) Vix_WCK_Ratio is defined by this equation: $Vix_WCK_Ratio = Vix_WCK_RF / Max(f(t))$

11.3 Output Slew Rate

11.3.1 Single Ended Output Slew Rate

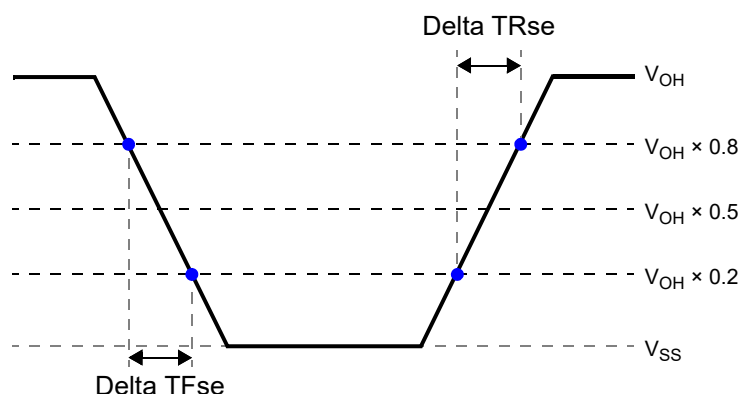


Figure 19. Single Ended Output Slew Rate Definition

[Table 45] Output Slew Rate (single-ended)

Parameter	Symbol	Value		Units
		Min	Max	
Single-ended Output Slew Rate	SRQse	1.5	-	V/ns
Output slew-rate matching Ratio (Rise to Fall)	-	70	130	-

Description:

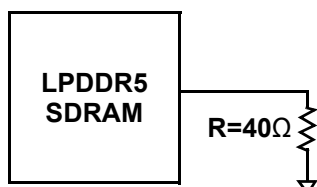
SR: Slew Rate

Q: Query Output (like in DQ, which stands for Data-in, Query-Output)

se: Single-ended Signals

NOTE :

1) These values are guaranteed by design. Assumption: Pull up/down driver=40ohm, Reference load=40ohm.



2) The ratio of pull-up to pull-down slew rate is specified for the same temperature and voltage, over the entire temperature and voltage range. For a given output, it represents the maximum difference between pull-up and pulldown drivers due to process variation. The equation is (Rise slew Rate)/(Fall Slew Rate) x 100.

3) The output slew rate for falling and rising edges is defined and measured between $V_{OH} \times 0.2$ and $V_{OH} \times 0.8$.

4) Slew rates are measured using a unit step signal which makes a full swing signal under average SSO conditions, with 50% of DQ signals per data byte switching. Because a high capacitance load reduces $V_{oh(ac)}$ at high frequency close to F_{max} , the signal using PRBS data pattern may not achieve a full swing at such conditions.

5) The parameters about Single-ended applies to $RDQS_t$ and $RDQS_c$ when either $RDQS_t$ or $RDQS_c$ is disabled.

11.3.2 Differential Output Slew Rate

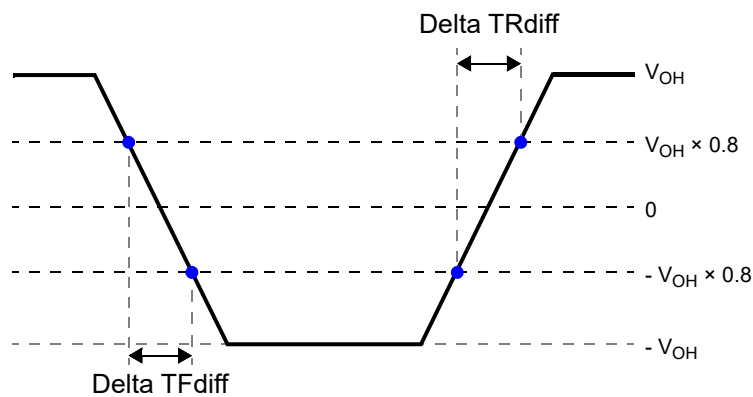


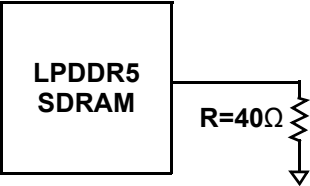
Figure 20. Differential Output Slew Rate Definition

[Table 46] Differential Output Slew Rate

Parameter	Symbol	Value		Units
		Min	Max	
Differential Output Slew Rate	SRQdiff	3.0	-	V/ns

Description:
SR: Slew Rate
Q: Query Output (like in DQ, which stands for Data-in, Query-Output)
diff: Differential Signals

NOTE :
1) These values are guaranteed by design. Assumption: Pull up/down driver=40ohm, Reference load=40ohm.



2) The output slew rate for falling and rising edges is defined and measured between VOH x 0.8 and VOH x 0.8.
3) Slew rates are measured using a unit step signal which makes a full swing signal under average SSO conditions, with 50% of DQ signals per data byte switching. Because a high capacitance load reduces Voh(ac) at high frequency close to Fmax, the signal using PRBS data pattern may not achieve a full swing at such conditions.

11.4 Driver Output Timing Reference load

These "Timing Reference Loads" are not intended as a precise representation of any particular system environment or a depiction of the actual load presented by a production tester. System designers should use IBIS or other simulation tools to correlate the timing reference load to a system environment. Manufacturers correlate to their production test conditions, generally one or more coaxial transmission lines terminated at the tester electronics.

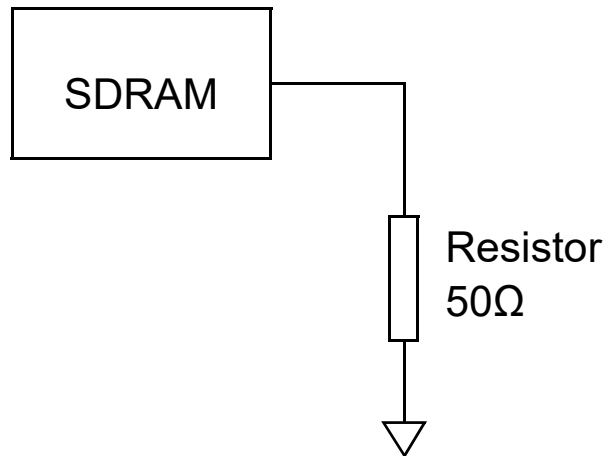


Figure 21. Driver Output Reference Load for Timing

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vivianlan@golden-supreme.com

11.5 Single Ended WCK

11.5.1 Single Ended WCK input definitions

The minimum input voltage need to satisfy both $V_{inse_WCK_SE_High}$ and $V_{inse_WCK_SE_Low}$ specification at input receiver and their measurement period is $1t_{WCK}$. $V_{inse_WCK_SE}$ is the peak to peak voltage centered on $V_{DDQ}/2$ and $V_{inse_WCK_SE_High}$ and $V_{inse_WCK_SE_Low}$ is max and min peak voltage from $V_{DDQ}/2$.

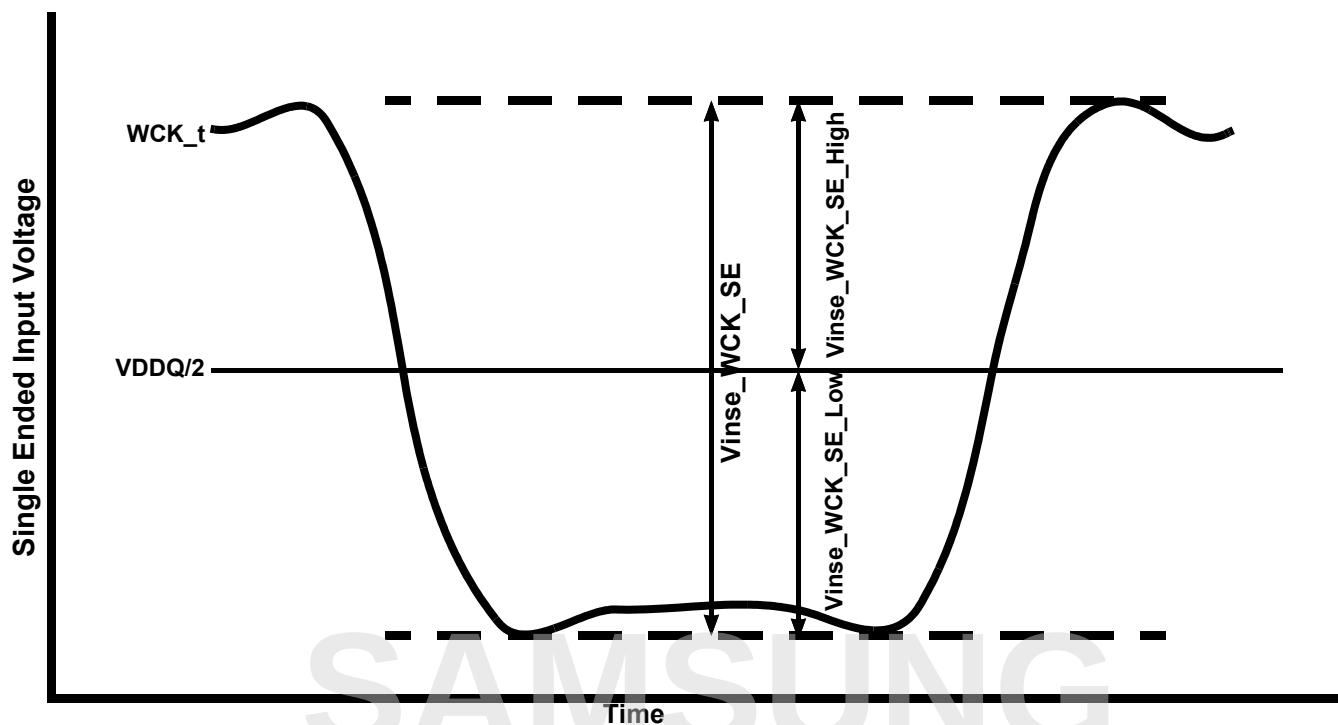


Figure 22. Single Ended Mode WCK input Voltage

11.5.2 Single Ended Mode WCK Pulse Definitions

Single Ended Mode WCK pulse definitions are shown in Figure 23.

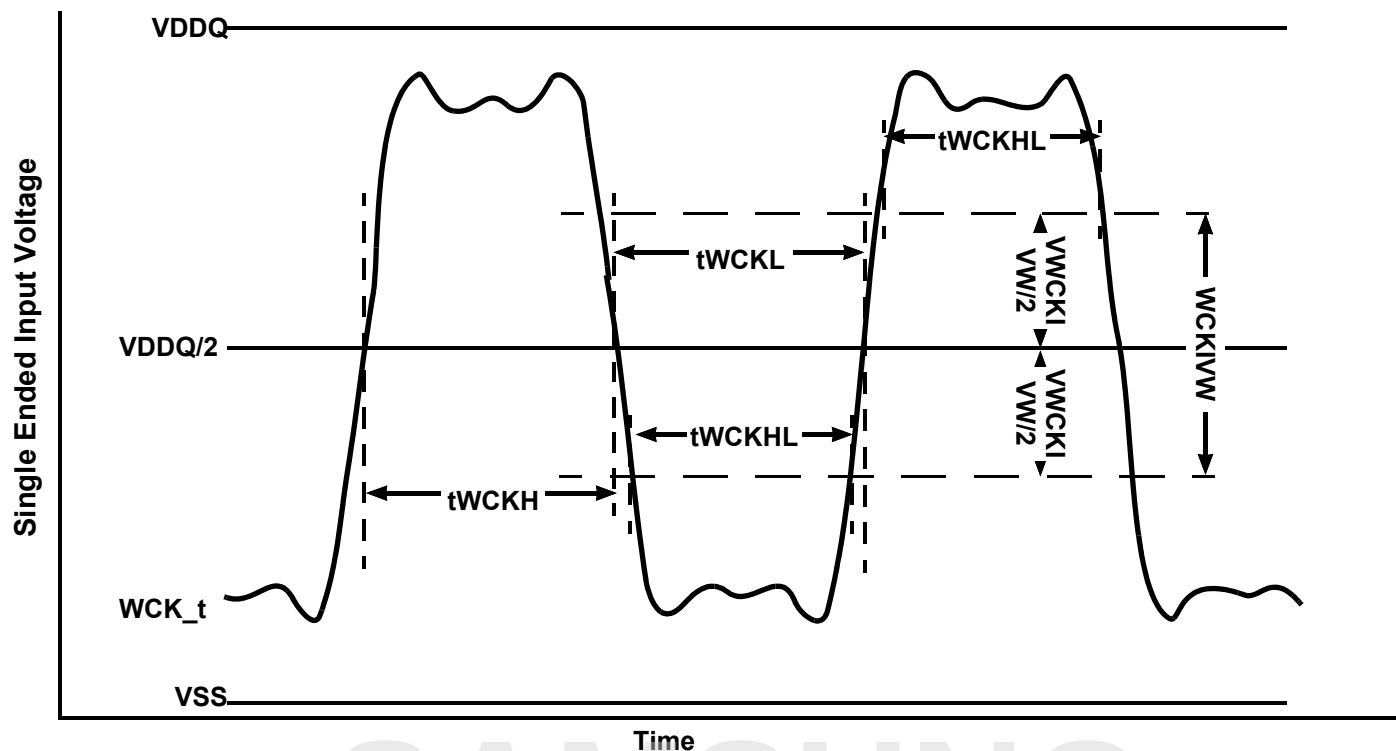


Figure 23. Single Ended Mode WCK pulse

[Table 47] Single Ended WCK parameters

Parameter	Symbol	Min/Max	WCK Frequency	Unit	Note
			800MHz		
WCK Single ended input voltage	Vinse_WCK_SE	Min	220	mV	
WCK Single Ended Input Voltage High	Vinse_WCK_SE_High	Min	110	mV	
WCK Single Ended Input Voltage Low	Vinse_WCK_SE_Low	Min	110	mV	
WCK single ended timing window	VWCKIVW	Min	180	mV	
Clock Single Ended WCK Pulse	tWCKHL	Min	0.23	tWCK (avg)	
WCK single ended Slew Rate	SRIWCKSE	Min	1	V/ns	1
		Max	7		

NOTE :

1) Single ended slew rate is measured at $VDDQ/2 - VWCKIVW/2$ and $VDDQ/2 + VWCKIVW/2$.

11.6 Single Ended CK

11.6.1 Single Ended CK input definitions

The minimum input voltage needs to satisfy both $V_{inse_CK_SE_High}$ and $V_{inse_CK_SE_Low}$ specification at input receiver and their measurement period is $1t_{CK}$. $V_{inse_CK_SE}$ is the peak to peak voltage centered on $V_{DDQ}/2$ and $V_{inse_CK_SE_High}$ and $V_{inse_CK_SE_Low}$ is max and min peak voltage from $V_{DDQ}/2$.

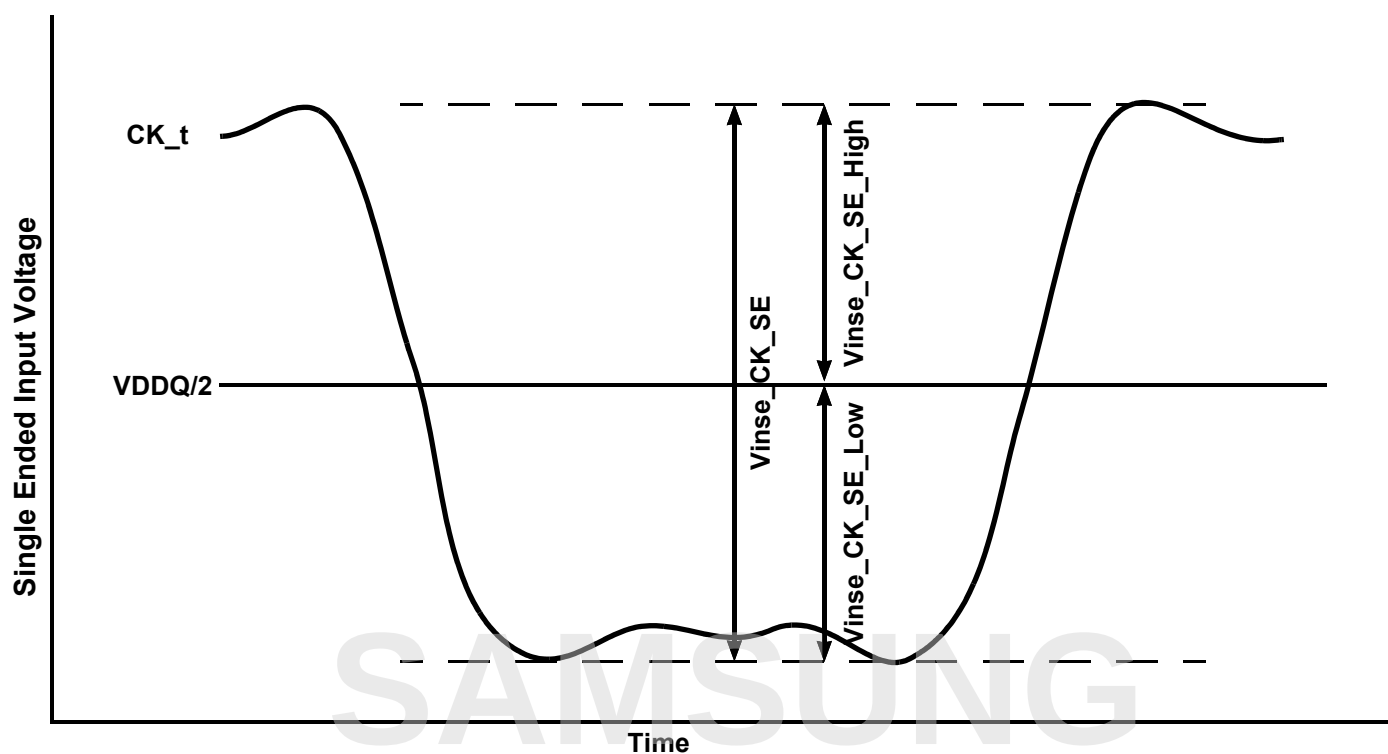


Figure 24. Single Ended Mode CK input Voltage

11.6.2 Single Ended Mode CK Pulse Definitions

Single Ended Mode CK pulse definitions are defined as shown in Table 48.

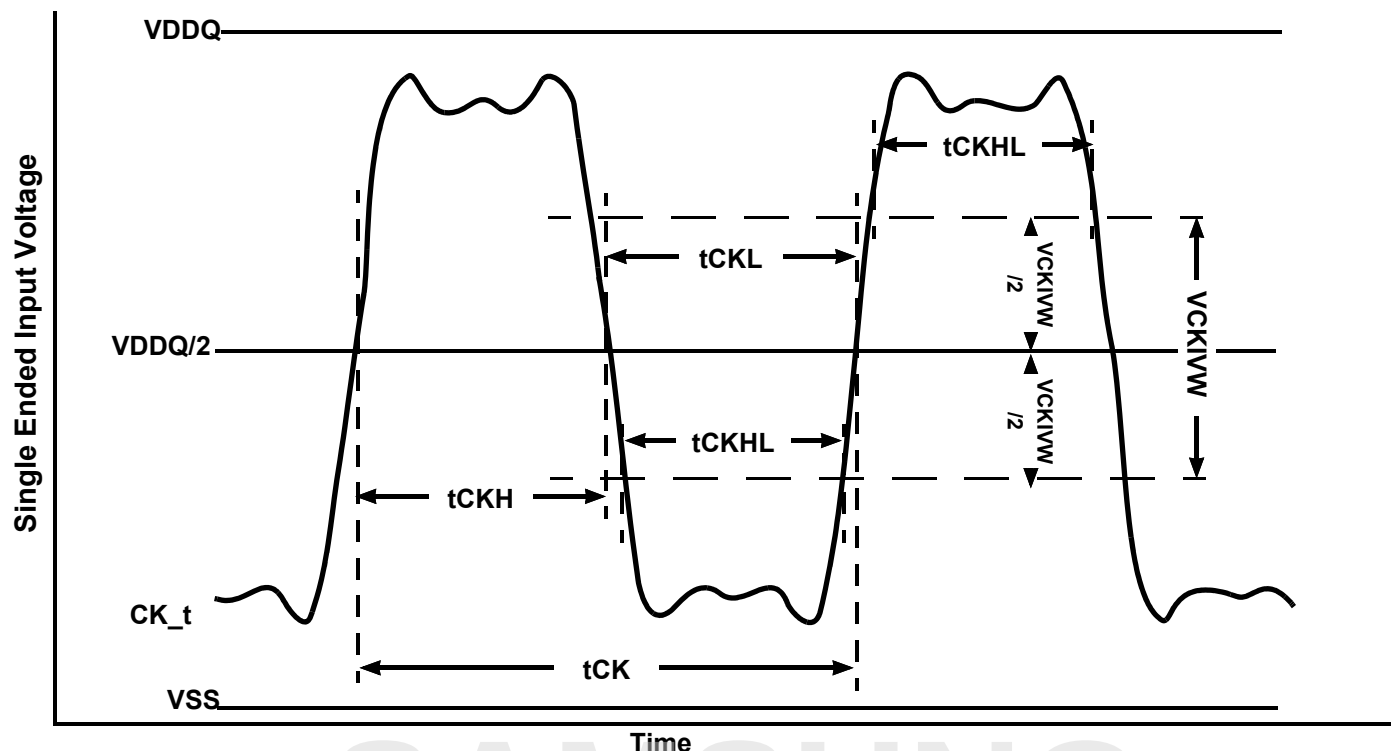


Figure 25. Single Ended Mode CK pulse

[Table 48] Single Ended CK parameters

Parameter	Symbol	Min/Max	CK Frequency	Unit	Note
			400MHz		
CK Single ended input voltage	Vinse_CK_SE	Min	220	mV	
CK Single Ended Input Voltage High	Vinse_CK_SE_High	Min	110	mV	
CK Single Ended Input Voltage Low	Vinse_CK_SE_Low	Min	110	mV	
CK single ended timing window	VCKIVW	Min	190	mV	
Clock Single Ended CK Pulse	tCKHL	Min	0.26	tCK(avg)	
CK single ended Slew Rate	SRICKSE	Min	1	V/ns	1
		Max	7		

NOTE :

1) Single ended slew rate is measured at $VDDQ/2 - VCKIVW/2$ and $VDDQ/2 + VCKIVW/2$.

11.7 Pull Up/Pull Down Driver Characteristics and Calibration

All output pins (DQ, DMI, RDQS_t and RDQS_c) driver characteristics and calibration are defined following.

[Table 49] Pull-down Driver Characteristics, with ZQ Calibration

$R_{ONPD,NOM}$	Resistor	Min	Nom	Max	Unit
40 Ohm	R_{ON40PD}	0.9	1.0	1.1	$R_{ZQ/6}$
48 Ohm	R_{ON48PD}	0.9	1.0	1.1	$R_{ZQ/5}$
60 Ohm	R_{ON60PD}	0.9	1.0	1.1	$R_{ZQ/4}$
80 Ohm	R_{ON80PD}	0.9	1.0	1.1	$R_{ZQ/3}$
120 Ohm	$R_{ON120PD}$	0.9	1.0	1.1	$R_{ZQ/2}$
240 Ohm	$R_{ON240PD}$	0.9	1.0	1.1	$R_{ZQ/1}$

NOTE :

- 1) All values are after ZQ Calibration, see Table 51. Without ZQ Calibration R_{ONPD} values are $\pm 30\%$.
- 2) R_{ONPD} limits are defined at same voltage and temperature as at the time ZQ calibration was done.

[Table 50] Pull-up Characteristics, with ZQ Calibration

$VOH_{PU, nom}$	VOH, nom (mV)	Min	Nom	Max	Unit
$VDDQ \times 0.5$	250	0.90	1.0	1.10	VOH, nom

NOTE :

- 1) All values are after ZQ Calibration. Without ZQ Calibration $V_{OH(nom)}$ values are $\pm 30\%$
- 2) $V_{OH, nom}$ (mV) values are based on a nominal $VDDQ = 0.5V$, $VDD2H = 1.05V$
- 3) VOH_{PU} limits are defined at same voltage and temperature as at the time ZQ calibration was done.
- 4) VOH_{PU} limits are defined for load termination matching the SOC ODT setting (in MR17 OP[2:0]) selected at the time ZQ calibration was done. If the selected SOC ODT setting MR17 OP[2:0]=000_B, then the VOH_{PU} limits are not defined.
- 5) VOH_{PU} limits are defined as DC levels when all DQ drivers are driving high.
- 6) Assumption of SOC ODT is typical for $V_{OHPU, nom}$ definition.

[Table 51] Valid Calibration Points

$VOH_{PU, nom}$	ODT Value					
	240	120	80	60	48	40
$VDDQ \times 0.5$	VALID	VALID	VALID	VALID	VALID	VALID

NOTE :

- 1) Once the output is calibrated for a given $V_{OH(nom)}$ calibration point, the ODT value may be changed without recalibration.

[Table 52] Un-terminated Pull Up Characteristics

RONUNPU,nom	Resistor	Min	Nom	Max	Unit
40 Ohm	R _{ON40UNPU}	0.7	1.0	1.3	RZQ/6
48 Ohm	R _{ON48UNPU}	0.7	1.0	1.3	RZQ/5
60 Ohm	R _{ON60UNPU}	0.7	1.0	1.3	RZQ/4
80 Ohm	R _{ON80UNPU}	0.7	1.0	1.3	RZQ/3
120 Ohm	R _{ON120UNPU}	0.7	1.0	1.3	RZQ/2
240 Ohm	R _{ON240UNPU}	0.7	1.0	1.3	RZQ/1

NOTE :

1) R_{ONUNPU} is defined at V_{OH} = V_{DDQ}/2.

[Table 53] Pull-down Driver Characteristics in Enhanced DVFSC Mode

R _{ONPDED,NOM}	Resistor	Min	Nom	Max	Unit
40 Ohm	R _{ON40PDED}	0.45	1.0	1.75	RZQ/6
48 Ohm	R _{ON48PDED}	0.45	1.0	1.75	RZQ/5
60 Ohm	R _{ON60PDED}	0.45	1.0	1.75	RZQ/4
80 Ohm	R _{ON80PDED}	0.45	1.0	1.75	RZQ/3
120 Ohm	R _{ON120PDED}	0.45	1.0	1.75	RZQ/2
240 Ohm	R _{ON240PDED}	0.45	1.0	1.75	RZQ/1

[Table 54] Un-terminated Pull Up Characteristics in Enhanced DVFSC mode

R _{ONUNPUED,NOM}	Resistor	Min	Nom	Max	Unit
40 Ohm	R _{ON40UNPUED}	0.45	1.0	1.75	RZQ/6
48 Ohm	R _{ON48UNPUED}	0.45	1.0	1.75	RZQ/5
60 Ohm	R _{ON60UNPUED}	0.45	1.0	1.75	RZQ/4
80 Ohm	R _{ON80UNPUED}	0.45	1.0	1.75	RZQ/3
120 Ohm	R _{ON120UNPUED}	0.45	1.0	1.75	RZQ/2
240 Ohm	R _{ON240UNPUED}	0.45	1.0	1.75	RZQ/1

NOTE :

1) R_{ONUNPUED} is defined at V_{OH} = V_{DDQ}/2.

11.8 Output Driver and Termination Resistance Temperature and Voltage Sensitivity

If temperature and/or voltage change after calibration, the tolerance limits widen according to the Tables shown below.

[Table 55] Worst Case Output Driver and Termination Resistance

Resistor	Definition Point	Min	Max	Unit	Notes
R_{ONPD}	$0.5 \times V_{DDQ}$	$R_{onPD}(nom) \times (0.90 - (dR_{on}dT(max) \times \Delta T) - (dR_{on}(max)dV2 \times \Delta V2) - (dR_{on}(max)dVQ \times \Delta VQ))$	$R_{onPD}(nom) \times (1.10 + (dR_{on}(max)dT \times \Delta T) + (dR_{on}(max)dV2 \times \Delta V2) + (dR_{on}(max)dVQ \times \Delta VQ))$	ohm	1,2,3
R_{TT}	$0.5 \times V_{DDQ}$	$R_{TT}(nom) \times (0.90 - (dR_{on}(max)dT \times \Delta T) - (dR_{on}(max)dV2 \times \Delta V2) - (dR_{on}(max)dVQ \times \Delta VQ))$	$R_{TT}(nom) \times (1.10 + (dR_{on}(max)dT \times \Delta T) + (dR_{on}(max)dV2 \times \Delta V2) + (dR_{on}(max)dVQ \times \Delta VQ))$	ohm	1,2,3
R_{ONUNPU}	$0.5 \times V_{DDQ}$	$R_{onUNPU}(nom) \times (0.70 - (dR_{on}UNdT(max) \times \Delta T) - (dR_{on}(max) UNdV2 \times \Delta V2) - (dR_{on}(max) UNdVQ \times \Delta VQ))$	$R_{onUNPU}(nom) \times (1.30 + (dR_{on}(max) UNdT \times \Delta T) + (dR_{on}(max) UNdV2 \times \Delta V2) + (dR_{on}(max) UNdVQ \times \Delta VQ))$	ohm	2,3
R_{TTCS}	$0.5 \times V_{DDQ}$	$R_{TTCS}(nom) \times (0.90 - (dR_{on}(max)dT \times \Delta T) - (dR_{on}(max)dV2 \times \Delta V2) - (dR_{on}(max)dVQ \times \Delta V2))$	$R_{TTCS}(nom) \times (1.10 + (dR_{on}(max)dT \times \Delta T) + (dR_{on}(max)dV2 \times \Delta V2) + (dR_{on}(max)dVQ \times \Delta V2))$	ohm	1,2,3
R_{ONPDED}	$0.5 \times V_{DDQ}$	$R_{onPDED}(nom) \times (0.45 - (dR_{on}PDEDdT(max) \times \Delta T) - (dR_{on}(max)PDEdV2 \times \Delta V2) - (dR_{on}(max)PDEdVQ \times \Delta VQ))$	$R_{onPDED}(nom) \times (1.75 - (dR_{on}PDEDdT(max) \times \Delta T) - (dR_{on}(max)PDEdV2 \times \Delta V2) - (dR_{on}(max)PDEdVQ \times \Delta VQ))$	ohm	1,2,3,4
R_{TTED}	$0.5 \times V_{DDQ}$	$R_{TTED}(nom) \times (0.45 - (dR_{on}PDEdDdT(max) \times \Delta T) - (dR_{on}(max)PDEdV2 \times \Delta V2) - (dR_{on}(max)PDEdVQ \times \Delta VQ))$	$R_{TTED}(nom) \times (1.75 - (dR_{on}PDEdDdT(max) \times \Delta T) - (dR_{on}(max)PDEdV2 \times \Delta V2) - (dR_{on}(max)PDEdVQ \times \Delta VQ))$	ohm	1,2,3,4
$R_{ONUNPUED}$	$0.5 \times V_{DDQ}$	$R_{onUNPUED}(nom) \times (0.45 - (dR_{on}UNEDdT(max) \times \Delta T) - (dR_{on}(max) UNEDdV2 \times \Delta V2) - (dR_{on}(max) UNEDdVQ \times \Delta VQ))$	$R_{onUNPUED}(nom) \times (1.75 - (dR_{on}UNEDdT(max) \times \Delta T) - (dR_{on}(max) UNEDdV2 \times \Delta V2) - (dR_{on}(max) UNEDdVQ \times \Delta VQ))$	ohm	1,2,3,4

NOTE :

- 1) $\Delta T = T - T(@ \text{ Calibration})$, $\Delta V2 = V_{DD2H} - V_{DD2H}(@ \text{ Calibration})$, $\Delta VQ = V_{DDQ} - V_{DDQ}(@ \text{ Calibration})$
- 2) $dR_{on}dT$, $dR_{on}dV2$, $dR_{on}dVQ$, dR_{TTdV2} , dR_{TTdVQ} , dR_{TTdT} , $dR_{on}UNdT$, $dR_{on}UNdV2$ and $dR_{on}UNdVQ$, $dR_{on}PDEdT$, $dR_{on}PDEdV2$, $dR_{on}PDEdVQ$, $dR_{on}UNEDdT$, $dR_{on}UNEDdV2$ and $dR_{on}UNEDdVQ$ are not subject to production test but are verified by design and characterization.
- 3) V_{DD1} , V_{DD2H} , V_{DD2L} , V_{DD2HQ} and V_{DDQ} must be nominal during measurement.
- 4) R_{ONPDED} , R_{TTED} and $R_{ONUNPUED}$ apply at Enhanced DVFS mode is enabled.

[Table 56] Worst Case Output high voltage

Voltage	Min	Max	Unit	Notes
V_{OHPU}	$VOHPU(nom) \times (0.90 - (dVOH(max)dT \times \Delta T) - (dVOH(max)dV2 \times \Delta V2) - (dVOH(max)dVQ \times \Delta VQ))$	$VOHPU(nom) \times (1.10 + (dVOH(max)dT \times \Delta T) + (dVOH(max)dV2 \times \Delta V2) + (dVOH(max)dVQ \times \Delta VQ))$	V	1,2,3,4

NOTE :

- 1) $\Delta T = T - T(@ \text{ Calibration})$, $\Delta V2 = V_{DD2H} - V_{DD2H}(@ \text{ Calibration})$, $\Delta VQ = V_{DDQ} - V_{DDQ}(@ \text{ Calibration})$
- 2) $dVOHdT$, $dVOHdV2$ and $dVOHdVQ$ are not subject to production test but are verified by design and characterization.
- 3) Refer to 11.7 Pull Up/Pull Down Driver Characteristics and Calibration.
- 4) V_{DD1} , V_{DD2H} , V_{DD2HQ} and V_{DDQ} must be nominal during measurement.

[Table 57] Output Driver and Termination Resistance Temperature and Voltage Sensitivity

Symbol	Parameter	Min	Max	Unit	Notes
dR _{ON} dT	R _{ON} Temperature Sensitivity	0.00	0.75	%/°C	
dR _{ON} dV2	R _{ON} V _{DD2H} Voltage Sensitivity	0.00	0.50	%/mV	
dR _{ON} dVQ	R _{ON} V _{DDQ} Voltage Sensitivity	0.00	0.20	%/mV	
dVOHdT	V _{OH} Temperature Sensitivity	0.00	0.75	%/°C	
dVOHdV2	V _{OH} V _{DD2H} Voltage Sensitivity	0.00	0.35	%/mV	
dVOHdVQ	V _{OH} V _{DDQ} Voltage Sensitivity	0.00	0.35	%/mV	
dR _{TT} dT	R _{TT} Temperature Sensitivity	0.00	0.75	%/°C	
dR _{TT} dV2	R _{TT} V _{DD2H} Voltage Sensitivity	0.00	0.50	%/mV	
dR _{TT} dVQ	R _{TT} V _{DDQ} Voltage Sensitivity	0.00	0.20	%/mV	
dRONUNdT	Un-terminated R _{ON} Temperature Sensitivity	0.00	0.75	%/°C	
dRONUNdV2	Un-terminated R _{ON} V _{DD2H} Voltage Sensitivity	0.00	0.75	%/mV	
dRONUNdVQ	Un-terminated R _{ON} V _{DDQ} Voltage Sensitivity	0.00	0.75	%/mV	
dRonPDEDdT	R _{ON} Temperature Sensitivity	0.00	0.75	%/°C	1
dRonPDEDdV2	R _{ON} V _{DD2L} Voltage Sensitivity	0.00	0.60	%/mV	1
dRonPDEDdVQ	R _{ON} V _{DDQ} Voltage Sensitivity	0.00	0.20	%/mV	1
dRonUNEDdT	Un-terminated R _{ON} Temperature Sensitivity	0.00	0.75	%/°C	1
dRonUNEDdV2	Un-terminated R _{ON} V _{DD2L} Voltage Sensitivity	0.00	0.75	%/mV	1
dRonUNEDdVQ	Un-terminated R _{ON} V _{DDQ} Voltage Sensitivity	0.00	0.75	%/mV	1

NOTE :

1) These parameters apply at Enhanced DVFSC mode is enabled.

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12.0 INPUT/OUTPUT CAPACITANCE

[Table 58] Input / Output Capacitance

Parameter	Symbol		LPDDR5X	Units	Notes
Input Capacitance, CK_t and CK_c	CCK	Min	TBD	pF	1,2
		Max	TBD	pF	1,2
Input Capacitance delta, CK_t and CK_c	CDCK	Min	TBD	pF	1,2,3
		Max	TBD	pF	1,2,3
Input Capacitance, WCK_t and WCK_c	CWCK	Min	TBD	pF	1,2
		Max	TBD	pF	1,2
Input Capacitance delta, WCK_t and WCK_c	CDWCK	Min	TBD	pF	1,2,4
		Max	TBD	pF	1,2,4
Input Capacitance, All other input-only pins except CS	CI1	Min	TBD	pF	1,2,5
		Max	TBD	pF	1,2,5
Cin, CS0/CS1	CI2	Min	TBD	pF	1,2,5
		Max	TBD	pF	1,2,5
Input Capacitance delta, All other input-only pins except CS	CDI1	Min	TBD	pF	1,2,6
		Max	TBD	pF	1,2,6
Cdelta, CS0/CS1	CDI2	Min	TBD	pF	1,2,6
		Max	TBD	pF	1,2,6,10
Input/output Capacitance, DQ and DMI	CIO	Min	TBD	pF	1,2,7
		Max	TBD	pF	1,2,7
Input/output Capacitance delta, DQ and DMI	CDIO	Min	TBD	pF	1,2,9
		Max	TBD	pF	1,2,9
Output Capacitance, RDQS_t and RDQS_c	COO	Min	TBD	pF	1,2
		Max	TBD	pF	1,2
Output Capacitance delta, RDQS_t and RDQS_c	CDOO	Min	TBD	pF	1,2,8
		Max	TBD	pF	1,2,8
Input/output capacitance, ZQ pin	CZQ	Min	TBD	pF	1,2
		Max	TBD	pF	1,2

NOTE :

- 1) This parameter applies to both die and package.
- 2) This parameter is not subject to production test. It is verified by design and characterization. The capacitance is measured according to JEP147 (Procedure for measuring input capacitance using a vector network analyzer (VNA) with VDD1, VDD2, VDDQ, VSS applied and all other pins floating.)
- 3) Absolute value of CCK_t - CCK_c.
- 4) Absolute value of CWCK_t - CWCK_c.
- 5) CI applies to CS, CA0~CA6.
- 6) CDI = CI - Average(CInputn, CCK_t, CCK_c)
- 7) DMI loading matches DQ.
- 8) Absolute value of CRDQS_t - CRDQS_c.
- 9) CDIO = CIO - Average(CDQn, CDMI) in byte-lane.
- 10) CDI2 = CI2 - 0.25 × (CCK_t + CCK_c).

13.0 IDD SPECIFICATION PARAMETERS AND TEST CONDITIONS

13.1 IDD Measurement Conditions

The following definitions are used within the IDD measurement tables unless stated otherwise:

- LOW: $V_{IN} \leq V_{IL}(DC)_{MAX}$
- HIGH: $V_{IN} \geq V_{IH}(DC)_{MIN}$
- STABLE: Inputs are stable at a HIGH or LOW level
- SWITCHING: See Table 59 through Table 67

[Table 59] Definition of Switching for CA Input Signals

CK _t edge	R1	F1	R2	F2	R3	F3	R4	F4	R5	F5	R6	F6	R7	F7	R8	F8
CS	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
CA0	H	L	L	L	L	H	H	H	H	L	L	L	L	H	H	H
CA1	H	H	H	L	L	L	L	H	H	H	H	L	L	L	L	H
CA2	H	L	L	L	L	H	H	H	H	L	L	L	L	H	H	H
CA3	H	H	H	L	L	L	L	H	H	H	H	L	L	L	L	H
CA4	H	L	L	L	L	H	H	H	H	L	L	L	L	H	H	H
CA5	H	H	H	L	L	L	L	H	H	H	H	L	L	L	L	H
CA6	H	L	L	L	L	H	H	H	H	L	L	L	L	H	H	H

NOTE :

- 1) CS must always be driven LOW.
- 2) 50% of CA bus is changing between HIGH and LOW once per clock for the CA bus.
- 3) The pattern is used continuously during IDD measurement for IDD values that require switching on the CA bus.

[Table 60] CA Pattern for IDD4R@ BG mode

Clock Cycle #	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5	CA6	Note
R0	High	Read	H	L	L	L	L	L	L	1
F0	Low		L	H	H	L	L	L	L	
R1	High	DES	L	L	L	L	L	L	L	
F1	Low		L	L	L	L	L	L	L	
R2	High	Read	H	L	L	L	L	L	L	2
F2	Low		L	H	L	H	L	L	L	
R3	Low	CAS (B3)	L	L	H	H	L	L	L	
F3	Low		L	L	L	L	L	L	H	
R4	High	Read	H	L	L	H	H	H	H	3
F4	Low		L	H	H	L	H	H	L	
R5	High	CAS (B3)	L	L	H	H	L	L	L	
F5	Low		L	L	L	L	L	L	H	
R6	High	Read	H	L	L	H	H	H	H	4
F6	Low		L	H	L	H	H	H	L	
R7	Low	DES	L	L	L	L	L	L	L	
F7	Low		L	L	L	L	L	L	L	

NOTE :

- 1) Pattern A is applied to DQs.
- 2) Pattern B is applied to DQs.
- 3) Pattern A' is applied to DQs.
- 4) Pattern B' is applied to DQs.
- 5) The pattern above is applied when WCK to CK frequency ratio is 4:1 and bank organization is BG mode.

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR HEADQUARTERS OF SAMSUNG ELECTRONICS.

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[Table 61] CA Pattern for IDD4R @ 16B mode

Clock Cycle #	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5	CA6	Note
R0	High	Read	H	L	L	L	L	L	L	1
F0	Low		L	H	H	L	L	L	L	
R1	High	CAS (B3)	L	L	H	H	L	L	L	
F1	Low		L	L	L	L	L	L	H	
R2	High	Read	H	L	L	H	H	H	H	2
F2	Low		L	H	H	L	H	H	L	
R3	Low	DES	L	L	L	L	L	L	L	
F3	Low		L	L	L	L	L	L	L	

NOTE :

- 1) Pattern A is applied to DQs.
2) Pattern B is applied to DQs.

3) The pattern above is applied when WCK to CK frequency ratio is 4:1 and bank organization is 16B mode. In the case WCK:CK=2:1, the number of DES commands is increased to match tCCD.

[Table 62] CA Pattern for IDD4W @ BG mode

Clock Cycle #	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5	CA6	Note
R0	High	WRITE	L	H	H	L	L	L	L	1
F0	Low		L	H	H	L	L	L	L	
R1	Low	DES	L	L	L	L	L	L	L	
F1	Low		L	L	L	L	L	L	L	
R2	High	WRITE	L	H	H	L	L	L	L	2
F2	Low		L	H	L	H	L	L	L	
R3	Low	DES	L	L	L	L	L	L	L	
F3	Low		L	L	L	L	L	L	L	
R4	High	WRITE	L	H	H	H	H	H	H	1
F4	Low		L	H	H	L	H	H	L	
R5	Low	DES	L	L	L	L	L	L	L	
F5	Low		L	L	L	L	L	L	L	
R6	High	WRITE	L	H	H	H	H	H	H	2
F6	Low		L	H	L	H	H	H	L	
R7	Low	DES	L	L	L	L	L	L	L	
F7	Low		L	L	L	L	L	L	L	

NOTE :

- 1) Pattern A is applied to DQs.
2) Pattern B is applied to DQs.

3) The pattern above is applied when WCK to CK frequency ratio is 4:1 and bank organization is BG mode.

[Table 63] CA Pattern for IDD4W @ 16B mode

Clock Cycle #	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5	CA6	Note
R0	High	WRITE	L	H	H	L	L	L	L	1
F0	Low		L	H	H	L	L	L	L	
R1	Low	DES	L	L	L	L	L	L	L	
F1	Low		L	L	L	L	L	L	L	
R2	High	WRITE	L	H	H	H	H	H	H	2
F2	Low		L	H	H	L	H	H	L	
R3	Low	DES	L	L	L	L	L	L	L	
F3	Low		L	L	L	L	L	L	L	

NOTE :

1) Pattern A is applied to DQs.

2) Pattern B is applied to DQs.

3) The pattern above is applied when WCK to CK frequency ratio is 4:1 and bank organization is 16B mode. In the case WCK:CK=2:1, the number of DES commands is increased to match tCCD.

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[Table 64] Data Pattern for IDD4R @ DBI Off

DQs – IDD4R DBI OFF											# of 1's
Type	BL	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
Pattern A	BL0	1	1	1	1	1	1	1	1	0	8
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	0	0	0	0
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	1	1	0	2
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	1	1	1	1	1	1	0	0	0	6
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	1	1	1	1	1	1	1	1	0	8
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	0	0	0	0
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	1	1	0	2
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	1	1	1	1	1	1	0	0	0	6
	BL15	1	1	1	1	0	0	0	0	0	4
Pattern B	BL0	1	1	1	1	1	1	0	0	0	6
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	1	1	0	2
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	0	0	0	0
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	1	1	1	1	1	1	1	1	0	8
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	1	1	0	2
	BL9	0	0	0	0	1	1	1	1	0	4
	BL10	1	1	1	1	1	1	0	0	0	6
	BL11	1	1	1	1	0	0	0	0	0	4
	BL12	1	1	1	1	1	1	1	1	0	8
	BL13	1	1	1	1	0	0	0	0	0	4
	BL14	0	0	0	0	0	0	0	0	0	0
	BL15	0	0	0	0	1	1	1	1	0	4

[Table 64] Data Pattern for IDD4R @ DBI Off

DQs – IDD4R DBI OFF											# of 1's
Type	BL	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
Pattern A'	BL0	1	1	1	1	1	1	1	1	0	8
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	0	0	0	0
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	1	1	0	2
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	1	1	1	1	1	1	0	0	0	6
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	1	1	1	1	1	1	1	1	0	8
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	0	0	0	0
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	1	1	0	2
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	1	1	1	1	1	1	0	0	0	6
	BL15	1	1	1	1	0	0	0	0	0	4
Pattern B'	BL0	0	0	0	0	0	0	1	1	0	2
	BL1	0	0	0	0	1	1	1	1	0	4
	BL2	1	1	1	1	1	1	0	0	0	6
	BL3	1	1	1	1	0	0	0	0	0	4
	BL4	1	1	1	1	1	1	1	1	0	8
	BL5	1	1	1	1	0	0	0	0	0	4
	BL6	0	0	0	0	0	0	0	0	0	0
	BL7	0	0	0	0	1	1	1	1	0	4
	BL8	1	1	1	1	1	1	0	0	0	6
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	1	1	0	2
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	0	0	0	0
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	1	1	1	1	1	1	1	1	0	8
	BL15	1	1	1	1	0	0	0	0	0	4
# of 1's		32	32	32	32	32	32	32	32	0	

NOTE :

- 1) Pattern A' is defined by B3 ordering change based on pattern A.
- 2) Pattern B' is defined by B3 ordering change based on pattern B.

[Table 65] Data Pattern for IDD4W @ DBI Off

DQs – IDD4W DBI OFF											# of 1's
Type	BL	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
Pattern A	BL0	1	1	1	1	1	1	1	1	0	8
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	0	0	0	0
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	1	1	0	2
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	1	1	1	1	1	1	0	0	0	6
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	1	1	1	1	1	1	1	1	0	8
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	0	0	0	0
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	1	1	0	2
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	1	1	1	1	1	1	0	0	0	6
	BL15	1	1	1	1	0	0	0	0	0	4
Pattern B	BL0	1	1	1	1	1	1	0	0	0	6
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	1	1	0	2
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	0	0	0	0
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	1	1	1	1	1	1	1	1	0	8
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	1	1	0	2
	BL9	0	0	0	0	1	1	1	1	0	4
	BL10	1	1	1	1	1	1	0	0	0	6
	BL11	1	1	1	1	0	0	0	0	0	4
	BL12	1	1	1	1	1	1	1	1	0	8
	BL13	1	1	1	1	0	0	0	0	0	4
	BL14	0	0	0	0	0	0	0	0	0	0
	BL15	0	0	0	0	1	1	1	1	0	4
# of 1's		16	16	16	16	16	16	16	16	0	

[Table 66] Data Pattern for IDD4R @ DBI On

DQs – IDD4R DBI ON											# of 1's
Type	BL	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
Pattern A	BL0	0	0	0	0	0	0	0	0	1	1
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	0	0	0	0
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	1	1	0	2
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	0	0	0	0	0	0	1	1	1	3
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	0	0	1	1
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	0	0	0	0
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	1	1	0	2
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	0	0	0	0	0	0	1	1	1	3
	BL15	1	1	1	1	0	0	0	0	0	4
Pattern B	BL0	0	0	0	0	0	0	1	1	1	3
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	1	1	0	2
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	0	0	0	0
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	0	0	0	0	0	0	0	0	1	1
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	1	1	0	2
	BL9	0	0	0	0	1	1	1	1	0	4
	BL10	0	0	0	0	0	0	1	1	1	3
	BL11	1	1	1	1	0	0	0	0	0	4
	BL12	0	0	0	0	0	0	0	0	1	1
	BL13	1	1	1	1	0	0	0	0	0	4
	BL14	0	0	0	0	0	0	0	0	0	0
	BL15	0	0	0	0	1	1	1	1	0	4

[Table 66] Data Pattern for IDD4R @ DBI On

DQs – IDD4R DBI ON											# of 1's
Type	BL	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
Pattern A'	BL0	0	0	0	0	0	0	0	0	1	1
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	0	0	0	0
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	1	1	0	2
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	0	0	0	0	0	0	1	1	1	3
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	0	0	1	1
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	0	0	0	0
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	1	1	0	2
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	0	0	0	0	0	0	1	1	1	3
	BL15	1	1	1	1	0	0	0	0	0	4
Pattern B'	BL0	0	0	0	0	0	0	1	1	0	2
	BL1	0	0	0	0	1	1	1	1	0	4
	BL2	0	0	0	0	0	0	1	1	1	3
	BL3	1	1	1	1	0	0	0	0	0	4
	BL4	0	0	0	0	0	0	0	0	1	1
	BL5	1	1	1	1	0	0	0	0	0	4
	BL6	0	0	0	0	0	0	0	0	0	0
	BL7	0	0	0	0	1	1	1	1	0	4
	BL8	0	0	0	0	0	0	1	1	1	3
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	1	1	0	2
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	0	0	0	0
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	0	0	0	0	0	0	0	0	1	1
	BL15	1	1	1	1	0	0	0	0	0	4
	# of 1's	16	16	16	16	16	16	32	32	16	

NOTE :

- 1) Pattern A' is defined by B3 ordering change based on pattern A.
- 2) Pattern B' is defined by B3 ordering change based on pattern B.

[Table 67] Data Pattern for IDD4W @ DBI On

DQs – IDD4W DBI ON											# of 1's
Type	BL	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
Pattern A	BL0	0	0	0	0	0	0	0	0	1	1
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	0	0	0	0
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	1	1	0	2
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	0	0	0	0	0	0	1	1	1	3
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	0	0	1	1
	BL9	1	1	1	1	0	0	0	0	0	4
	BL10	0	0	0	0	0	0	0	0	0	0
	BL11	0	0	0	0	1	1	1	1	0	4
	BL12	0	0	0	0	0	0	1	1	0	2
	BL13	0	0	0	0	1	1	1	1	0	4
	BL14	0	0	0	0	0	0	1	1	1	3
	BL15	1	1	1	1	0	0	0	0	0	4
Pattern B	BL0	0	0	0	0	0	0	1	1	1	3
	BL1	1	1	1	1	0	0	0	0	0	4
	BL2	0	0	0	0	0	0	1	1	0	2
	BL3	0	0	0	0	1	1	1	1	0	4
	BL4	0	0	0	0	0	0	0	0	0	0
	BL5	0	0	0	0	1	1	1	1	0	4
	BL6	0	0	0	0	0	0	0	0	1	1
	BL7	1	1	1	1	0	0	0	0	0	4
	BL8	0	0	0	0	0	0	1	1	0	2
	BL9	0	0	0	0	1	1	1	1	0	4
	BL10	0	0	0	0	0	0	1	1	1	3
	BL11	1	1	1	1	0	0	0	0	0	4
	BL12	0	0	0	0	0	0	0	0	1	1
	BL13	1	1	1	1	0	0	0	0	0	4
	BL14	0	0	0	0	0	0	0	0	0	0
	BL15	0	0	0	0	1	1	1	1	0	4
# of 1's		8	8	8	8	8	8	16	16	8	

13.2 IDD Specifications

[Table 68] LPDDR5X IDD Specification Parameters and Operating Conditions

Parameter / Condition	Symbol	Power Supply	Note
Operating one bank active-Precharge current: $t_{CK} = t_{CKmin}$; $t_{RC} = t_{RCmin}$; CS is LOW between valid commands; CA bus inputs are switching; Data bus inputs are stable RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable	IDD0 ₁	VDD1	9
	IDD0 _{2H}	VDD2H	9
	IDD0 _{2L}	VDD2L	9
	IDD0 _Q	VDDQ	3
Idle power-down standby current: $t_{CK} = t_{CKmin}$; Power-down entry command is issued CS is LOW; All banks are idle; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable	IDD2P ₁	VDD1	
	IDD2P _{2H}	VDD2H	
	IDD2P _{2L}	VDD2L	
	IDD2P _Q	VDDQ	3
Idle power-down standby current with clock stop: CK_t = LOW, CK_c = HIGH; CS is LOW; All banks are idle; CA bus inputs are stable; Data bus inputs are stable RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable	IDD2PS ₁	VDD1	
	IDD2PS _{2H}	VDD2H	
	IDD2PS _{2L}	VDD2L	
	IDD2PS _Q	VDDQ	3
Idle non power-down standby current: $t_{CK} = t_{CKmin}$; CS is LOW; All banks are idle; CA bus inputs are switching; Data bus inputs are stable RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable	IDD2N ₁	VDD1	
	IDD2N _{2H}	VDD2H	
	IDD2N _{2L}	VDD2L	
	IDD2N _Q	VDDQ	3
Idle non power-down standby current with clock stopped: CK_t = LOW; CK_c = HIGH; CS is LOW; All banks are idle; CA bus inputs are stable; Data bus inputs are stable RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable	IDD2NS ₁	VDD1	
	IDD2NS _{2H}	VDD2H	
	IDD2NS _{2L}	VDD2L	
	IDD2NS _Q	VDDQ	3
Active power-down standby current: $t_{CK} = t_{CKmin}$; CS is LOW; One bank is active; Power-down entry command is issued CA bus inputs are switching; Data bus inputs are stable RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable	IDD3P ₁	VDD1	9
	IDD3P _{2H}	VDD2H	9
	IDD3P _{2L}	VDD2L	9
	IDD3P _Q	VDDQ	3

[Table 68] LPDDR5X IDD Specification Parameters and Operating Conditions

Parameter / Condition	Symbol	Power Supply	Note
Active power-down standby current with clock stop: Power-down entry command is issued CK _t =LOW, CK _c =HIGH; CS is LOW; One bank is active; CA bus inputs are stable; Data bus inputs are stable RDQS _t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD3PS ₁	VDD1	9
	IDD3PS _{2H}	VDD2H	9
	IDD3PS _{2L}	VDD2L	8, 9
	IDD3PS _Q	VDDQ	4
Active non-power-down standby current: t _{CK} = t _{CKmin} ; CS is LOW; One bank is active; CA bus inputs are switching; Data bus inputs are stable RDQS _t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD3N ₁	VDD1	9
	IDD3N _{2H}	VDD2H	9
	IDD3N _{2L}	VDD2L	8, 9
	IDD3N _Q	VDDQ	4
Operating burst READ current @ BG Mode t _{CK} = t _{CKmin} ; t _{WCK} =t _{WCKmin} ; CS is LOW between valid commands; One bank in each bank group 1 and 2 is active; BL = 16 or 32; RL = RLmin; CA bus inputs are switching; 50% data change each burst transfer ODT disabled	IDD4R ₁	VDD1	
	IDD4R _{2H}	VDD2H	
	IDD4R _{2L}	VDD2L	8
	IDD4R _Q	VDDQ	5
Operating burst READ current @ 16Bank Mode t _{CK} = t _{CKmin} ; t _{WCK} =t _{WCKmin} ; CS is LOW between valid commands; One bank is active; BL = 16 or 32 @ 16bank mode; RL = RLmin; CA bus inputs are switching; 50% data change each burst transfer ODT disabled	IDD4R ₁	VDD1	
	IDD4R _{2H}	VDD2H	
	IDD4R _{2L}	VDD2L	8
	IDD4R _Q	VDDQ	5
Operating burst READ current @ 16Bank Mode with DVFSC or E-DVFSC t _{CK} = t _{CKmin} ; t _{WCK} =t _{WCKmin} ; CS is LOW between valid commands; One bank is active; BL = 16 or 32 @ 16bank mode; RL = RLmin; CA bus inputs are switching; 50% data change each burst transfer ODT disabled DVFSC or E-DVFSC mode is enabled	IDD4R _{DVFSC1}	VDD1	13
	IDD4R _{DVFSC2H}	VDD2H	13
	IDD4R _{DVFSC2L}	VDD2L	8,13,14
	IDD4R _{DVFSCQ}	VDDQ	5,13
Operating burst WRITE current @ BG Mode t _{CK} = t _{CKmin} ; t _{WCK} =t _{WCKmin} ; CS is LOW between valid commands; One bank in each bank group 1 and 2 is active; BL = 16 or 32; WL = WLmin; CA bus inputs are switching; 50% data change each burst transfer RDQS _t is stable (if Link ECC is enabled) ODT disabled	IDD4W ₁	VDD1	
	IDD4W _{2H}	VDD2H	
	IDD4W _{2L}	VDD2L	8
	IDD4W _Q	VDDQ	4

[Table 68] LPDDR5X IDD Specification Parameters and Operating Conditions

Parameter / Condition	Symbol	Power Supply	Note
Operating burst WRITE current @ 16Bank Mode $t_{CK} = t_{CKmin}$; $t_{WCK} = t_{WCKmin}$; CS is LOW between valid commands; One bank is active; BL = 16 or 32 @ 16bank mode; WL = WLmin; CA bus inputs are switching; 50% data change each burst transfer RDQS_t is stable (if Link ECC is enabled) ODT disabled	IDD4W ₁	VDD1	
	IDD4W _{2H}	VDD2H	
	IDD4W _{2L}	VDD2L	8
	IDD4W _Q	VDDQ	4
Operating burst WRITE current @ 16Bank Mode with DVFSC or E-DVFSC $t_{CK} = t_{CKmin}$; $t_{WCK} = t_{WCKmin}$; CS is LOW between valid commands; One bank is active; BL = 16 or 32 @ 16bank mode; WL = WLmin; CA bus inputs are switching; 50% data change each burst transfer ODT disabled DVFSC or E-DVFSC mode is enabled	IDD4W _{DVFSC1}	VDD1	13
	IDD4W _{DVFSC2H}	VDD2H	13
	IDD4W _{DVFSC2L}	VDD2L	8,13,14
	IDD4W _{DVFSCQ}	VDDQ	4,13
All-bank REFRESH Burst current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; $t_{RC} = t_{RFCabmin}$; Burst refresh; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD5 ₁	VDD1	
	IDD5 _{2H}	VDD2H	
	IDD5 _{2L}	VDD2L	8
	IDD5 _Q	VDDQ	4
All-bank REFRESH Average current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; $t_{RC} = t_{REFI}$; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD5AB ₁	VDD1	
	IDD5AB _{2H}	VDD2H	
	IDD5AB _{2L}	VDD2L	8
	IDD5AB _Q	VDDQ	4
All-bank REFRESH Burst current with E-DVFSC: $t_{CK} = 1.25ns$ (800MHz) CKR=2:1 CS is LOW between valid commands; $t_{RC} = t_{RFCabmin}$; Burst refresh; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is don't care. ODT disabled WCK inputs are stable and static Enhanced DVFSC mode is enabled	IDD5 _{ED1}	VDD1	
	IDD5 _{ED2H}	VDD2H	
	IDD5 _{ED2L}	VDD2L	8
	IDD5 _{EDQ}	VDDQ	4

[Table 68] LPDDR5X IDD Specification Parameters and Operating Conditions

Parameter / Condition	Symbol	Power Supply	Note
All-bank REFRESH Average current with E-DVFS: $t_{CK} = 1.25\text{ns}$ (800MHz) CKR=2:1 CS is LOW between valid commands; $t_{RC} = t_{REFI}$; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is don't care. ODT disabled WCK inputs are stable and static Enhanced DVFS mode is enabled	IDD5AB _{ED1}	VDD1	
	IDD5AB _{ED2H}	VDD2H	
	IDD5AB _{ED2L}	VDD2L	8
	IDD5AB _{EDQ}	VDDQ	4
Per-bank REFRESH Average current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; $t_{RC} = t_{REFI}/8$; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD5PB ₁	VDD1	
	IDD5PB _{2H}	VDD2H	
	IDD5PB _{2L}	VDD2L	8
	IDD5PB _Q	VDDQ	4
Per-bank REFRESH Average current with E-DVFS: $t_{CK} = 1.25\text{ns}$ (800MHz) CKR=2:1 CS is LOW between valid commands; $t_{RC} = t_{REFI}/8$; CA bus inputs are switching; Data bus inputs are stable; RDQS_t is don't care. ODT disabled WCK inputs are stable and static Enhanced DVFS mode is enabled	IDD5PB _{ED1}	VDD1	
	IDD5PB _{ED2H}	VDD2H	
	IDD5PB _{ED2L}	VDD2L	8
	IDD5PB _{EDQ}	VDDQ	4
Power down Self refresh current: CK_t=LOW, CK_c=HIGH; CS is LOW; CA bus inputs are stable; Data bus inputs are stable; Maximum 1x Self-Refresh Rate; ($\leq 85^\circ\text{C}$) Maximum TBDx Self-Refresh Rate; ($> 85^\circ\text{C}$) RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD6 ₁	VDD1	6,7,11
	IDD6 _{2H}	VDD2H	6,7,11
	IDD6 _{2L}	VDD2L	6,7,8,11
	IDD6 _Q	VDDQ	4,6,7,8,11
Deep sleep mode current: CK_t=LOW, CK_c=HIGH; CS is LOW; CA bus inputs are stable; Data bus inputs are stable; Maximum 1x Self-Refresh Rate; ($\leq 85^\circ\text{C}$) Maximum TBDx Self-Refresh Rate; ($> 85^\circ\text{C}$) RDQS_t is stable (if Link ECC is enabled) ODT disabled WCK inputs are stable and static	IDD6DS ₁	VDD1	6,7,12
	IDD6DS _{2H}	VDD2H	6,7,12
	IDD6DS _{2L}	VDD2L	6,7,8,12
	IDD6DS _Q	VDDQ	4,6,7,12

NOTE :

- 1) Published IDD values are the maximum of the distribution of the arithmetic mean.
- 2) ODT disabled. MR11 OP[6:4] = 000B
- 3) IDD current specifications are tested after the device is properly initialized.
- 4) Measured currents are the summation of VDDQ and VDD2H / VDD2L.
- 5) Guaranteed by design with output load = 5pF and RON = 40 ohm.
- 6) The 1x Self-Refresh Rate is the rate at which the LPDDR5 device is refreshed internally during Self-Refresh, before going into the elevated Temperature range.
- 7) This is the general definition that applies to full array Self Refresh.
- 8) When MR13 OP[7] is high, single VDD2 rail, VDD2L Current shall be added to VDD2H Current.
- 9) IDD values can be different according to the bank organization set by MR3 OP[4:3].
- 10) When DVFS is enabled, the minimum tCK shall be set by following DVFS operating frequency.
- 11) IDD6 25°C is guaranteed, IDD6 85°C is typical of the distribution of the arithmetic mean.
- 12) IDD6DS 25°C is guaranteed, IDD6DS 85°C is typical of the distribution of the arithmetic mean.
- 13) IDD values can be different according to the DVFS set by MR19 OP[1:0].
- 14) When LPDDR5X SDRAM(MR8 OP[1:0]=01B) supports both DVFS and E-DVFS mode(MR41 OP[2:1]=10B), this IDD current at mode which consumes bigger VDD2L current than the other

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13.3 IDD Spec Table

[Table 69] IDD Specifications for 16Gb LPDDR5X

Symbol		Power Supply	16Gb x16	Units
			LPDDR5X	
			7500Mbps	
IDD0	IDD0 ₁	VDD1	TBD	mA
	IDD0 _{2H}	VDD2H	TBD	mA
	IDD0 _{2L}	VDD2L	TBD	mA
	IDD0 _Q	VDDQ	TBD	mA
IDD2P	IDD2P ₁	VDD1	TBD	mA
	IDD2P _{2H}	VDD2H	TBD	mA
	IDD2P _{2L}	VDD2L	TBD	mA
	IDD2P _Q	VDDQ	TBD	mA
IDD2PS	IDD2PS ₁	VDD1	TBD	mA
	IDD2PS _{2H}	VDD2H	TBD	mA
	IDD2PS _{2L}	VDD2L	TBD	mA
	IDD2PS _Q	VDDQ	TBD	mA
IDD2N	IDD2N ₁	VDD1	TBD	mA
	IDD2N _{2H}	VDD2H	TBD	mA
	IDD2N _{2L}	VDD2L	TBD	mA
	IDD2N _Q	VDDQ	TBD	mA
IDD2NS	IDD2NS ₁	VDD1	TBD	mA
	IDD2NS _{2H}	VDD2H	TBD	mA
	IDD2NS _{2L}	VDD2L	TBD	mA
	IDD2NS _Q	VDDQ	TBD	mA
IDD3P	IDD3P ₁	VDD1	TBD	mA
	IDD3P _{2H}	VDD2H	TBD	mA
	IDD3P _{2L}	VDD2L	TBD	mA
	IDD3P _Q	VDDQ	TBD	mA
IDD3PS	IDD3PS ₁	VDD1	TBD	mA
	IDD3PS _{2H}	VDD2H	TBD	mA
	IDD3PS _{2L}	VDD2L	TBD	mA
	IDD3PS _Q	VDDQ	TBD	mA
IDD3N	IDD3N ₁	VDD1	TBD	mA
	IDD3N _{2H}	VDD2H	TBD	mA
	IDD3N _{2L}	VDD2L	TBD	mA
	IDD3N _Q	VDDQ	TBD	mA
IDD4R	IDD4R ₁	VDD1	TBD	mA
	IDD4R _{2H}	VDD2H	TBD	mA
	IDD4R _{2L}	VDD2L	TBD	mA
	IDD4R _Q	VDDQ	TBD	mA

[Table 69] IDD Specifications for 16Gb LPDDR5X

Symbol			Power Supply	16Gb x16	Units
				LPDDR5X	
				7500Mbps	
IDD4W	IDD4W ₁		VDD1	TBD	mA
	IDD4W _{2H}		VDD2H	TBD	mA
	IDD4W _{2L}		VDD2L	TBD	mA
	IDD4W _Q		VDDQ	TBD	mA
IDD5	IDD5 ₁		VDD1	TBD	mA
	IDD5 _{2H}		VDD2H	TBD	mA
	IDD5 _{2L}		VDD2L	TBD	mA
	IDD5 _Q		VDDQ	TBD	mA
IDD5AB	IDD5AB ₁		VDD1	TBD	mA
	IDD5AB _{2H}		VDD2H	TBD	mA
	IDD5AB _{2L}		VDD2L	TBD	mA
	IDD5AB _Q		VDDQ	TBD	mA
IDD5PB	IDD5PB ₁		VDD1	TBD	mA
	IDD5PB _{2H}		VDD2H	TBD	mA
	IDD5PB _{2L}		VDD2L	TBD	mA
	IDD5PB _Q		VDDQ	TBD	mA
IDD6	IDD6 ₁	25°C	VDD1	TBD	mA
		85°C		TBD	mA
	IDD6 _{2H}	25°C	VDD2H	TBD	mA
		85°C		TBD	mA
	IDD6 _{2L}	25°C	VDD2L	TBD	mA
		85°C		TBD	mA
	IDD6 _Q	25°C	VDDQ	TBD	mA
		85°C		TBD	mA
IDD6DS	IDD6DS ₁	25°C	VDD1	TBD	mA
		85°C		TBD	mA
	IDD6DS _{2H}	25°C	VDD2H	TBD	mA
		85°C		TBD	mA
	IDD6DS _{2L}	25°C	VDD2L	TBD	mA
		85°C		TBD	mA
	IDD6DS _Q	25°C	VDDQ	TBD	mA
		85°C		TBD	mA

[Table 70] IDD4R and IDD4W with DVFSC for 16Gb LPDDR5X

Symbol		Power Supply	16Gb x16	Units
			1600Mbps	
IDD4R @16Bank Mode with DVFSC	IDD4R _{DVFSC1}	VDD1	TBD	mA
	IDD4R _{DVFSC2H}	VDD2H	TBD	mA
	IDD4R _{DVFSC2L}	VDD2L	TBD	mA
	IDD4R _{DVFSCQ}	VDDQ	TBD	mA
IDD4W @16Bank Mode with DVFSC	IDD4W _{DVFSC1}	VDD1	TBD	mA
	IDD4W _{DVFSC2H}	VDD2H	TBD	mA
	IDD4W _{DVFSC2L}	VDD2L	TBD	mA
	IDD4W _{DVFSCQ}	VDDQ	TBD	mA

[Table 71] IDD4R and IDD4W with E-DVFSC for 16Gb LPDDR5X

Symbol		Power Supply	16Gb x16	Units
			3200Mbps	
IDD4R @16Bank Mode with E-DVFSC	IDD4R _{EDVFSC1}	VDD1	TBD	mA
	IDD4R _{EDVFSC2H}	VDD2H	TBD	mA
	IDD4R _{EDVFSC2L}	VDD2L	TBD	mA
	IDD4R _{EDVFSCQ}	VDDQ	TBD	mA
IDD4W @16Bank Mode with E-DVFSC	IDD4W _{EDVFSC1}	VDD1	TBD	mA
	IDD4W _{EDVFSC2H}	VDD2H	TBD	mA
	IDD4W _{EDVFSC2L}	VDD2L	TBD	mA
	IDD4W _{EDVFSCQ}	VDDQ	TBD	mA

[Table 72] IDD5, IDD5AB AND IDD5PB with E-DVFSC for 16Gb LPDDR5X

Symbol		Power Supply	16Gb x16	Units
			3200Mbps	
IDD5 with E-DVFSC	IDD5 _{ED1}	VDD1	TBD	mA
	IDD5 _{ED2H}	VDD2H	TBD	mA
	IDD5 _{ED2L}	VDD2L	TBD	mA
	IDD5 _{EDQ}	VDDQ	TBD	mA
IDD5AB with E-DVFSC	IDD5AB _{ED1}	VDD1	TBD	mA
	IDD5AB _{ED2H}	VDD2H	TBD	mA
	IDD5AB _{ED2L}	VDD2L	TBD	mA
	IDD5AB _{EDQ}	VDDQ	TBD	mA
IDD5PB with E-DVFSC	IDD5PB _{ED1}	VDD1	TBD	mA
	IDD5PB _{ED2H}	VDD2H	TBD	mA
	IDD5PB _{ED2L}	VDD2L	TBD	mA
	IDD5PB _{EDQ}	VDDQ	TBD	mA

14.0 ELECTRICAL CHARACTERISTICS AND AC TIMING

14.1 Clock Specification

The jitter specified is a random jitter meeting a Gaussian distribution. Input clocks violating the min/max values may result in malfunction of the LPDDR5 device.

14.1.1 Definition for $t_{CK(avg)}$ and n_{CK}

$t_{CK(avg)}$ is calculated as the average clock period across any consecutive 200 cycle window, where each clock period is calculated from rising edge to rising edge.

$$t_{CK(avg)} = \left(\sum_{j=1}^N t_{CK_j} \right) / N$$

where $N = 200$

Unit " $t_{CK(avg)}$ " represents the actual clock average $t_{CK(avg)}$ of the input clock under operation. Unit " n_{CK} " represents one clock cycle of the input clock, counting the actual clock edges. $t_{CK(avg)}$ may change by up to $\pm 1\%$ within a 100 clock cycle window, provided that all jitter and timing specs are met.

14.1.2 Definition for $t_{CK(abs)}$

$t_{CK(abs)}$ is defined as the absolute clock period, as measured from one rising edge to the next consecutive rising edge.

$t_{CK(abs)}$ is not subject to production test.

14.1.3 Definition for $t_{CH(avg)}$ and $t_{CL(avg)}$

$t_{CH(avg)}$ is defined as the average high pulse width, as calculated across any consecutive 200 high pulses.

$$t_{CH(avg)} = \left(\sum_{j=1}^N t_{CH_j} \right) / (N \times t_{CK(avg)})$$

where $N = 200$

$t_{CL(avg)}$ is defined as the average low pulse width, as calculated across any consecutive 200 low pulses.

$$t_{CL(avg)} = \left(\sum_{j=1}^N t_{CL_j} \right) / (N \times t_{CK(avg)})$$

where $N = 200$

14.1.4 Definition for tCH(abs) and tCL(abs)

t_{CH(abs)} is the absolute instantaneous clock high pulse width, as measured from one rising edge to the following falling edge.

t_{CL(abs)} is the absolute instantaneous clock low pulse width, as measured from one falling edge to the following rising edge.

Both t_{CH(abs)} and t_{CL(abs)} are not subject to production test.

14.1.5 Definition for tJIT(per)

t_{JIT(per)} is the single period jitter defined as the largest deviation of any signal t_{CK} from t_{CK(avg)}.

t_{JIT(per)} = Min/max of {t_{CKi} - t_{CK(avg)}} where i = 1 to 200}.

t_{JIT(per)}, act is the actual clock jitter for a given system.

t_{JIT(per)}, allowed is the specified allowed clock period jitter.

t_{JIT(per)} is not subject to production test.

14.1.6 Definition for tJIT(cc)

t_{JIT(cc)} is defined as the absolute difference in clock period between two consecutive clock cycles.

t_{JIT(cc)} = Max of |{t_{CK(i+1)} - t_{CK(i)}}|.

t_{JIT(cc)} defines the cycle to cycle jitter.

t_{JIT(cc)} is not subject to production test.

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14.1.7 Refresh Requirement

Between SRX command and SRE command, at least one extra refresh command is required. After the DRAM Self-Refresh Exit command, in addition to the normal Refresh command at t_{REFI} interval, the LPDDR5 DRAM requires minimum of one extra Refresh command prior to Self-Refresh Entry command.

[Table 73] Refresh Requirement Parameters for BG mode or 16Bank mode

Refresh Requirements		Symbol	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb	Units	Notes
Number of banks per channel			16										
Refresh Window (t_{REFW}) (1x Refresh) ^{2),3)}		t_{REFW}	32										ms
Required Number of REFRESH Commands in a t_{REFW} window		R	8192	8192	8192	8192	8192	8192	8192	8192	8192		
Average Refresh Interval (1x Refresh) ²⁾	REFAb	t_{REFI}	3.906	3.906	3.906	3.906	3.906	3.906	3.906	3.906	3.906	us	
	REFpb	t_{REFipb}	488	488	488	488	488	488	488	488	488	ns	
Refresh Cycle Time (All Banks)		t_{RFCab}	130	180	180	210	210	280	280	380	380	ns	4
Refresh Cycle time (Per Bank)		t_{RFCpb}	120	120	120	120	120	140	140	190	190	ns	4
Per-bank Refresh to Per-bank Refresh different bank time		$t_{pbR2pbR}$	60	90	90	90	90	90	90	90	90	ns	4
REFpb to Activate command to different bank		$t_{pbR2act}$	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	ns	
Refresh Management Cycle time (All Banks)		t_{RFMab}	-	180	180	210	210	280	280	380	380	ns	4
Refresh Management Cycle time (Per Bank)		t_{RFMpb}	-	170	170	170	170	190	190	260	260	ns	4
Refresh Cycle Time (All Banks): Enhanced DVFS is Enabled		t_{RFCab}	-	200	200	230	230	300	300	410	410	ns	5
Refresh Cycle time (Per Bank): Enhanced DVFS is Enabled		t_{RFCpb}	-	130	130	130	130	150	150	205	205	ns	5
Per-bank Refresh to Per-bank Refresh different bank time: Enhanced DVFS is Enabled		$t_{pbR2pbR}$	-	100	100	100	100	100	100	100	100	ns	5
Refresh Management Cycle time (All Banks): Enhanced DVFS is Enabled		t_{RFMab}	-	200	200	230	230	300	300	410	410	ns	5
Refresh Management Cycle time (Per Bank): Enhanced DVFS is Enabled		t_{RFMpb}	-	170	170	170	170	190	190	260	260	ns	5

NOTE :

- 1) Refresh for each channel is independent of the other channel in a package. Power delivery in the user's system should be verified to make sure the DC operating conditions are maintained when multiple channels are refreshed simultaneously.
- 2) 1x refresh rate ($t_{REFW}=32ms$) is supported at all temperatures at or below 85°C Tcase. If MR4 OP[4:0] indicates a refresh rate of greater than 1x is supported, t_{REFW} can be extended.
- 3) Refer to MR4 OP[4:0] (MR4 definition) for detailed Refresh Rate and its multipliers.
- 4) Applied when Enhanced DVFS is Disabled.
- 5) Applied when Enhanced DVFS is Enabled.

14.1.8 Clock Timing

[Table 74] Clock AC Timings for 5/10/67/133MHz

Parameter	Symbol	5MHz		10MHz		67MHz		133MHz		Units
		Min	Max	Min	Max	Min	Max	Min	Max	
Average clock period	$t_{CK(avg)}$	200	200	100	200	14.93	200	7.5	200	ns
Average High pulse width	$t_{CH(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Average Low pulse width	$t_{CL(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Absolute clock period	$t_{CK(abs)}$	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	ns
Absolute High clock pulse width	$t_{CH(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Absolute Low clock pulse width	$t_{CL(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Clock period jitter	$t_{JIT(per)}$	-11200	11200	-5600	5600	-840	840	-430	430	ps
Maximum Clock Jitter between consecutive cycles	$t_{JIT(cc)}$	-	22400	-	11200	-	1680	-	860	ps

[Table 75] Clock AC Timings for 200/267/344/400MHz

Parameter	Symbol	200MHz		267MHz		344MHz		400MHz		Units
		Min	Max	Min	Max	Min	Max	Min	Max	
Average clock period	$t_{CK(avg)}$	5	200	3.75	200	2.9	200	2.5	200	ns
Average High pulse width	$t_{CH(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Average Low pulse width	$t_{CL(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Absolute clock period	$t_{CK(abs)}$	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	ns
Absolute High clock pulse width	$t_{CH(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Absolute Low clock pulse width	$t_{CL(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Clock period jitter	$t_{JIT(per)}$	-280	280	-210	210	-170	170	-140	140	ps
Maximum Clock Jitter between consecutive cycles	$t_{JIT(cc)}$	-	560	-	420	-	340	-	280	ps

[Table 76] Clock AC Timings for 467/533/600/688

Parameter	Symbol	467MHz		533MHz		600MHz		688MHz		Units
		Min	Max	Min	Max	Min	Max	Min	Max	
Average clock period	$t_{CK(avg)}$	2.15	200	1.875	200	1.667	200	1.453	200	ns
Average High pulse width	$t_{CH(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Average Low pulse width	$t_{CL(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Absolute clock period	$t_{CK(abs)}$	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	$t_{CK(avg)}$ MIN + $t_{JIT(per)}$ MIN	-	ns
Absolute High clock pulse width	$t_{CH(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Absolute Low clock pulse width	$t_{CL(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Clock period jitter	$t_{JIT(per)}$	-120	120	-110	110	-95	95	-85	85	ps
Maximum Clock Jitter between consecutive cycles	$t_{JIT(cc)}$	-	240	-	220	-	190	-	170	ps

[Table 77] Clock AC Timings for 750/800/937.5MHz

Parameter	Symbol	750MHz		800MHz		937.5MHz		Units
		Min	Max	Min	Max	Min	Max	
Average clock period	$t_{CK(avg)}$	1.333	200	1.25	200	1066.7 ps	200ns	ns
Average High pulse width	$t_{CH(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Average Low pulse width	$t_{CL(avg)}$	0.46	0.54	0.46	0.54	0.46	0.54	$t_{CK(avg)}$
Absolute clock period	$t_{CK(abs)}$	$t_{CK(avg) MIN} + t_{JIT(per) MIN}$	-	$t_{CK(avg) MIN} + t_{JIT(per) MIN}$	-	$t_{CK(avg) MIN} + t_{JIT(per) MIN}$	-	ns
Absolute High clock pulse width	$t_{CH(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Absolute Low clock pulse width	$t_{CL(abs)}$	0.43	0.57	0.43	0.57	0.43	0.57	$t_{CK(avg)}$
Clock period jitter	$t_{JIT(per)}$	-75	75	-70	70	-60	60	ps
Maximum Clock Jitter between consecutive cycles	$t_{JIT(cc)}$	-	150	-	140	-	120	ps

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14.2 Write Clock Specification

The jitter specified is a random jitter meeting a Gaussian distribution. Input write clocks violating the min/max values may result in malfunction of the LPDDR5 device.

14.2.1 Definition for tWCK(avg) and nWCK

t_{WCK(avg)} is calculated as the average write clock period across any consecutive 200 cycle window, where each write clock period is calculated from rising edge to rising edge.

$$tWCK(avg) = \left(\sum_{j=1}^N tWCKj \right) / N$$

where $N = 200$

Unit "t_{WCK(avg)}" represents the actual write clock average t_{WCK(avg)} of the input write clock under operation. Unit "nWCK" represents one write clock cycle of the input write clock, counting the actual write clock edges. t_{WCK(avg)} may change by up to +/-1% within a 100 write clock cycle window, provided that all jitter and timing specs are met.

14.2.2 Definition for tWCK(abs)

t_{WCK(abs)} is defined as the absolute write clock period, as measured from one rising edge to the next consecutive rising edge.

t_{WCK(abs)} is not subject to production test.

14.2.3 Definition for tWCKH(avg) and tWCKL(avg)

t_{WCKH(avg)} is defined as the average high pulse width, as calculated across any consecutive 200 high pulses.

$$tWCKH(avg) = \left(\sum_{j=1}^N tWCKHj \right) / (N \times tWCK(avg))$$

where $N = 200$

t_{WCKL(avg)} is defined as the average low pulse width, as calculated across any consecutive 200 low pulses.

$$tWCKL(avg) = \left(\sum_{j=1}^N tWCKLj \right) / (N \times tWCK(avg))$$

where $N = 200$

14.2.4 Definition for tWCKH(abs) and tWCKL(abs)

t_{WCKH(abs)} is the absolute instantaneous write clock high pulse width, as measured from one rising edge to the following falling edge.

t_{WCKL(abs)} is the absolute instantaneous write clock low pulse width, as measured from one falling edge to the following rising edge.

Both t_{WCKH(abs)} and t_{WCKL(abs)} are not subject to production test.

14.2.5 Definition for tJIT(per)

t_{JIT(per)} is the single period jitter defined as the largest deviation of any signal t_{WCK} from t_{WCK(avg)}.

t_{JIT(per)} = Min/max of {t_{WCKi} - t_{WCK(avg)} where i = 1 to 200}.

t_{JIT(per)}, act is the actual write clock jitter for a given system.

t_{JIT(per)}, allowed is the specified allowed write clock period jitter.

t_{JIT(per)} is not subject to production test.

14.2.6 Definition for tJIT(cc)

t_{JIT(cc)} is defined as the absolute difference in write clock period between two consecutive write clock cycles.

t_{JIT(cc)} = Max of |{t_{WCK(i+1)} - t_{WCK(i)}}|.

t_{JIT(cc)} defines the cycle to cycle jitter.

t_{JIT(cc)} is not subject to production test.

14.2.7 Definition for tERR(2per)

t_{ERR(2per)} is defined as the cumulative error across 2 consecutive cycles from t_{WCK(avg)}. t_{ERR(2per)} is not subject to production test.

$$tERR(nper) = \left(\sum_{j=1}^{i+n-1} tWCKj \right) - n \times tWCK(avg)$$

where N = 2

14.2.8 Definition for tERR(3per)

t_{ERR(3per)} is defined as the cumulative error across 3 consecutive cycles from t_{WCK(avg)}. t_{ERR(3per)} is not subject to production test.

$$tERR(nper) = \left(\sum_{j=1}^{i+n-1} tWCKj \right) - n \times tWCK(avg)$$

where N = 3

14.2.9 Definition for tERR(4per)

t_{ERR(4per)} is defined as the cumulative error across 4 consecutive cycles from t_{WCK(avg)}. t_{ERR(4per)} is not subject to production test.

$$tERR(nper) = \left(\sum_{j=1}^{i+n-1} tWCKj \right) - n \times tWCK(avg)$$

where N = 4

14.2.10 Write Clock Timing

[Table 78] Write Clock AC Timings for 266/533/800/1067MHz

Parameter	Symbol	266MHz		533MHz		800MHz		1067MHz		Units
		Min	Max	Min	Max	Min	Max	Min	Max	
Write Clock Timing										
Average Write Clock period	t _{WCK} (avg)	3.76	50	1.877	50	1.25	50	0.938	50	ns
Average High pulse width	t _{WCKH} (avg)	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	t _{WCK} (avg)
Average Low pulse width	t _{WCKL} (avg)	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	t _{WCK} (avg)
Absolute Write Clock period	t _{WCK} (abs)	t _{WCK} (avg) MIN + t _{JIT} (per) MIN	-	t _{WCK} (avg) MIN + t _{JIT} (per) MIN	-	t _{WCK} (avg) MIN + t _{JIT} (per) MIN	-	t _{WCK} (avg) MIN + t _{JIT} (per) MIN	-	ns
Absolute High Write Clock pulse width	t _{WCKH} (abs)	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t _{WCK} (avg)
Absolute Low Write Clock pulse width	t _{WCKL} (abs)	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t _{WCK} (avg)
Write Clock period jitter t _{JIT} (per)	WCK period jitter t _{JIT} (per)	-190	190	-95	95	-70	70	-56	56	ps
Maximum Write Clock Jitter between consecutive cycles	t _{JIT} (cc)	-	380	-	190	-	140	-	112	ps
Cumulative error across 2 cycles	t _{ERR} (2per)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps
Cumulative error across 3 cycles	t _{ERR} (3per)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps
Cumulative error across 4 cycles	t _{ERR} (4per)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps

[Table 79] Write Clock AC Timings for 1375/1600/1867/2134MHz

Parameter	Symbol	1375MHz		1600MHz		1867MHz		2134MHz		Units
		Min	Max	Min	Max	Min	Max	Min	Max	
Write Clock Timing										
Average Write Clock period	t _{WCK(avg)}	0.728	50	0.625	50	0.536	50	0.469	50	ns
Average High pulse width	t _{WCKH(avg)}	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	t _{WCK} (avg)
Average Low pulse width	t _{WCKL(avg)}	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	t _{WCK} (avg)
Absolute Write Clock period	t _{WCK(abs)}	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	ns
Absolute High Write Clock pulse width	t _{WCKH(abs)}	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t _{WCK} (avg)
Absolute Low Write Clock pulse width	t _{WCKL(abs)}	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t _{WCK} (avg)
Write Clock period jitter t _{JIT(per)}	WCK period jitter t _{JIT(per)}	-46	46	-40	40	-34	34	-30	30	ps
Maximum Write Clock Jitter between consecutive cycles	t _{JIT(cc)}	-	92	-	80	-	68	-	60	ps
Cumulative error across 2 cycles	t _{ERR(2per)}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps
Cumulative error across 3 cycles	t _{ERR(3per)}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps
Cumulative error across 4 cycles	t _{ERR(4per)}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps

[Table 80] Write Clock AC Timings for 2400/2750/3000/3200MHz

Parameter	Symbol	2400MHz		2750MHz		3000MHz		3200MHz		Units
		Min	Max	Min	Max	Min	Max	Min	Max	
Write Clock Timing										
Average Write Clock period	t _{WCK(avg)}	0.417	50	0.364	50	0.334	50	0.313	50	ns
Average High pulse width	t _{WCKH(avg)}	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	t _{WCK(avg)}
Average Low pulse width	t _{WCKL(avg)}	0.46	0.54	0.46	0.54	0.46	0.54	0.46	0.54	t _{WCK(avg)}
Absolute Write Clock period	t _{WCK(abs)}	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	t _{WCK} (avg) MIN + t _{JIT(per)} MIN	-	ns
Absolute High Write Clock pulse width	t _{WCKH(abs)}	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t _{WCK(avg)}
Absolute Low Write Clock pulse width	t _{WCKL(abs)}	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t _{WCK(avg)}
Write Clock period jitter t _{JIT(per)}	WCK period jitter t _{JIT(per)}	-28	28	-26	26	-24	24	-22	22	ps
Maximum Write Clock Jitter between consecutive cycles	t _{JIT(cc)}	-	56	-	52	-	48	-	44	ps
Cumulative error across 2 cycles	t _{ERR(2per)}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps
Cumulative error across 3 cycles	t _{ERR(3per)}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps
Cumulative error across 4 cycles	t _{ERR(4per)}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	ps

[Table 81] Write Clock AC Timings for 3750MHz

Parameter	Symbol	3750MHz		Units
		Min	Max	
Write Clock Timing				
Average Write Clock period	t _{WCK(avg)}	266.7ps	50ns	-
Average High pulse width	t _{WCKH(avg)}	0.46	0.54	t _{WCK(avg)}
Average Low pulse width	t _{WCKL(avg)}	0.46	0.54	t _{WCK(avg)}
Absolute Write Clock period	t _{WCK(abs)}	t _{WCK(avg)} MIN + t _{JIT(per)} MIN	-	ns
Absolute High Write Clock pulse width	t _{WCKH(abs)}	0.43	0.57	t _{WCK(avg)}
Absolute Low Write Clock pulse width	t _{WCKL(abs)}	0.43	0.57	t _{WCK(avg)}
Write Clock period jitter t _{JIT(per)}	WCK period jitter t _{JIT(per)}	-19	19	ps
Maximum Write Clock Jitter between consecutive cycles	t _{JIT(cc)}	-	38	ps
Cumulative error across 2 cycles	t _{ERR(2per)}	TBD	TBD	ps
Cumulative error across 3 cycles	t _{ERR(3per)}	TBD	TBD	ps
Cumulative error across 4 cycles	t _{ERR(4per)}	TBD	TBD	ps

14.2.11 tWCK2DQ AC Parameters

[Table 82] tWCK2DQ AC parameters

Parameter	Symbol	Data Rate		Min/Max	Units	Notes
		≤ 6400Mbps	7500Mbps			
DQ to WCK input offset	t _{WCK2DQI_HF}	300/700	250/600	Min/Max	ps	
	t _{WCK2DQI_LF}	300/900	N/A	Min/Max	ps	
	t _{WCK2DQI_HF_L}	300/900	N/A	Min/Max	ps	4
WCK to DQ input offset temperature variation (Write)	t _{WCK2DQI_temp_HF}	0.6	0.5	Max	ps/°C	
	t _{WCK2DQI_temp_LF}	0.7	N/A	Max	ps/°C	
	t _{WCK2DQI_temp_HF_L}	0.7	N/A	Max	ps/°C	4
WCK to DQ input offset voltage variation (Write)	t _{WCK2DQI_volt_HF}	25	25	Max	ps/50mV	
	t _{WCK2DQI_volt_LF}	50	N/A	Max	ps/50mV	
	t _{WCK2DQI_volt_HF_L}	50	N/A	Max	ps/50mV	4
DQ to WCK output offset	t _{WCK2DQO_HF}	650/1600	650/1600	Min/Max	ps	5
	t _{WCK2DQO_LF}	650/1900	N/A	Min/Max	ps	5
	t _{WCK2DQO_HF_L}	650/1900	N/A	Min/Max	ps	
WCK to DQ output offset temperature variation (Read)	t _{WCK2DQO_temp_HF}	1.5	1.4	Max	ps/°C	
	t _{WCK2DQO_temp_LF}	1.8	N/A	Max	ps/°C	
	t _{WCK2DQO_temp_HF_L}	1.8	N/A	Max	ps/°C	4
WCK to DQ output offset voltage variation (Read)	t _{WCK2DQO_volt_HF}	3.0	2.5	Max	ps/mV	
	t _{WCK2DQO_volt_LF}	5.0	N/A	Max	ps/mV	
	t _{WCK2DQO_volt_HF_L}	5.0	N/A	Max	ps/mV	4

NOTE :

1) HF means high frequency and LF means low frequency.

2) ' _LF' is used up to 3200Mbps.

3) WCK2DQ AC parameters (HF/LF) can be selectable by MR18 OP[3].

4) This parameter value is applied when MR18 OP[3] and MR19 OP[1:0] set as follows at the same time.

- WCK Frequency Mode=High frequency mode: MR18 OP[3]=1_B.- Enhanced DVFS mode is enabled: MR19 OP[1:0]=10_B.

5) Depending on the frequency of operation, the value may change within the specified range. Hence, the system should undergo re-training to align with the updated timing for proper operation following a frequency change.

6) ' _HF_L' is used up to 3200Mbps.

7) Depending on the frequency of operation, the value exhibited by the SDRAM may change within the specified range (Informative).

[Table 83] WCK to CK/DQ offset Rank to Rank Variation

Item	Symbol	WCK Freq Mode	Min/ Max	Data Rate		Units	Note
				6400	7500		
WCK to CK offset rank to rank variation	$t_{WCK2CK_rank2rank}$	All Modes	Min	0	0	ps	1,2,3
			Max	100	90	ps	
WCK to DQ Input offset rank to rank variation	$t_{WCK2DQI_rank2rank}$	High Freq Mode	Min	0	0	ps	1,2,3, 4,5
			Max	150	140	ps	
		Low Freq Mode	Min	0	N/A	ps	
			Max	250	N/A	ps	
WCK to DQ Output offset rank to rank variation	$t_{WCK2DQO_rank2rank}$	High Freq Mode	Min	0	0	ps	1,2,3, 4,5
			Max	400	400	ps	
		Low Freq Mode	Min	0	N/A	ps	
			Max	650	N/A	ps	

NOTE :

- 1) The same voltage and temperature are applied to $t_{WCK2CK_rank2rank}$, $t_{WCK2DQI_rank2rank}$, $t_{WCK2DQO_rank2rank}$ AC parameters.
- 2) $t_{WCK2CK_rank2rank}$, $t_{WCK2DQI_rank2rank}$ and $t_{WCK2DQO_rank2rank}$ AC parameters are applied to multi-ranks DQ bytes per channel within a package consisting of the same design dies.
- 3) $t_{WCK2CK_rank2rank}$, $t_{WCK2DQI_rank2rank}$ and $t_{WCK2DQO_rank2rank}$ AC parameters are applied to multi byte mode dies DQ bytes per channel which share the same CK input within a package consisting of the same design dies.
- 4) MR18 OP[3]=0_B WCK Low Freq mode, MR18 OP[3]=1_B WCK High Freq mode.
- 5) WCK Low Freq mode is valid when Write Clock frequency is at equal or less than 1600MHz.

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